

PRACTICE SHEET

1. From past experience it is known that an investor will invest in security A with a probability of 0.6, will invest in security B with a probability 0.3 and will invest in both A and B with a probability of 0.2. What is the probability that an investor will invest neither in A nor in B?
(a) 0.7 (b) 0.28
(c) 0.3 (d) 0.4
2. Five coins whose faces are marked 2, 3 are thrown. What is the probability of obtaining a total of 12?
(a) 1/16 (b) 3/16
(c) 5/16 (d) 7/16
3. A coin is tossed three times. What is the probability of getting head and tail (HTH) or tail and head (THT) alternatively?
(a) 1/4 (b) 1/5
(c) 1/6 (d) 1/8
4. The probability that a student passes in mathematics is 4/9 and that he passes in physics is 2/5. Assuming that passing in mathematics and physics are independent of each other, what is the probability that he passes in mathematics but fails in physics?
(a) $\frac{4}{15}$ (b) $\frac{8}{45}$
(c) $\frac{26}{45}$ (d) $\frac{19}{45}$
5. From a pack of 52 cards, two cards are drawn, the first being replaced before the second is drawn. What is the probability that the first is a diamond and the second is a king?
(a) $\frac{1}{4}$ (b) $\frac{4}{13}$
(c) $\frac{1}{52}$ (d) $\frac{4}{15}$
6. What is the probability of having a knave and a queen when two cards are drawn from a pack of 52 cards?
(a) $\frac{16}{663}$ (b) $\frac{2}{663}$
(c) $\frac{4}{663}$ (d) $\frac{8}{663}$
7. Consider the following statement:
"The mean of a binomial distribution is 3 and variance is 4."
Which of the following is correct regarding the statement?
(a) It is always true
(b) It is sometimes true
(c) It is never true
(d) No conclusion can be drawn
8. In throwing of two dice, what is the number of exhaustive events?
(a) 6 (b) 12
(c) 36 (d) 18
9. What is the probability of getting five heads and seven tails in 12 flips of a balanced coin?
(a) $C(12, 5)/(2^5)$ (b) $C(12, 5)/(2^7)$
(c) $C(12, 5)/(2^{12})$ (d) $C(12, 7)/(2^6)$
10. In a lottery, 16 tickets are sold and 4 prizes are awarded. If a person buys 4 tickets, what is the probability of his winning a prize?
(a) $\frac{4}{16^4}$ (b) $\frac{175}{256}$
(c) $\frac{1}{4}$ (d) $\frac{81}{256}$
11. If A and B are any two events such that $P(A \cup B) = 3/4$, $P(A \cap B) = 1/4$ and $P(\bar{A}) = 2/3$, where $\bar{A}$ stands for the complementary event of A, then what is $P(B)$?
(a) 1/3 (b) 2/3
(c) 1/9 (d) 2/9
12. A card is drawn from a pack of 52 cards and a gambler bets that it is a spade or an ace. Which one of the following are the odds against his winning this bet?
(a) 13 to 4 (b) 4 to 13
(c) 9 to 4 (d) 4 to 9
13. A can hit a target 4 times in 5 shots:
B can hit a target 3 times in 4 shots:
C can hit a target 2 times in 3 shots:
All the three fire a shot each. What is the probability that two shots are at least hit?
(a) 1/6 (b) 3/5
(c) 5/6 (d) 1/3
14. A box contains 10 identical electronic components of which 4 are defective. If 3 components are selected at random from the box in succession, without replacing the units already drawn, what is the probability that two of the selected components are defective?
(a) 1/5 (b) 5/24
(c) 3/10 (d) 1/40
15. Each of A and B tosses two coins. What is the probability that they get equal number of heads?
(a) $\frac{3}{16}$ (b) $\frac{5}{16}$
(c) $\frac{4}{16}$ (d) $\frac{6}{16}$
16. Six text books numbered 1, 2, 3, 4, 5 and 6 are arranged at random. What is the probability that the text books 2 and 3 will occupy consecutive places?
(a) 1/2 (b) 1/3
(c) 1/4 (d) 1/6
17. What is the probability that in a family of 4 children there will be at least one boy?
(a) $\frac{15}{16}$ (b) $\frac{3}{8}$
(c) $\frac{1}{16}$ (d) $\frac{7}{8}$
18. In a school there are 40% science students and the remaining 60% are arts students. It is known that 5% of the science students are girls and 10% of the arts students are girls. One

student selected at random is a girl. What is the probability that she is an arts student?

- (a) $1/3$ (b) $3/4$
(c) $1/5$ (d) $3/5$

19. Given $P(A \cup B) = \frac{5}{6}$, $P(A \cap B) = \frac{1}{3}$ and $P(\bar{B}) = \frac{1}{2}$. What

is $P(\bar{A})$?

- (a) $1/6$ (b) $1/3$
(c) $2/3$ (d) $1/2$

20. The outcomes of 5 tosses of a coin are recorded in a single sequences as H (head) and T (tail) for each toss. What is the number of elementary events in the sample space?

- (a) 5 (b) 10
(c) 25 (d) 32

21. Which of the following numbers is nearest to the probability that three randomly selected persons are born on three different days of the week?

- (a) 0.7 (b) 0.6
(c) 0.5 (d) 0.4

22. The chance of winning the race of the horse A is $1/5$ and that of horse B is $1/6$. What is the probability that the race will be won by A or B?

- (a) $1/30$ (b) $1/3$
(c) $11/30$ (d) $1/15$

23. What is the probability of two persons being born on the same day (ignoring date)?

- (a) $1/49$ (b) $1/365$
(c) $1/7$ (d) $2/7$

24. Consider the following statements related to the nature of Bayes' theorem

1. Bayes' theorem is a formula for computation of a conditional probability.

2. Bayes' theorem modifies an assumed probability of an event in the light of a related event which is observed.

Which of the statements given above is / are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

25. If A, B, C are any three arbitrary events, then which one of the following expressions shows that both A and B occur but not C?

- (a) $A \cap \bar{B} \cap \bar{C}$ (b) $A \cap B \cap \bar{C}$
(c) $A \cap B \cap C$ (d) $A \cap \bar{B} \cap C$

26. By Baye's theorem, which one of the following probabilities is calculated?

- (a) Prior probability
(b) Likelihood probability
(c) Posterior probability
(d) Conditional probability

27. Given that $P(A) = 1/3$, $P(B) = 1/4$, $P(A/B) = 1/6$, then what is $P(B/A)$ equal to?

- (a) $1/4$ (b) $1/8$
(c) $3/4$ (d) $1/2$

28. If A and B are two mutually exclusive and exhaustive events with $P(B) = 3P(A)$, then what is the value of $P(\bar{B})$?

- (a) $3/4$ (b) $1/4$
(c) $1/3$ (d) $2/3$

29. Two dice are thrown. What is the probability that the sum of the face equals or exceeds 10?

- (a) $1/12$ (b) $1/4$
(c) $1/3$ (d) $1/6$

30. When a card is drawn from a well shuffled pack of cards, what is the probability of getting a Queen?

- (a) $2/13$ (b) $1/13$
(c) $1/26$ (d) $1/52$

31. What is the probability of having 53 Sundays or 53 Mondays in a leap year?

- (a) $2/7$ (b) $3/7$
(c) $4/7$ (d) $5/7$

32. Three digital numbers are formed using the digits 0,2,4,6,8, A number is chosen at random out of these numbers. What is the probability that the number has the same digit?

- (a) $1/16$ (b) $1/25$
(c) $16/25$ (d) $1/645$

33. A lot of 4 white and 4 red balls is randomly divided into two halves. What is the probability that there will be 2 red and 2 white balls in each half?

- (a) $18/35$ (b) $3/35$
(c) $1/2$ (d) $2/25$

34. Two balls are selected from a box containing 2 blue and 7 red balls. What is the probability that at least one ball is blue?

- (a) $\frac{2}{9}$ (b) $\frac{7}{9}$
(c) $\frac{5}{12}$ (d) $\frac{7}{12}$

35. The probability of guessing a correct answer is $\frac{x}{12}$. If the

probability of not guessing the correct answer is $\frac{2}{3}$, then

what is x equal to?

- (a) 2 (b) 3
(c) 4 (d) 6

36. An observed event B can occur after one of the three events, A_1, A_2 . If $P(A_1) = P(A_2) = 0.4$, $P(A_3) = 0.2$ and $P(B/A_1) = 0.25$, $P(B/A_2) = 0.4$, $P(B/A_3) = 0.125$, what is the probability of A_1 after observing B?

- (a) $\frac{1}{3}$ (b) $\frac{6}{19}$
(c) $\frac{20}{57}$ (d) $\frac{2}{5}$

37. The probability distribution of random variable X with two missing probabilities p_1 and p_2 is given below:

X	P(X)
1	K
2	P_1
3	$4k$
4	P_2
5	$2k$

It is further given that $P(x \leq 2) = 0.25$ and $P(x \geq 4) = 0.35$. Consider the following statements.

I. $p_1 = p_2$ II. $p_2 + p_2 = P(X=3)$

Which of the above statements is/are correct?

- (a) Only I (b) Only II

- (c) Both I and II (d) Neither I nor II
38. The mean and variance of a binomial distribution are 8 and 4 respectively. What is $P(X=1)$ equal to?
- (a) $\frac{1}{2^{12}}$ (b) $\frac{1}{2^8}$
(c) $\frac{1}{2^6}$ (d) $\frac{1}{2^4}$
39. Three letters are randomly selected from the 26 capital letters of the English alphabet. What is the probability that the letter 'A' will not be included in the choice?
- (a) $\frac{1}{2}$ (b) $\frac{23}{26}$
(c) $\frac{12}{13}$ (d) $\frac{25}{26}$
40. A coin is tossed 10 times. The number of heads minus the number of tails in 10 tosses is considered as the outcome of the experiment. What is the number of points in the sample space?
- (a) 10 (b) 11
(c) 21 (d) 99
41. The probabilities of two events A and B are given as $P(A)=0.8$ and $P(B)=0.7$. What is the minimum value of $P(A \cap B)$?
- (a) 0 (b) 0.1
(c) 0.5 (d) 1
42. Two numbers X and Y are simultaneously drawn from the set $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$. What is the conditional probability of exactly one of the two numbers X and Y being even, given $(X + Y) = 15$?
- (a) 1 (b) $\frac{3}{4}$
(c) $\frac{1}{2}$ (d) $\frac{1}{4}$
43. If $P(A) = \frac{1}{3}$, $P(B) = \frac{3}{4}$ and $P(A \cup B) = \frac{11}{12}$, then what is $P(B/A)$?
- (a) $\frac{1}{6}$ (b) $\frac{4}{9}$
(c) $\frac{1}{2}$ (d) $\frac{1}{3}$
44. For a binomial distribution $B(n, p)$, $np=4$ and variance $npq = \frac{4}{3}$. What is the probability $P(x \geq 5)$ equal to?
- (a) $(\frac{2}{3})^6$ (b) $2^5/3^6$
(c) $(\frac{1}{3})^6$ (d) $28/36$

ANSWER KEY

1.	c	2.	c	3.	a	4.	a	5.	c	6.	c	7.	c	8.	c	9.	c	10.	c
11.	b	12.	c	13.	c	14.	c	15.	b	16.	b	17.	a	18.	b	19.	b	20.	b
21.	b	22.	c	23.	b	24.	c	25.	b	26.	d	27.	b	28.	b	29.	d	30.	b
31.	b	32.	b	33.	a	34.	a	35.	c	36.	c	37.	d	38.	a	39.	b	40.	b
41.	c	42.	a	43.	c	44.	d												

Solutions

Sol.1. (c)

As given $P(A) = 0.6$, $P(B) = 0.3$ and $P(A \cap B) = 0.2$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= 0.6 + 0.3 - 0.2 = 0.9 - 0.2 = 0.7$$

$$\text{So, } P(\text{neither in A nor in B}) = 1 - P(A \cup B)$$

$$= 1 - 0.7 = 0.3$$

Sol.2. (c)

Let E be the event of total of 12.

$E = (2, 2, 2, 3, 3), (2, 2, 3, 3, 2), (2, 3, 3, 2, 2), (3, 3, 2, 2, 2), (3, 2, 3, 2, 2), (3, 2, 2, 3, 2), (3, 2, 2, 2, 3), (2, 3, 2, 3, 2), (2, 3, 2, 2, 3), (2, 2, 3, 2, 3)$

$$n(E) = 10$$

Sample space contains total possibility $= 2^5 = 32$

$$\text{Hence, } n(S) = 32$$

$$\text{So, } P(E) = \frac{n(E)}{n(S)} = \frac{10}{32} = \frac{5}{16}$$

Sol.3. (a)

Total possible outcomes, $S = \{HHH, HHT, HTH, THT, TTH, THH, TTT, HTT\}$ and desired outcomes $E = \{HTH, THT\}$

$$\Rightarrow n(E) = 2 \text{ and } n(S) = 8$$

Hence, required probability =

$$P(E) = \frac{n(E)}{n(S)} = \frac{2}{8} = \frac{1}{4}$$

Sol.4. (a)

Probability of passing in mathematics $= \frac{4}{9}$

Probability of passing in physics $= \frac{2}{3}$

Probability of failure in physics =

$$1 - \frac{2}{3} = \frac{1}{3}$$

Given that both the events are independent.

$$\text{Required probability} = \frac{4}{9} \times \frac{1}{3} = \frac{4}{27}$$

Sol.5. (c)

Probability of getting a diamond,

$$P(D) = \frac{13}{52} = \frac{1}{4}$$

$$\text{and probability to king } P(K) = \frac{13}{52} = \frac{1}{4}$$

So, required probability $= P(D) \cdot P(K)$

$$= \frac{1}{4} \times \frac{1}{4} = \frac{1}{16}$$

Sol.6. (c)

$$\text{Required probability} = \frac{4}{52} \times \frac{4}{51}$$

[Since, card is not replaced after first draw]

$$= \frac{4}{13 \times 51} = \frac{4}{663}$$

Sol.7. (c)

Given, that $np = 3$ and $npq = 4$

Where p is the probability of success in one trial and q is the probability of failure and n is the number of trials

$$\Rightarrow q = \frac{4}{3} \text{ and this is not possible.}$$

Thus, the given statement is never true.

Sol.8. (c)

A dice has six faces. So, in throwing of two dice, the number of exhaustive events is $6 \times 6 = 36$.

Sol.9. (c)

Number of ways of selecting 5 heads out of total 12 flips $= {}^{12}C_5$

Probability of getting one head in a coin $= \frac{1}{2}$

Also, probability of getting one tail in a coin = $1/2$

$$\text{Probability of getting 5 head} = \left(\frac{1}{2}\right)^5$$

$$\text{Probability of getting 7 tails} = \left(\frac{1}{2}\right)^7$$

So, required probability

$$= 12C_5 \left(\frac{1}{2}\right)^5 \left(\frac{1}{2}\right)^7 = 12C_5 \left(\frac{1}{2}\right)^{12} = \frac{12C_5}{2^{12}}$$

Sol.10. (c)

16 tickets are sold and 4 prizes are awarded. A person buys 4 tickets, then required probability

$$= \frac{4}{16} = \frac{1}{4}$$

Sol.11. (b)

As given, $P(A \cup B) = \frac{3}{4}$, $P(A \cap B) = \frac{1}{4}$ and

$$P(\bar{A}) = \frac{2}{3}$$

$$P(\bar{A}) = 1 - P(A) = \frac{2}{3} \Rightarrow P(A) = \frac{1}{3}$$

We know that

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\Rightarrow \frac{3}{4} = \frac{1}{3} + P(B) - \frac{1}{4}$$

$$\Rightarrow 1 - \frac{1}{3} = P(B) \Rightarrow \frac{2}{3} = P(B)$$

Sol.12. (c)

$$\text{Probability of getting a spade} = \frac{13}{52}$$

$$\text{Probability of an ace} = \frac{4}{52}$$

and probability of getting a spade ace =

$$\frac{1}{52}$$

$\therefore$ Required probability

$$= \frac{13}{52} + \frac{4}{52} - \frac{1}{52} = \frac{16}{52} = \frac{4}{13}$$

Odds against his winning

$$= \frac{1 - \frac{4}{13}}{\frac{4}{13}} = \frac{\frac{9}{13}}{\frac{4}{13}} = \frac{9}{4}$$

Sol.13. (c)

Probability of no one hitting the target

$$= \frac{1}{5 \times 4 \times 3} = \frac{1}{60}$$

Probability of one hitting the target

$$= \frac{4+3+2}{60} = \frac{9}{60}$$

$\therefore$ Probability of maximum one hit

$$= \frac{1}{60} + \frac{9}{60} = \frac{10}{60} = \frac{1}{6}$$

Probability that two shots are hit at least is

$$\text{the required probability} = 1 - \frac{1}{6} = \frac{5}{6}$$

Sol.14. (c)

Total number of selecting 3 components out of 10 = $10C_3$. Out of 3 selected

components two defective pieces can be selected in $4C_2$ ways and one non -

defective piece will be selected in $6C_1$

ways, hence,

Required probability

$$= \frac{{}^6C_1 \times {}^4C_2}{{}^{10}C_3} = \frac{6 \times 6 \times 6}{10 \times 9 \times 8} = \frac{3}{10}$$

Sol.15. (b)

If both get one head then it is $\frac{1}{4} \times \frac{1}{4}$ and if

both get two heads then it is $\frac{1}{2} \times \frac{1}{2}$

$\Rightarrow$ Prob (getting same number of heads)

$$\frac{1}{4} \times \frac{1}{4} + \frac{1}{2} \times \frac{1}{2} = \frac{1}{16} + \frac{1}{4} = \frac{5}{16}$$

Sol.16. (b)

Total number of possible arrangements n (s) = 6!

Since 2 and 3 occupy consecutive places so, they are grouped together.

So, there will be 5! Such arrangements. But 2 and 3 can be arranged in themselves in 2! Ways

$$\text{Required Probability} = \frac{5! \times 2!}{6!} = \frac{2}{6} = \frac{1}{3}$$

Sol.17. (a)

Total possibility of 4 children, either girl or boy is $2^4 = 16$. Out of these there is one possibility in which there will be no boy and only girls. So, total possibility of at least one boy is $16 - 1 = 15$

$$\Rightarrow P(\text{at least one boy}) = \frac{15}{16}$$

Sol.18. (b)

Let there be 100 students.

So, there are 40 students of science and 60 students of arts.

5% of 40 = 2 science students (girls)

10% of 60 = 6 science students (girls)

Total girls students = 8

If a girl is chosen then $P(\text{arts}) = 6/8 = 3/4$

Sol.19. (b)

As given: $P(A \cup B) = 5/6$, $P(A \cap B) = 1/3$

$$\text{and } P(\bar{B}) = \frac{1}{2}$$

$$P(B) = 1 - P(\bar{B}) = 1 - \frac{1}{2} = \frac{1}{2}$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\Rightarrow 5/6 = P(A) + 1/2 - 1/3$$

$$\Rightarrow 5/6 = 1 - P(\bar{A}) + 1/2 - 1/3$$

$$\Rightarrow P(\bar{A}) = 1 + 1/2 - 1/3 - 5/6$$

$$= \frac{6+3-2-5}{6} = \frac{2}{6} = \frac{1}{3}$$

Sol.20. (b)

A coin has two faces and is tossed 5 times. So, number of elements in the sample space = 10.

Sol.21. (b)

There are 7 days in a week. If 1st person's birth day falls on any day out of 7. So, probability is $7/7$. Since birthday of second person will fall on any of the remaining six days then its probability = $6/7$. And, birthday of 3rd person will fall on any of the remaining 5 days so, its probability = $5/7$.

$\Rightarrow$ Probability that all three persons will have different day as their birthday.

$$= \frac{7}{7} \times \frac{6}{7} \times \frac{5}{7} = \frac{30}{49} = 0.612 \approx 0.60$$

Sol.22. (c)

Let P (A) be the probability that the race will be won by A and P(B) be the probability that the race will be won by B.

$$\therefore P(A) = \frac{1}{5} \text{ and } P(B) = \frac{1}{6}$$

$\therefore$ Probability that the race will be won by

$$A \text{ or } B = P(A) + P(B) = \frac{1}{5} + \frac{1}{6} = \frac{11}{30}$$

Sol.23. (b)

$$\text{Required probability} = \frac{365}{365} \times \frac{1}{365} = \frac{1}{365}$$

Sol.24. (c)

Baye's theorem says

If $A_1, A_2, \dots, A_n$ are (n) mutually exclusive and exhaustive events in sample space S and E is any event in S intersecting events.

A_i ($i = 1, 2, \dots, n$) such that $P(E) \neq 0$ then

$$P(A_i/E) = \frac{P(E/A_i)}{P(E)}$$

$$= \frac{P(A_i)P(E/A_i)}{\sum_{i=1}^n P(A_i)P(E/A_i)}$$

Thus, both statement 1 and 2 are correct.

Sol.25. (b)

If A, B, C are any three arbitrary events

then only expression $A \cap B \cap \bar{C}$ will show that both A and B occur but not C.

Sol.26. (d)

By Baye's theorem, we know that, conditional probability is calculated.

Sol.27. (b)

$$\text{Given, } P(A) = 1/3, P(B) = 1/4, P\left(\frac{A}{B}\right) = \frac{1}{6}$$

$$\text{But } P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)}$$

$$\Rightarrow \frac{1}{6} = \frac{P(A \cap B)}{\frac{1}{4}}$$

$$\Rightarrow P(A \cap B) = \frac{1}{24}$$

$$\therefore P\left(\frac{B}{A}\right) = \frac{P(A \cap B)}{P(A)} = \frac{1/24}{1/3} = \frac{1}{8}$$

Sol.28. (b)

A and B are mutually exclusive and exhaustive events with

$$P(A \cap B) = 0, P(A \cup B) = 1$$

We know that

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\Rightarrow 1 = P(A) + 3P(A)$$

$$\Rightarrow P(A) = 1/4 \therefore P(B) = 3/4$$

$$\text{Hence, } P(\bar{B}) = 1 - P(B) = 1 - \frac{3}{4} = \frac{1}{4}$$

Sol.29. (d)

Let E be the sum of the faces equals or exceeds

Then, $E = \{(5,5), (4,6), (6,4), (5,6), (6,5), (6,6)\}$

$$\therefore n(E) = 6$$

$$\text{Hence, } P(E) = \frac{n(E)}{n(S)} = \frac{6}{36} = \frac{1}{6}$$

Sol.30. (b)

Since, $n(S) = 52$ and $n(E) = 4$

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{4}{52} = \frac{1}{13}$$

Sol.31. (b)

A leap year has 366 days, in which 2 days may be any one of the following pairs.

(Sunday, Monday), (Monday, Tuesday), (Tuesday, Wednesday), (Wednesday, Thursday), (Thursday, Friday), (Friday, Saturday), (Saturday, Sunday)

Sol.32. (b)

Favorable numbers = {222, 444, 666, 888}

Total digit numbers = $4 \times 5 \times 5$

$$\therefore \text{Required probability} = \frac{4}{4 \times 25} = \frac{1}{25}$$

Sol.33. (a)

$$\frac{{}^4C_2 \cdot {}^4C_2}{{}^4C_0 \cdot {}^4C_4 + {}^2 \cdot {}^4C_1 \cdot {}^4C_3 + {}^4C_2 \cdot {}^4C_2} = \frac{36}{70} = \frac{18}{35}$$

Sol.34. (a)

Required probability

= P(One balls is blue) + P(both balls are blue)

$$= \frac{2}{9} \times \frac{7}{8} + \frac{2}{9} \times \frac{1}{8} = \frac{14}{72} + \frac{2}{72} = \frac{16}{72} = \frac{2}{9}$$

Sol.35. (c)

$$\frac{x}{12} + \frac{2}{3} = 1$$

$$\Rightarrow x = 4$$

Sol.36. (c)

Required probability = $P(A_1/B)$

$$= \frac{P(A_1) P(B/A_1)}{P(A_1) P(B/A_1) + P(A_2) P(B/A_2) + P(A_3) P(B/A_3)}$$

$$= \frac{0.4 \times 0.25}{0.4 \times 0.25 + 0.4 \times 0.4 + 0.2 \times 0.125}$$

$$= \frac{0.1}{0.1 + 0.16 + 0.025} = \frac{0.1}{0.285} = \frac{20}{57}$$

Sol.37. (d)

$$\therefore P(X \leq 2) = 0.25$$

$$\Rightarrow P(X=1)P + P(X=2) = 0.25$$

$$\Rightarrow k + p_1 = 0.25 \Rightarrow p_1 = 0.25 - k$$

$$\text{And } P(X \geq 4) = 0.35$$

$$\Rightarrow P(X=4) + P(X=5) = 0.35$$

$$\Rightarrow p_2 + 2k = 0.35$$

$$\Rightarrow p_2 = 0.35 - 2k \Rightarrow p_1 \neq p_2$$

$$\text{and } p_1 + p_2 = 0.25 - k + 0.35 - 2k = 0.6 - 3k$$

$$(X = 3)$$

Hence, neither I nor II is correct.

Sol.38. (a)

Here, $np = 8$ and $npq = 4$

$$\Rightarrow q = \frac{npq}{np} = \frac{4}{8} = \frac{1}{2}$$

$$\text{and } p + q = 1$$

$$\Rightarrow p = 1 - \frac{1}{2} = \frac{1}{2}$$

$$\Rightarrow n \left(\frac{1}{2} \right) = 8$$

$$\Rightarrow n = 16$$

$$\text{Now, } P(X=1) = {}^{10}C_1 \left(\frac{1}{2} \right)^1 \left(\frac{1}{2} \right)^{16-1}$$

$$= \frac{16}{2^{15} \cdot 2} = \frac{1}{2^{12}}$$

Sol.39. (b)

$$\therefore \text{Required probability} = \frac{{}^{25}C_3}{{}^{26}C_3} = \frac{23}{26}$$

Sol.40. (b)

{(10,0), (9,1), (8,2), (7,3), (6,4), (5,5), (4,6), (3,7), (2,8), (1,9), (0,10)}

total 11 point in sample space

Sol.41. (c)

Since, $P(A \cup B) \leq 1$

$$\Rightarrow P(A) + P(B) - P(A \cap B) \leq 1$$

$$\Rightarrow 0.8 + 0.7 - P(A \cap B) \leq 1$$

$$\Rightarrow P(A \cap B) \geq 1.5 - 1$$

$$\Rightarrow P(A \cap B) \geq 0.5$$

Sol.42. (a)

$X + Y = \text{odd number}$

it is only possible when one of them is odd and other is even

so probability is 1

Sol.43. (c)

$$P\left(\frac{B}{A}\right) = \frac{P(A \cap B)}{P(A)} = \frac{P(A) + P(B) - P(A \cup B)}{P(A)}$$

$$= \frac{\frac{1}{2} + \frac{3}{4} - \frac{11}{12}}{\frac{1}{2}} = \frac{\frac{12}{12} + \frac{9}{12} - \frac{11}{12}}{\frac{1}{2}} = \frac{\frac{10}{12}}{\frac{1}{2}} = \frac{10}{6} = \frac{5}{3}$$

Sol.44. (d)

$$np = 4, npq = 4/3$$

$$q = 1/3 \text{ than } p = 2/3 \text{ and } n = 6$$

$$P(X \geq 5) = {}^6C_3 \left(\frac{2}{3} \right)^3 \left(\frac{1}{3} \right)^3 + {}^6C_6 \left(\frac{2}{3} \right)^6 \left(\frac{1}{3} \right)^0$$

$$= 6 \cdot \frac{2^5}{3^6} + \frac{2^6}{3^6} = 3 \cdot \frac{2^6}{3^6} + \frac{2^6}{3^6} = 4 \cdot \frac{2^6}{3^6} = \frac{2^8}{3^6}$$

NDA PYQ

1. A experiment consists of flipping a coin and then flipping it a second time if head occurs. If a tail occurs on the first flip, then a six-faced die is tossed once. Assuming that the outcomes are equally likely. What is the probability of getting one head and one tail?
 (a) $\frac{1}{4}$ (b) $\frac{1}{36}$
 (c) $\frac{1}{6}$ (d) $\frac{1}{8}$
[NDA (I) - 2011]
2. If A and B are events such that $P(A \cup B) = 0.5$, $P(\bar{B}) = 0.8$ and $P(A/B) = 0.4$, then what is $P(A \cap B)$ equal to:
 (a) 0.08 (b) 0.02
 (c) 0.8 (d) 0.2
[NDA (I) - 2011]
3. A box contains 6 distinct dolls. From this box, three dolls are randomly selected one by one with replacement. What is the probability of selecting 3 distinct dolls?
 (a) $\frac{5}{54}$ (b) $\frac{12}{25}$
 (c) $\frac{1}{20}$ (d) $\frac{5}{9}$
[NDA (I) - 2011]
4. There are 4 letters and 4 directed envelopes. These 4 letters are randomly inserted into the 4 envelopes. What is the probability that the letter are inserted into the corresponding envelopes?
 (a) $\frac{11}{12}$ (b) $\frac{23}{24}$
 (c) $\frac{1}{24}$ (d) none of these
[NDA (I) - 2011]
5. In an examination, there are 3 multiple-choice questions and each question has 4 choices. If a student randomly. Select an answer for all the three question, what is the probability that the students will not answer all the three questions correctly?
 (a) $\frac{1}{64}$ (b) $\frac{63}{64}$
 (c) $\frac{1}{12}$ (d) $\frac{11}{12}$
[NDA (I) - 2011]
6. If A and B are two mutually exclusive events, then what is $P\{AB\}$ equal to?
 (a) 0 (b) $P(A) + P(B)$
 (c) $P(A) P(B)$ (d) $P(A)P(B|A)$
[NDA-2011(1)]
7. Two letters are drawn at random from the word 'HOME'. What is the probability that both the letters are vowels?
 (a) $\frac{1}{6}$ (b) $\frac{5}{6}$
 (c) $\frac{3}{5}$ (d) $\frac{1}{3}$
[NDA (II) - 2011]
8. There is a point inside a circle. What is the probability that this point is close to the circumference than to the center?
 (a) $\frac{3}{4}$ (b) $\frac{1}{2}$
 (c) $\frac{1}{4}$ (d) $\frac{1}{3}$
[NDA (II) - 2011]
9. In a class of 125 students 70 passed in mathematics, 55 passed in statistics and 30 passed in both. What is the probability that a student selected at random from the class has passed in only one subject?
 (a) $\frac{12}{25}$ (b) $\frac{13}{25}$
 (c) $\frac{17}{25}$ (d) $\frac{8}{25}$
[NDA (II) - 2011]
10. A husband and wife appear in an interview for two vacancies in the same post. The probability of husband's selection is $\frac{1}{5}$ and that of wife selection is $\frac{1}{3}$. What is the probability that only one of them will be selected?
 (a) $\frac{1}{5}$ (b) $\frac{2}{5}$
 (c) $\frac{3}{5}$ (d) $\frac{4}{5}$
[NDA (II) - 2011]
11. In a random arrangement of the letters of the word UNIVERSITY. What is the probability that two I's do not come together?
 (a) $\frac{4}{5}$ (b) $\frac{1}{5}$
 (c) $\frac{1}{10}$ (d) $\frac{9}{10}$
[NDA (II) - 2011]
12. Three dice are thrown. What is the probability that the same number will appear on each of them?
 (a) $\frac{1}{6}$ (b) $\frac{1}{18}$
 (c) $\frac{1}{24}$ (d) $\frac{1}{36}$
[NDA-2012(1)]
13. What is the probability that a leap year selected at random contains 53 Mondays?
 (a) $\frac{1}{7}$ (b) $\frac{2}{7}$
 (c) $\frac{7}{366}$ (d) $\frac{26}{183}$
[NDA-2012(1)]
14. If four dice are thrown together, then what is the probability that the sum of the numbers appearing on them is 25?
 (a) 0 (b) $\frac{1}{2}$
 (c) 1 (d) $\frac{1}{1296}$
[NDA-2012(1)]
15. If $P(E)$ denotes the probability of an event E, then, E is called certain event, if
 (a) $P(E) = 0$ (b) $P(E) = 1$
 (c) $P(E)$ is either 0 or 1 (d) $P(E) = \frac{1}{2}$
[NDA (II) - 2012]
16. If A and B are two events, such that $P(A \cup B) = \frac{3}{4}$, $P(A \cap B) = \frac{1}{4}$ and $P(\bar{A}) = \frac{2}{3}$, where $\bar{A}$ is the complement of A, then what is the value of $P(B)$?
 (a) $\frac{1}{3}$ (b) $\frac{2}{3}$
 (c) $\frac{1}{9}$ (d) $\frac{2}{9}$
[NDA (II) - 2012]
17. Three coins are tossed simultaneously. What is the probability that they will fall two heads and one tail?
 (a) $\frac{1}{3}$ (b) $\frac{1}{2}$
 (c) $\frac{1}{4}$ (d) $\frac{3}{8}$
[NDA (II) - 2012]
18. Which one of the following is correct?
 (a) An event having no sample point is called an elementary event
 (b) An event having one sample point is called an elementary event.
 (c) An event having two sample points is called an elementary event.
 (d) An event having many sample points is called an elementary event.

- [NDA (II) - 2012]**
19. What is the most probable number of successes in 10 trials with probability of success $\frac{2}{3}$?
 (a) 10 (b) 7
 (c) 5 (d) 4
- [NDA (II) - 2012]**
- Direction (20 to 21):** An urn contains one black ball and one green ball. A second urn contains one white and one green ball. One ball is drawn at random from each urn.
20. What is the probability that both balls are of same colour?
 (a) $\frac{1}{2}$ (b) $\frac{1}{3}$
 (c) $\frac{1}{4}$ (d) $\frac{2}{3}$
- [NDA (II) - 2012]**
21. What is the probability of getting at least one green ball?
 (a) $\frac{1}{2}$ (b) $\frac{1}{3}$
 (c) $\frac{2}{3}$ (d) $\frac{3}{4}$
- [NDA (II) - 2012]**
- Direction (for next two):** Two dice each numbered from 1 to 6 are thrown together. Let A and B be two events given by:
 A: Even number on the first die.
 B: Number on the second die is greater than 4.
22. What is the value of $P(A \cup B)$?
 (a) $\frac{1}{2}$ (b) $\frac{1}{4}$
 (c) $\frac{2}{3}$ (d) $\frac{1}{6}$
- [NDA (II) - 2012]**
23. What is the value of $P(A \cap B)$?
 (a) $\frac{1}{2}$ (b) $\frac{1}{4}$
 (c) $\frac{2}{3}$ (d) $\frac{1}{6}$
- [NDA (II) - 2012]**
24. What is the probability that a leap year selected at random will contain 53 Mondays?
 (a) $\frac{2}{5}$ (b) $\frac{2}{7}$
 (c) $\frac{1}{7}$ (d) $\frac{5}{7}$
- [NDA (II) - 2012]**
25. In throwing a six faced die, let A be the event that an even number and B be the event that an odd number occurs and C be the event that a number greater than 3 occurs. Which one of the following is correct?
 (a) A and C are mutually exclusive
 (b) A and B are mutually exclusive
 (c) B and C are mutually exclusive
 (d) A, B and C are mutually exclusive
- [NDA (I) - 2013]**
26. Consider a random experiment of throwing together a die and two coins. The associated sample space has?
 (a) 8 points (b) 12 points
 (c) 24 points (d) 36 points
- [NDA (I) - 2013]**
27. If A and B are two events such that $P(A') = 0.4$, $P(B') = 0.3$, $P(A \cup B) = 0.9$, then what is the value of $P(\overline{A} \cap \overline{B})$?
 (a) 0.2 (b) 0.5
 (c) 0.6 (d) 0.7
- [NDA (I) - 2013]**
28. A fair coin is tossed repeatedly. The probability of getting a result in the fifth toss different from those obtained in the first four tosses is:
 (a) $\frac{1}{2}$ (b) $\frac{1}{32}$
 (c) $\frac{31}{32}$ (d) $\frac{1}{16}$
- [NDA (I) - 2013]**
29. What is the probability of getting a sum of 7 with two dice?
 (a) $\frac{1}{6}$ (b) $\frac{1}{3}$
 (c) $\frac{1}{12}$ (d) $\frac{5}{36}$
- [NDA (I) - 2013]**
30. Four coins are tossed simultaneously. What is the probability of getting exactly 2 heads?
 (a) $\frac{1}{2}$ (b) $\frac{1}{4}$
 (c) $\frac{1}{8}$ (d) $\frac{3}{8}$
- [NDA (I) - 2013]**
31. A bag contains 5 black and 3 white balls. Two balls are drawn at random one after the other without replacement. What is the probability that both are white?
 (a) $\frac{1}{28}$ (b) $\frac{1}{14}$
 (c) $\frac{3}{28}$ (d) $\frac{1}{21}$
- [NDA (I) - 2013]**
32. If X follows a binomial distribution with parameters $n=100$ and $p=\frac{1}{3}$, then $P(X=r)$ is maximum when:
 (a) $r=16$ (b) $r=32$
 (c) $r=33$ (d) $r=34$
- [NDA-2013(1)]**
33. Two numbers are successively drawn from the set $U = \{1, 2, 3, 4, 5, 6, 7, 8\}$, the second being drawn without replacing the first. The number of elementary events in the sample is:
 (a) 64 (b) 56
 (c) 32 (d) 14
- [NDA-2013(1)]**
34. The binomial distribution has:
 (a) only one parameter (b) two parameter
 (c) three parameter (d) four parameter
- [NDA-2013(1)]**
35. In a binomial distribution, the occurrence and the non-occurrence of an event are equally likely and the mean is 6. The number of trials required is:
 (a) 15 (b) 12
 (c) 10 (d) 6
- [NDA (II) - 2013]**
36. A die is tossed twice. What is the probability of getting a sum of 10?
 (a) $\frac{1}{18}$ (b) $\frac{1}{6}$
 (c) $\frac{1}{12}$ (d) $\frac{5}{12}$
- [NDA (II) - 2013]**
37. Three dice are thrown. What is the probability of getting as triplet?
 (a) $\frac{1}{6}$ (b) $\frac{1}{18}$
 (c) $\frac{1}{36}$ (d) $\frac{1}{72}$
- [NDA (II) - 2013]**
38. What is the number of outcomes when a coin is tossed and then a die is rolled only in case of head is shown on the coin?
 (a) 6 (b) 7
 (c) 8 (d) 5
- [NDA (II) - 2013]**
39. Which one of the following may be the parameter of a binomial distribution?
 (a) $np = 2$, $npq = 4$ (b) $n = 4$, $p = \frac{3}{2}$
 (c) $n = 8$, $p = 1$ (d) $np = 10$, $npq = 8$
- [NDA (II) - 2013]**

40. If A and B are independent events such that $P(A) = \frac{1}{5}$
 $P(A \cup B) = \frac{7}{10}$, then what is $P(\bar{B})$ equal to?
 (a) $\frac{2}{7}$ (b) $\frac{3}{7}$
 (c) $\frac{3}{8}$ (d) $\frac{7}{9}$ [NDA (II) - 2013]
41. Consider the following statements
 I. If A and B are exhaustive events, then their union is the sample space.
 II. If A and B are exhaustive events, then their intersection must be an empty event.
 Which of the above statements is/are correct?
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II [NDA (II) - 2013]
42. If $P(A) = \frac{2}{3}$, $P(B) = \frac{2}{5}$ and $P(A \cup B) - P(A \cap B) = \frac{2}{5}$, then what is $P(A \cap B)$ equal to?
 (a) $\frac{3}{5}$ (b) $\frac{5}{11}$
 (c) $\frac{1}{3}$ (d) $\frac{3}{11}$ [NDA (II) - 2013]
43. If three events, A, B, C are mutually exclusive, then which one of the following is correct?
 (a) $P(A \cup B \cup C) = 0$ (b) $P(A \cup B \cup C) = 1$
 (c) $P(A \cap B \cap C) = 0$ (d) $P(A \cap B \cap C) = 1$ [NDA-2013(2)]
44. What is the probability that there are 5 Mondays in the month of February 2016?
 (a) 0 (b) $\frac{1}{7}$
 (c) $\frac{2}{7}$ (d) none of the above [NDA-2013(2)]
45. In a relay race, there are six teams A, B, C, D, E and F. What is the probability that A, B, C finish first, second, third respectively?
 (a) $\frac{1}{2}$ (b) $\frac{1}{12}$
 (c) $\frac{1}{60}$ (d) $\frac{1}{120}$ [NDA-2013(2)]
46. It has been found that, if A and B play a game 12 times, A wins 6 times, B wins 4 times and they draw twice. A and B take part in a service of 3 games. The probability that they win alternately, is:
 (a) $\frac{5}{12}$ (b) $\frac{5}{36}$
 (c) $\frac{19}{27}$ (d) $\frac{5}{27}$ [NDA (I) - 2014]
47. A box contains 3 white and 2 black balls. Two balls are drawn at random one after the other. If the balls are not replaced, what is the probability that both the balls are black?
 (a) $\frac{2}{5}$ (b) $\frac{1}{5}$
 (c) $\frac{1}{10}$ (d) None of these [NDA (I) - 2014]
- Direction (for next four):** Number X is randomly selected from the set of odd numbers and Y is randomly selected from the set of even numbers of the set {1, 2, 3, 4, 5, 6, 7}. Let, $Z = (X + Y)$.
48. What is $P(Z = 5)$ equal to?
 (a) $\frac{1}{2}$ (b) $\frac{1}{3}$
 (c) $\frac{1}{4}$ (d) $\frac{1}{6}$
49. What is $P(Z = 10)$ equal to?
 (a) 0 (b) $\frac{1}{2}$
 (c) $\frac{1}{3}$ (d) $\frac{1}{5}$ [NDA (I) - 2014]
50. What is $P(Z > 11)$ equal to ?
 (a) 0 (b) $\frac{1}{4}$
 (c) $\frac{1}{6}$ (d) $\frac{1}{12}$ [NDA (I) - 2014]
51. What is $P(Z \text{ is the product of two prime numbers})$ equal to?
 (a) 0 (b) $\frac{1}{2}$
 (c) $\frac{1}{4}$ (d) None of these [NDA (I) - 2014]
52. Suppose A and B are two events. Event B has occurred and it is known that $P(B) < 1$. What is $P(A/B)$ equal to?
 (a) $\frac{P(A) - P(B)}{1 - P(B)}$ (b) $\frac{P(A) - P(AB)}{1 - P(B)}$
 (c) $\frac{P(A) + P(B')}{1 - P(B)}$ (d) None of these [NDA (II) - 2014]
53. The probability that in a random arrangement of the letters of the word 'UNIVERSITY', the two I's do not come together is:
 (a) $\frac{4}{5}$ (b) $\frac{1}{5}$
 (c) $\frac{1}{10}$ (d) $\frac{9}{10}$ [NDA (II) - 2014]
54. There are 4 white and 3 black balls in a box. In another box, there are 3 white and 4 black balls. An unbiased die is rolled. If it shows a number less than or equal to 3, then a ball is drawn from the second box, otherwise from the first box. If the balls drawn is black, then the probability that the ball was drawn from the first box, is:
 (a) $\frac{1}{2}$ (b) $\frac{6}{7}$
 (c) $\frac{4}{7}$ (d) $\frac{3}{7}$ [NDA (II) - 2014]
55. Two students X and Y appeared in an examination. The probability that X will qualify the examination is 0.05 and Y will qualify the examination is 0.10. The probability that both will qualify the examination is 0.02. What is the probability that only one of them will qualify the examination?
 (a) 0.15 (b) 0.14
 (c) 0.12 (d) 0.11 [NDA (II) - 2014]
56. A fair coin is tossed four times. What is the probability that at most three tails occur?
 (a) $\frac{7}{8}$ (b) $\frac{15}{16}$
 (c) $\frac{13}{16}$ (d) $\frac{3}{4}$ [NDA (II) - 2014]
- For the next four (04) items that follow:**
 Consider events A, B, C, D, E of the sample space $S = \{n : n \text{ is an integer such that } 10 \leq n \leq 20\}$ given by:
 A is the set of all even numbers
 B is the set of all prime numbers
 $C = \{15\}$
 D is the set of all integers ≤ 16
 E is the set of all double digit numbers expressible as a power of 2.
57. A, B and D are:
 (a) Mutually exclusive events but not exhaustive events
 (b) Exhaustive events but not mutually exclusive events
 (c) Mutually exclusive and exhaustive events
 (d) Elementary events [NDA-2014(2)]

58. A, B and C are:
(a) Mutually exclusive events but not exhaustive events
(b) Exhaustive events but not mutually exclusive events
(c) Mutually exclusive and exhaustive events
(d) Elementary events
[NDA-2014(2)]
59. B and C are:
(a) Mutually exclusive events but not exhaustive events
(b) Exhaustive events but not mutually exclusive events
(c) Mutually exclusive and exhaustive events
(d) Elementary events
[NDA-2014(2)]
60. C and E are
(a) Mutually exclusive events but not exhaustive events
(b) Exhaustive events but not mutually exclusive events
(c) Mutually exclusive and exhaustive events
(d) Elementary events
[NDA-2014(2)]
61. For any two events A and B, which one of the following holds?
(a) $P(A \cap B) \leq P(A) \leq P(A \cup B) \leq P(A) + P(B)$
(b) $P(A \cup B) \leq P(A) \leq P(A \cap B) \leq P(A) + P(B)$
(c) $P(A \cup B) \leq P(B) \leq P(A \cap B) \leq P(A) + P(B)$
(d) $P(A \cap B) \leq P(B) \leq P(A) + P(B) \leq P(A \cup B)$
[NDA-2014(2)]
62. Two similar boxes B_i ($i = 1, 2$) contain $(i + 1)$ red and $(5 - i - 1)$ black balls. One box is chosen at random and two balls are drawn randomly. What is the probability that both the balls are of different colours?
(a) $1/2$ (b) $3/10$
(c) $2/3$ (d) $3/5$
[NDA (I) - 2015]
63. In an examination, the probability of a candidate solving a question is $1/2$. Out of given 5 questions in the examination, what is the probability that the candidate was able to solve at least 2 question?
(a) $1/64$ (b) $3/16$
(c) $1/2$ (d) $13/16$
[NDA (I) - 2015]
64. If $A \subseteq B$, then which one of the following is not correct?
(a) $P(A \cap \bar{B}) = 0$ (b) $P(A|B) = \frac{P(A)}{P(B)}$
(c) $P(B|A) = \frac{P(B)}{P(A)}$ (d) $P(A|(A \cup B)) = \frac{P(A)}{P(B)}$
[NDA (I) - 2015]
65. The mean and the variance in a binomial distribution are found to be 2 and 1 respectively. The probability $P(X = 0)$ is:
(a) $1/2$ (b) $1/4$
(c) $1/8$ (d) $1/16$
[NDA (I) - 2015]
66. Two men hit at a target with probabilities $1/2$ and $1/3$, respectively. What is the probability that exactly one of them hits the target?
(a) $1/2$ (b) $1/3$
(c) $1/6$ (d) $2/3$
[NDA (I) - 2015]
67. If A and B are two events such that $P(A \cup B) = 3/4$, $P(A \cap B) = 1/4$ and $P(\bar{A}) = \frac{2}{3}$, then what is $P(B)$ equal to?
(a) $1/3$ (b) $2/3$ (c) $1/8$ (d) $2/9$
[NDA (I) - 2015]
68. In throwing of two dice, the number of exhaustive events that '5' will never appear on any one of the dice is?
(a) 5 (b) 18
(c) 25 (d) 36
[NDA (I) - 2015]
69. Two cards are drawn successively without replacement from a well shuffled pack of 52 cards. The probability of drawing two aces is:
(a) $\frac{1}{26}$ (b) $\frac{1}{221}$
(c) $\frac{4}{223}$ (d) $\frac{1}{13}$
[NDA (I) - 2015]
70. An unbiased coin is tossed until the first head appears or until four tosses are completed, whichever happens earlier. Which of the following statements is/are correct?
1. The probability that no head is observed is $1/16$.
2. The probability that the experiment ends with three tosses is $1/8$.
Select the correct answer using the codes given below:
(a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2
[NDA (II) - 2015]
71. A bag contains 4 white and 2 black balls and another bag contains 3 white and 5 black balls. If one ball is drawn from each bag, then the probability that one ball is white and one ball is black is:
(a) $5/24$ (b) $13/24$
(c) $1/4$ (d) $2/3$
[NDA (II) - 2015]
72. A problem in statistics is given to three students A, B and C whose chances of solving it independently are $\frac{1}{2}$, $\frac{1}{3}$ and $\frac{1}{4}$ respectively. The probability that the problem will be solved is:
(a) $1/12$ (b) $11/12$
(c) $1/2$ (d) $3/4$
[NDA (II) - 2015]
73. What is the probability that the sum of any two different single digit natural numbers is a prime number?
(a) $5/27$ (b) $7/18$
(c) $1/3$ (d) None of these
[NDA (II) - 2015]
74. Three digits are chosen at random from 1, 2, 3, 4, 5, 6, 7, 8 and 9 without repeating any digit. What is the probability that the product is odd?
(a) $\frac{2}{3}$ (b) $\frac{7}{48}$
(c) $\frac{5}{42}$ (d) $\frac{5}{108}$
[NDA (II) - 2015]
75. Two events A and B are such that $P(\text{not } B) = 0.8$, $P(A \cup B) = 0.5$ and $P(A|B) = 0.4$. Then $P(A)$ is equal to:
(a) 0.28 (b) 0.32
(c) 0.38 (d) None of these
[NDA (II) - 2015]

76. If mean and variance of a Binomial variate X are 2 and 1 respectively, then the probability that X takes a value greater than 1 is:
 (a) $\frac{2}{3}$ (b) $\frac{4}{5}$
 (c) $\frac{7}{8}$ (d) $\frac{11}{16}$ [NDA (II) - 2015]
77. Seven unbiased coins are tossed 128 times. In how many throws would you find at least three heads?
 (a) 99 (b) 102
 (c) 103 (d) 104 [NDA (II) - 2015]
78. A coin tossed five times. What is the probability that heads are observed more than three times?
 (a) $\frac{3}{16}$ (b) $\frac{5}{16}$
 (c) $\frac{1}{2}$ (d) $\frac{3}{32}$ [NDA (II) - 2015]
79. If $x \in [0, 5]$, then what is the probability that $x^2 - 3x + 2 \geq 0$?
 (a) $\frac{4}{5}$ (b) $\frac{1}{5}$
 (c) $\frac{2}{5}$ (d) $\frac{3}{5}$ [NDA (II) - 2015]
80. An insurance company insured 2000 scooter drivers, 4000 car drivers and 6000 truck drivers. The probabilities of an accident involving a scooter driver, car driver and a truck driver are 0.01, 0.03 and 0.15 respectively. One of the insured persons meets with an accident. The probability that the person is a scooter driver is:
 (a) $\frac{1}{52}$ (b) $\frac{3}{52}$
 (c) $\frac{15}{52}$ (d) $\frac{19}{52}$ [NDA (II) - 2015]
81. A coin is tossed 5 times. The probability that tail appears an odd number of times, is:
 (a) $\frac{1}{2}$ (b) $\frac{1}{3}$
 (c) $\frac{2}{5}$ (d) $\frac{1}{5}$ [NDA (II) - 2015]
82. A point is chosen at random inside a rectangle measuring 6 inches by 5 inches. What is the probability that the randomly selected point is at least one inch from the edge of the rectangle?
 (a) $\frac{2}{3}$ (b) $\frac{1}{3}$
 (c) $\frac{1}{4}$ (d) $\frac{2}{5}$ [NDA (I) - 2016]
83. A fair coin is tossed 100 times. What is the probability of getting tails an odd number of times?
 (a) $\frac{1}{2}$ (b) $\frac{3}{8}$
 (c) $\frac{1}{4}$ (d) $\frac{1}{8}$ [NDA-2016(1)]
84. Three dice are thrown simultaneously. What is the probability that the sum on the three faces is at least 5?
 (a) $\frac{17}{18}$ (b) $\frac{53}{54}$
 (c) $\frac{103}{108}$ (d) $\frac{215}{216}$ [NDA (I) - 2016]
85. Two independent events A and B have $P(A) = \frac{1}{3}$ and $P(B) = \frac{3}{4}$. What is the probability that exactly one of the two events A or B occurs?
 (a) $\frac{1}{4}$ (b) $\frac{5}{6}$
 (c) $\frac{5}{12}$ (d) $\frac{7}{12}$ [NDA (I) - 2016]
86. A coin is tossed three times. What is the probability of getting head and tail alternately?
 (a) $\frac{1}{8}$ (b) $\frac{1}{4}$
 (c) $\frac{1}{2}$ (d) $\frac{3}{4}$ [NDA (I) - 2016]
87. A card is drawn from a well-shuffled deck of 52 cards. What is the probability that it is queen of spade?
 (a) $\frac{1}{52}$ (b) $\frac{1}{13}$
 (c) $\frac{1}{4}$ (d) $\frac{1}{8}$ [NDA (I) - 2016]
88. If two dice are thrown, then what is the probability that the sum on the two faces is greater than or equal to 4?
 (a) $\frac{13}{18}$ (b) $\frac{5}{6}$
 (c) $\frac{11}{12}$ (d) $\frac{35}{36}$ [NDA (I) - 2016]
89. A certain type of missile hits the target with probability $p = 0.3$. What is the least number of missiles should be fired so that there is at least an 80% probability that the target is hit?
 (a) 5 (b) 6
 (c) 7 (d) None of these [NDA (I) - 2016]
90. For two mutually exclusive events A and B , $P(A) = 0.2$ and $P(\bar{A} \cap B) = 0.3$. What is $P(A | (A \cup B))$ equal to?
 (a) $\frac{1}{2}$ (b) $\frac{2}{5}$
 (c) $\frac{2}{7}$ (d) $\frac{2}{3}$ [NDA (I) - 2016]
91. What is the probability of 5 Sunday in the month of December?
 (a) $\frac{1}{7}$ (b) $\frac{2}{7}$
 (c) $\frac{3}{7}$ (d) None of these [NDA (I) - 2016]
92. A salesman has a 70% chance to sell a product to any customer. The behaviour of successive customers is independent. If two customers A and B enter, what is the probability that the salesman will sell the product to customer A or B ?
 (a) 0.98 (b) 0.91
 (c) 0.70 (d) 0.49 [NDA (II) - 2016]

93. A special dice with numbers 1, -1, 2, -2, 0 and 3 is thrown thrice. What is the probability that the sum of the numbers occurring on the upper face is zero?

(a) $\frac{1}{72}$ (b) $\frac{1}{8}$
(c) $\frac{7}{12}$ (d) $\frac{25}{216}$

[NDA (II) - 2016]

94. There is 25% chance that it rains on any particular day. What is probability that there is at least one rainy day within a period of 7 days?

(a) $1 - \left(\frac{1}{4}\right)^7$ (b) $\left(\frac{1}{4}\right)^7$
(c) $\left(\frac{3}{4}\right)^7$ (d) $1 - \left(\frac{3}{4}\right)^7$

[NDA (II) - 2016]

95. A student appears for tests I, II and III. The student is considered successful if he passes in test 1, II or I, III or all the three. The probabilities of the student passing in test I, II and III are m, n and $\frac{1}{2}$ respectively. If the probability of the

student to be successful is $\frac{1}{2}$, then which one of the following is correct?

(a) $m(1+n)=1$ (b) $n(1+m)=1$
(c) $m=1$ (d) $mn=1$

[NDA (II) - 2016]

96. Three candidates solve a question. Odds in favour of the correct answer are 5:2, 4:3 and 3:4 respectively for the three candidates. What is the probability that at least two of them solve the question correctly?

(a) $\frac{209}{243}$ (b) $\frac{134}{343}$
(c) $\frac{149}{343}$ (d) $\frac{60}{343}$

[NDA (II) - 2016]

97. A medicine is known to be 75% effective to cure a patient. If the medicine is given to 5 patients, what is the probability that at least one patient is cured by this medicine?

(a) $1/1024$ (b) $243/1024$
(c) $1023/1024$ (d) $781/1024$

[NDA (II) - 2016]

98. For two events, A and B, it is given that $P(A)=\frac{3}{5}$, $P(B)=\frac{3}{10}$

and $P(A|B)=\frac{2}{3}$. If $\bar{A}$ and $\bar{B}$ are the complementary events

of A and B, then what is $P(\bar{A} | \bar{B})$ equal to?

(a) $\frac{3}{7}$ (b) $\frac{3}{4}$
(c) $\frac{1}{3}$ (d) $\frac{4}{7}$

[NDA (II) - 2016]

99. A machine has three parts, A, B and C, whose chances of being defective are 0.02, 0.10 and 0.05 respectively. The machine stops working if any one of the parts becomes

defective. What is the probability that the machine will not stop working?

(a) 0.06 (b) 0.16
(c) 0.84 (d) 0.94

[NDA (II) - 2016]

100. Three independent events, A_1 , A_2 and A_3 occur with probabilities $P(A_i)=\frac{1}{1+i}$, $i=1, 2, 3$. What is the probability that at least one of the three events occurs?

(a) $\frac{1}{4}$ (b) $\frac{2}{3}$
(c) $\frac{3}{4}$ (d) $\frac{1}{24}$

[NDA (II) - 2016]

101. In a series of 3 one-day cricket matches between teams A and B of a college, the probability of team A winning or drawings are $\frac{1}{3}$ and $\frac{1}{6}$ respectively. If a win, lose or draw gives 2, 0 and 1 point respectively, then what is the probability that team A will score 5 points in the series?

(a) $\frac{17}{18}$ (b) $\frac{11}{12}$
(c) $\frac{1}{12}$ (d) $\frac{1}{18}$

[NDA (II) - 2016]

102. Let the random variable X follow $B(6,p)$. If $16P(X=4) = P(X=2)$, then what is the value of p?

(a) $\frac{1}{3}$ (b) $\frac{1}{4}$
(c) $\frac{1}{5}$ (d) $\frac{1}{6}$

[NDA (II) - 2016]

103. For two dependent events A and B, it is given that $P(A) = 0.2$ and $P(B) = 0.5$. If $A \subseteq B$, then the values of conditional probabilities $P(A|B)$ and $P(B|A)$ are respectively:

(a) $\frac{2}{5}, \frac{3}{5}$

(b) $\frac{2}{5}, 1$

(c) $1, \frac{2}{5}$

(d) Information is insufficient

[NDA (I) - 2017]

104. A card is drawn from a well-shuffled ordinary deck of 52 cards. What is the probability that it is an ace?

(a) $\frac{1}{13}$ (b) $\frac{2}{13}$
(c) $\frac{3}{13}$ (d) $\frac{1}{52}$

[NDA (I) - 2017]

105. Consider the following statements:

1. Two events are mutually exclusive if the occurrence of one event prevents the occurrence of the other.

2. The probability of the union of two mutually exclusive events is the sum of their individual probabilities.

Which of the above statements is/are correct?

- (a) Only 1
(c) Both 1 and 2

- (b) Only 2
(d) Neither 1 nor 2

[NDA (I) - 2017]

106. If two fair dice are thrown, then what is the probability that the sum is either 8 or 9?

- (a) $\frac{1}{6}$
(c) $\frac{3}{4}$

- (b) $\frac{1}{4}$
(d) $\frac{5}{6}$

[NDA (I) - 2017]

107. Let A and B are two mutually exclusive events with $P(A) = \frac{1}{3}$ and $P(B) = \frac{1}{4}$. What is the value of $P(\bar{A} \cap \bar{B})$?

- (a) $\frac{1}{6}$
(c) $\frac{1}{3}$

- (b) $\frac{1}{4}$
(d) $\frac{5}{12}$

[NDA (I) - 2017]

108. A committee of two persons is constituted from two men and two women. What is the probability that the committee will have only women?

- (a) $\frac{1}{6}$
(c) $\frac{1}{2}$

- (b) $\frac{1}{3}$
(d) $\frac{2}{3}$

[NDA (I) - 2017]

109. A question is given to three students A, B and C whose chances of solving it are $\frac{1}{2}$, $\frac{1}{3}$ and $\frac{1}{4}$ respectively. What is the probability that the question will be solved?

- (a) $\frac{1}{24}$
(c) $\frac{3}{4}$

- (b) $\frac{1}{4}$
(d) $\frac{23}{24}$

[NDA (I) - 2017]

110. A point is chosen at random inside a circle. What is the probability that the point is closer to the centre of the circle than to its boundary?

- (a) $\frac{1}{5}$
(c) $\frac{1}{3}$

- (b) $\frac{1}{4}$
(d) $\frac{1}{2}$

[NDA (I) - 2017]

111. The mean and standard deviation of a binomial distribution are 12 and 2 respectively. What is the number of trials?

- (a) 2
(c) 18

- (b) 12
(d) 24

[NDA-2017(1)]

112. Let the sample space consists of non negative integers up to 50, X denote the numbers which are multiples of 3 and Y denote the odd numbers. Which of the following is/are correct?

1. $P(X) = \frac{8}{25}$
2. $P(Y) = \frac{1}{2}$

Select the correct answer using the code given below:

- (a) 1 only
(c) both 1 and 2

- (b) 2 only
(d) neither 1 nor 2

[NDA-2017(2)]

113. Five sticks of length 1, 3, 5, 7 and 9 feet are given. Three of these sticks are selected at random. What is the probability that the selected sticks can form a triangle?

- (a) 0.5
(c) 0.3

- (b) 0.4
(d) 0

[NDA (II) - 2017]

114. A committee of two persons is selected from two men and two women. The probability that the committee will have exactly one women is:

- (a) $\frac{1}{6}$
(c) $\frac{1}{3}$

- (b) $\frac{2}{3}$
(d) $\frac{1}{2}$

[NDA (II) - 2017]

115. For two events A and B, $P(A) = \frac{1}{2}$ let $P(A \cup B) = \frac{2}{3}$ and

$P(A \cap B) = \frac{1}{6}$. What is $P(\bar{A} \cap \bar{B})$ equal to?

- (a) $\frac{1}{6}$
(c) $\frac{1}{3}$

- (b) $\frac{1}{4}$
(d) $\frac{1}{2}$

[NDA (II) - 2017]

116. Let A and B be two events with $P(A) = \frac{1}{3}$, $P(B) = \frac{1}{6}$ and

$P(A \cap B) = \frac{1}{12}$. What is $P(B | \bar{A})$ equal to?

- (a) $\frac{1}{5}$
(c) $\frac{1}{8}$

- (b) $\frac{1}{7}$
(d) $\frac{1}{10}$

[NDA (II) - 2017]

117. In a binomial distribution, the mean is $\frac{2}{3}$ and the variance is

$\frac{5}{9}$. What is the probability that $X = 2$?

- (a) $\frac{5}{36}$
(c) $\frac{25}{216}$

- (b) $\frac{25}{36}$
(d) $\frac{25}{54}$

[NDA (II) - 2017]

118. The probability that a ship safely reaches a port is $\frac{1}{3}$. The

probability that out of 5 ships, at least 4 ships would arrive safely is:

- (a) $\frac{1}{243}$
(c) $\frac{11}{243}$

- (b) $\frac{10}{243}$
(d) $\frac{13}{243}$

[NDA (II) - 2017]

119. What is the probability that at least two persons out of a group of three persons were born in the same month (disregard years)?
 (a) $\frac{33}{144}$ (b) $\frac{17}{72}$
 (c) $\frac{1}{144}$ (d) $\frac{2}{9}$ [NDA (II) - 2017]
120. If $P(B) = \frac{3}{4}$, $P(A \cap B \cap \bar{C}) = \frac{1}{3}$ and $(\bar{A} \cap B \cap \bar{C}) = \frac{1}{3}$, then what is $P(B \cap C)$ equal to?
 (a) $\frac{1}{12}$ (b) $\frac{3}{4}$
 (c) $\frac{1}{15}$ (d) $\frac{1}{9}$ [NDA (II) - 2017]
121. Let a die be loaded in such a way that even faces are twice likely to occur as the odd faces. What is the probability that a prime number will show up when the die is tossed?
 (a) $\frac{1}{3}$ (b) $\frac{2}{3}$
 (c) $\frac{4}{9}$ (d) $\frac{5}{9}$ [NDA (II) - 2017]
122. In a multiple choice test, an examinee either knows the correct answer with probability p , or guesses with probability $1-p$. The probability of answering a question correctly is $\frac{1}{m}$, if he or she merely guesses. If the examinee answers a question correctly, the probability that he or she really knows the answer is:
 (a) $\frac{mp}{1+mp}$ (b) $\frac{mp}{1+(m-1)p}$
 (c) $\frac{(m-1)p}{1+(m-1)p}$ (d) $\frac{(m-1)p}{1+mp}$ [NDA (II) - 2017]
123. In a Binomial distribution, the mean is three times its variance. What is the probability of exactly 3 successes out of 5 trials?
 (a) $80/243$ (b) $40/243$
 (c) $20/243$ (d) $10/243$ [NDA-2018(1)]
124. Consider the following statements:
 1. $P(\bar{A} \cup B) = P(\bar{A}) + P(B) - P(\bar{A} \cap B)$
 2. $P(A \cap \bar{B}) = P(B) - P(A \cap B)$
 3. $P(A \cap B) = P(B)P(A|B)$
 Which of the above statements are correct?
 (a) 1 and 2 only (b) 1 and 3 only
 (c) 2 and 3 only (d) 1, 2 and 3 [NDA-2018(1)]
125. The probabilities that a student will solve question A and question B are 0.4 and 0.5 respectively. What is the probability that he solves at least one of the two questions?
 (a) 0.6 (b) 0.7
 (c) 0.8 (d) 0.9
126. A committee of three has to be chosen from a group of 4 men and 5 women. If the selection is made at random, what is the probability that exactly two members are men?
 (a) $5/14$ (b) $1/21$
 (c) $3/14$ (d) $8/21$ [NDA-2018(1)]
127. Consider the following in respect of two events A and B.
 I. $P(A \text{ occurs but not } B) = P(A) - P(B)$ if $B \subset A$
 II. $P(A \text{ alone or } B \text{ alone occurs}) = P(A) + P(B) - P(A \cap B)$
 III. $P(A \cup B) = P(A) + P(B)$ if A and B are mutually exclusive.
 Which of the above is/are correct?
 (a) I only (b) I and III only
 (c) 2 and 3 only (d) 1 and 2 only [NDA (I) - 2018]
128. One bag contains 3 white and 2 black balls, another bag contains 5 white and 3 black balls, if a bag is chosen at random and a ball is drawn from it, what is the chance that it is white?
 (a) $3/8$ (b) $49/80$
 (c) $8/13$ (d) $1/2$ [NDA (I) - 2018]
129. A box has ten chits numbered 0, 1, 2, 3, ..., 9. First, one chit is drawn at random and kept aside. From the remaining, a second chit is drawn at random. What is the probability that the second chit drawn is "9"?
 (a) $1/10$
 (b) $1/9$
 (c) $1/90$
 (d) None of the above [NDA (I) - 2018]
130. If A and B are two events such that $2P(A) = 3P(B)$, where $0 < P(B) < P(A) < 1$, then which of the following is correct?
 (a) $P(A|B) < P(B|A) < P(A \cap B)$
 (b) $P(A \cap B) < P(B|A) < P(A|B)$
 (c) $P(B|A) < P(A|B) < P(A \cap B)$
 (d) $P(A \cap B) < P(A|B) < P(B|A)$ [NDA (I) - 2018]
131. Two fair dice are rolled. What is the probability of getting a sum of 7?
 (a) $1/36$ (b) $1/6$
 (c) $7/12$ (d) $5/12$ [NDA (I) - 2018]
132. Three dice having digits 1, 2, 3, 4, 5 and 6 on their faces are marked I, II and III and rolled. Let x, y and z represent the number on die-I die-II and die-III respectively. What is the number of possible outcomes such that $x > y > z$?
 (a) 14 (b) 16
 (c) 18 (d) 20 [NDA (II) - 2018]
133. Two integers x and y are chosen with replacement from the set (0, 1, 2, ..., 10). The probability that $|x-y| > 5$ is:
 (a) $\frac{6}{11}$ (b) $\frac{35}{121}$
 (c) $\frac{30}{121}$ (d) $\frac{25}{121}$ [NDA (II) - 2018]
134. If the probability of simultaneous occurrence of two events A and B is p and the probability that exactly one of A, B occurs is q, then which of the following is/are correct?
 $I. P(\bar{A}) + P(\bar{B}) = 2 - 2p - q$

$$II. P(\overline{A \cap B}) = 1 - p - q$$

Select the correct answer using the code given below:

- (a) I only (b) II only
(c) Both I and II (d) Neither 1 nor 2

[NDA (II) - 2018]

135. Three groups of children contain 3 girls and 1 boy; 2 girls and 2 boys; 1 girl and 3 boys. One child is selected at random from each group. The probability that the three selected consists of 1 girl and 2 boys is:

- (a) $\frac{13}{32}$ (b) $\frac{9}{32}$
(c) $\frac{3}{32}$ (d) $\frac{1}{32}$

[NDA (II) - 2018]

136. 8 coins are tossed simultaneously. The probability of getting at least 6 heads is:

- (a) $\frac{7}{64}$ (b) $\frac{57}{64}$
(c) $\frac{37}{256}$ (d) $\frac{229}{256}$

[NDA (II) - 2018]

137. In a bolt factory, machines X, Y, Z manufacture bolt that are respectively 25%, 35% and 40% of the factor's total output. The machines X, Y, Z respectively produce 2%, 4% and 5% defective bolts. A bolt is drawn at random from the product and is found to be defective. What is the probability that it was manufactured by machine X?

- (a) $\frac{5}{39}$ (b) $\frac{11}{39}$
(c) $\frac{20}{39}$ (d) $\frac{34}{39}$

[NDA (II) - 2018]

138. Let A, B and C be three mutually exclusive and exhaustive events associated with a random experiment. If $P(B) = 1.5 P(A)$ and $P(C) = 0.5 P(B)$, then $P(A)$ is equal to:

- (a) $\frac{3}{4}$ (b) $\frac{4}{13}$
(c) $\frac{2}{3}$ (d) $\frac{1}{2}$

[NDA (II) - 2018]

139. If two dice are thrown and at least one of the dice shows 5, then the probability that the sum is 10 or more is:

- (a) $\frac{1}{6}$ (b) $\frac{4}{11}$
(c) $\frac{3}{11}$ (d) $\frac{2}{11}$

[NDA (II) - 2018]

140. If A, B, C are three events, then what is the probability that at-least two of these events occur together?

- (a) $P(A \cap B) + P(B \cap C) + P(C \cap A)$
(b) $P(A \cap B) + P(B \cap C) + P(C \cap A) - P(A \cap B \cap C)$
(c) $P(A \cap B) + P(B \cap C) + P(C \cap A) - 2P(A \cap B \cap C)$
(d) $P(A \cap B) + P(B \cap C) + P(C \cap A) - 3P(A \cap B \cap C)$

[NDA (I) - 2019]

141. Two independent events A and B are such that $P(A \cup B) = \frac{2}{3}$, and $P(A \cap B) = \frac{1}{6}$. If $P(B) < P(A)$, then what is $P(B)$ equal to?

- (a) $\frac{1}{4}$ (b) $\frac{1}{3}$
(c) $\frac{1}{2}$ (d) $\frac{1}{6}$

[NDA (I) - 2019]

142. If two fair dice are rolled then what is the conditional probability that the first dice shows 6 given that the sum of numbers on the dice is 8.

- (a) $\frac{1}{3}$ (b) $\frac{1}{4}$
(c) $\frac{1}{5}$ (d) $\frac{1}{6}$

[NDA (I) - 2019]

143. Two symmetric dice flipped with each dice having two sides painted red, two painted black, one painted yellow and the other painted white, What is the probability that both land on the same colour?

- (a) $\frac{3}{18}$ (b) $\frac{2}{9}$
(c) $\frac{5}{18}$ (d) $\frac{1}{3}$

[NDA (I) - 2019]

144. There are n socks in a drawer, of which 3 socks are red. If 2 of the socks are chosen randomly and the probability that both selected socks are red is $\frac{1}{2}$, then what is the value of n?

- (a) 3 (b) 4
(c) 5 (d) 6

[NDA (I) - 2019]

145. Two cards are chosen at random from a deck of 52 playing cards. What is the probability that both of them have the same value?

- (a) $\frac{1}{17}$ (b) $\frac{3}{17}$
(c) $\frac{5}{17}$ (d) $\frac{7}{17}$

[NDA (I) - 2019]

146. In eight throws of a die, 5 or 6 is considered a success. The mean and standard deviation of total number of successes is respectively given by

- (a) $\frac{8}{3}, \frac{16}{9}$ (b) $\frac{8}{3}, \frac{4}{3}$
(c) $\frac{4}{3}, \frac{4}{3}$ (d) $\frac{4}{3}, \frac{16}{9}$

[NDA (I) - 2019]

147. A and B are two events such that A' and B' are mutually exclusive. If $P(A) = 0.5$ and $P(B) = 0.6$, then what is the value of $P(A/B)$?

- (a) $\frac{1}{5}$ (b) $\frac{1}{6}$
(c) $\frac{2}{5}$ (d) $\frac{1}{3}$

[NDA (I) - 2019]

148. What is the probability that an interior point in a circle is closer to the centre than to the circumference?
 (a) $\frac{1}{4}$
 (b) $\frac{1}{2}$
 (c) $\frac{3}{4}$
 (d) It cannot be determined
[NDA (I) - 2019]
149. If A and B are two events, then what is the probability of occurrence of either A or event B?
 (a) $P(A) + P(B)$
 (b) $P(A \cup B)$
 (c) $P(A \cap B)$
 (d) $P(A)P(B)$
[NDA (I) - 2019]
150. From a deck of cards, cards are taken out with replacement. What is the probability that the fourteenth card taken out is an ace?
 (a) $\frac{1}{51}$
 (b) $\frac{4}{51}$
 (c) $\frac{1}{52}$
 (d) $\frac{1}{13}$
[NDA (I) - 2019]
151. If A and B are two events such that $P(A) = 0.5$, $P(B) = 0.6$ and $P(A \cap B) = 0.4$, then what is $P(\overline{A \cup B})$ equal to?
 (a) 0.9
 (b) 0.7
 (c) 0.5
 (d) 0.3
[NDA (I) - 2019]
152. A person is given to three students A, B and C whose probabilities of solving the problem are $\frac{1}{2}$, $\frac{3}{4}$ and $\frac{1}{4}$ respectively. What probability that the problem will be they all solve the problem independently
 (a) $\frac{29}{32}$
 (b) $\frac{27}{32}$
 (c) $\frac{25}{32}$
 (d) $\frac{23}{32}$
[NDA (I) - 2019]
153. A pair of fair dice is rolled. What is the probability that the second dice lands on a higher value than does the first?
 (a) $\frac{1}{4}$
 (b) $\frac{1}{6}$
 (c) $\frac{5}{12}$
 (d) $\frac{5}{18}$
[NDA (I) - 2019]
154. A fair coin is tossed and an unbiased dice rolled together. What is the probability of getting a 2 or 4 or 6 along with head?
 (a) $\frac{1}{2}$
 (b) $\frac{1}{3}$
 (c) $\frac{1}{4}$
 (d) $\frac{1}{6}$
[NDA (I) - 2019]
155. A coin is biased so that heads comes up thrice as likely as tails. For three independent tosses of a coin, what is the probability of getting at most two tails.
 (a) 0.16
 (b) 0.48
 (c) 0.58
 (d) 0.98
[NDA (II) - 2019]
156. A bag contains 20 books out of which 5 are defective. If 3 of the books are selected at random and removed from the bag in succession without replacement, then what is the probability that all three books are defective?
 (a) 0.009
 (b) 0.016
 (c) 0.026
 (d) 0.047
[NDA (II) - 2019]
157. If a coin is tossed till the first head appears, then what will be the sample space?
 (a) {H}
 (b) {TH}
 (c) {T, HT, HHT, HHHT, ...}
 (d) {H, TH, TTH, TTTH,}
[NDA (II) - 2019]
158. Two dices are thrown simultaneously. What is the probability that the sum of the numbers appearing on them is a prime number?
 (a) $\frac{5}{12}$
 (b) $\frac{1}{2}$
 (c) $\frac{7}{12}$
 (d) $\frac{2}{3}$
[NDA (II) - 2019]
159. If 5 of a company's 10 delivery trucks do not meet emission standards and 3 of them are chosen for inspection, then what is the probability that none of the trucks chosen will meet emission standards?
 (a) $\frac{1}{8}$
 (b) $\frac{3}{8}$
 (c) $\frac{1}{12}$
 (d) $\frac{1}{4}$
[NDA (II) - 2019]
160. There are three coins in a box. One is a two-headed; another is a fair coin; and third is biased coin that comes up heads 75% of time. When one of the three coins is selected at random and flipped, it shows heads. What is the probability that it was the two-headed coin?
 (a) $\frac{2}{9}$
 (b) $\frac{1}{3}$
 (c) $\frac{4}{9}$
 (d) $\frac{5}{9}$
[NDA (II) - 2019]
161. Consider the following statements:
 I. If A and B are mutually exclusive events, then it is possible that $P(A) = P(B) = 0.6$
 II. If A and B are any two events such that $P(A/B) = 1$, then $P(\overline{B} / \overline{A}) = 1$.
 which of the above statement is/are corrects?
 (a) 1 only
 (b) 2 only
 (c) Both 1 and 2
 (d) Neither 1 nor 2
[NDA (II) - 2019]
162. If a fair die is rolled 4 times, then what is the probability that there are exactly 2 sixes?
 (a) $\frac{5}{216}$
 (b) $\frac{25}{216}$
 (c) $\frac{125}{216}$
 (d) $\frac{175}{216}$
[NDA (II) - 2019]
163. What is the probability that February of a leap year selected at random, will have five Sundays?
 (a) $\frac{1}{5}$
 (b) $\frac{1}{7}$
 (c) $\frac{2}{7}$
 (d) 1
[NDA 2020]
164. A husband and wife appear in an interview for two vacancies for the same The post. Probability of the husband's selection is $\frac{1}{7}$ and that of the wife's selection is $\frac{1}{5}$. If the events are independent, then the probability of which one of the following is $\frac{11}{35}$?
 (a) At least one of them will be selected
 (b) Only one of them will be selected

- (c) None of them will be selected
(d) Both of them will be selected

[NDA 2020]

165. A dealer has a stock of 15 gold coins out of which 6 are counterfeits. A person randomly picks 4 of the 15 gold coins. What is the probability that all the coins picked will be counterfeits?

- (a) $\frac{1}{91}$ (b) $\frac{4}{91}$
(c) $\frac{6}{91}$ (d) $\frac{15}{91}$

[NDA 2020]

166. A committee of 3 is to be formed from a group of 2 boys and 2 girls. What is the probability that the committee consists of 2 boys and 1 girl?

- (a) $\frac{2}{3}$ (b) $\frac{1}{4}$
(c) $\frac{3}{4}$ (d) $\frac{1}{2}$

[NDA 2020]

167. In a lottery of 10 tickets numbered 1 to 10, two tickets are simultaneously. What is the probability that both the tickets drawn have prime numbers?

- (a) $\frac{1}{15}$ (b) $\frac{1}{2}$
(c) $\frac{2}{15}$ (d) $\frac{1}{5}$

[NDA 2020]

168. Consider a random variable X which follows Binomial distribution with parameters $n = 10$ and $P = \frac{1}{5}$. Then $Y = 10 - X$ follows Binomial distribution with parameters n and P respectively given by

- (a) 5, $\frac{1}{5}$ (b) 5, $\frac{2}{5}$
(c) 10, $\frac{3}{5}$ (d) 10, $\frac{4}{5}$

[NDA 2020]

169. If A and B are two events such that $P(A) = 0.6$, $P(B) = 0.5$ and $P(A \cap B) = 0.4$, then consider the following statements :

1. $P(\bar{A} \cup B) = 0.9$
2. $P(\bar{B} / \bar{A}) = 0.6$

Which of the above statements is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA 2020]

170. Three cooks X, Y and Z bake a special kind of cake, and with respective probabilities 0.02, 0.03 and 0.05, it fails to rise. In the restaurant where they work, X bakes 50%, Y bakes 30% and Z bakes 20% of cakes. What is the proportion of failures caused by X?

- (a) $\frac{9}{29}$ (b) $\frac{10}{29}$
(c) $\frac{19}{28}$ (d) $\frac{28}{29}$

[NDA 2020]

171. If three dice are rolled under the condition that no two dice show the same face then what is the probability that one of the faces is having the number 6?

- (a) $\frac{5}{6}$ (b) $\frac{5}{9}$
(c) $\frac{1}{2}$ (d) $\frac{5}{12}$

[NDA 2020]

172. If $P(A \cup B) = \frac{5}{6}$, $P(A \cap B) = \frac{1}{3}$ and $P(\text{not } A) = \frac{1}{2}$, then which one of the following is not correct ?

- (a) $P(B) = \frac{2}{3}$
(b) $P(A \cap B) = P(A)P(B)$
(c) $P(A \cup B) > P(A) + P(B)$
(d) $P(\text{not } A \text{ and not } B) = P(\text{not } A)P(\text{not } B)$

[NDA 2020]

173. Let two events A and B be such that $P(A) = L$ and $P(B) = M$. Which one of the following is correct?

- (a) $P(A/B) < \frac{L+M-1}{M}$

(b) $P(A/B) > \frac{L+M-1}{M}$

(c) $P(A/B) \geq \frac{L+M-1}{M}$

(d) $P(A/B) = \frac{L+M-1}{M}$

[NDA (I) 2021]

174. If $P(A \cup B) = \frac{5}{6}$, $P(A \cap B) = \frac{1}{3}$ and $P(\bar{A}) = \frac{1}{2}$, then which of

the following is/are correct ?

1. A and B are independent events.
2. A and B are mutually exclusive events.

Select the correct answer using the code given below.

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA (I) 2021]

175. A coin is tossed twice. If E and F denote occurrence of head on first toss and second toss respectively, then what is $P(E \cup F)$ equal to ?

- (a) $\frac{1}{4}$ (b) $\frac{1}{2}$
(c) $\frac{3}{4}$ (d) $\frac{1}{3}$

[NDA (I) 2021]

176. In a binomial distribution, the mean is $\frac{2}{3}$ and variance is $\frac{5}{9}$. What is the probability that random variable $X = 2$?

- (a) $\frac{5}{36}$ (b) $\frac{25}{36}$
(c) $\frac{25}{54}$ (d) $\frac{25}{216}$

[NDA (I) 2021]

177. If A and B are two events such that $P(A) = \frac{3}{4}$ and $P(B) = \frac{5}{8}$, then consider the following statements :

1. The minimum value of $P(A \cup B)$ is $\frac{3}{4}$
2. The maximum value of $P(A \cap B)$ is $\frac{5}{8}$

Which of the above statements is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA (I) 2021]

178. Consider the following relations for two events E and F.

1. $P(E \cap F) \geq P(E) + P(F) - 1$
2. $P(E \cup F) = P(E) + P(F) + P(E \cap F)$
3. $P(E \cup F) \leq P(E) + P(F)$

Which of the above relations is/are correct?

- (a) 1 only (b) 3 only
(c) 1 and 3 only (d) 1, 2 and 3

[NDA (II) 2021]

179. If $P(A|B) < P(A)$, then which one of the following is correct?

- (a) $P(B|A) < P(B)$ (b) $P(B|A) > P(B)$
(c) $P(B|A) = P(B)$ (d) $P(B|A) > P(A)$

[NDA (II) 2021]

180. A problem is given to three students A, B and C, whose probabilities of solving the problem independently are $\frac{1}{2}$, $\frac{3}{4}$ and p respectively. If the probability that the problem can be solved is $\frac{29}{32}$, then what is the value of p ?

- (a) $\frac{2}{5}$ (b) $\frac{2}{3}$
(c) $\frac{1}{3}$ (d) $\frac{1}{4}$

[NDA (II) 2021]

181. In a cricket match, a batsman hits a six 8 times out of 60 balls he plays. What is the probability that on a ball played he does not hit a six?

- (a) $\frac{2}{3}$ (b) $\frac{1}{15}$

(c) $\frac{2}{15}$

(d) $\frac{13}{15}$

[NDA (II) 2021]

182. 3-digit numbers are formed using the digit 1, 3, 7 without repetition of digits. A number is randomly selected. What is the probability that the number is divisible by 3?

(a) 0

(b) $\frac{1}{3}$

(c) $\frac{1}{4}$

(d) $\frac{1}{8}$

[NDA (II) 2021]

183. What is the probability that the roots of the equation $x^2 + x + n = 0$ are real, where $n \in \mathbb{N}$ and $n < 4$?

(a) 0

(b) $\frac{1}{4}$

(c) $\frac{1}{3}$

(d) $\frac{1}{2}$

[NDA (II) 2021]

184. If A and B are two events such that $P(\text{not } A) = \frac{7}{10}$, $P(\text{not } B) = \frac{3}{10}$ and $P(A|B) = \frac{3}{14}$, then what is $P(B|A)$ equal to?

(a) $\frac{11}{14}$

(b) $\frac{9}{14}$

(c) $\frac{1}{4}$

(d) $\frac{1}{2}$

[NDA (II) 2021]

185. Seven white balls and three black balls are randomly placed in a row. What is the probability that no two black balls are placed adjacently?

(a) $\frac{7}{15}$

(b) $\frac{8}{15}$

(c) $\frac{11}{15}$

(d) $\frac{13}{15}$

[NDA (II) 2021]

186. Two digits out of 1, 2, 3, 4, 5 are chosen at random and multiplied together. What is the probability that the last digit in the product appears as 0?

(a) $\frac{1}{10}$

(b) $\frac{1}{5}$

(c) $\frac{2}{5}$

(d) $\frac{4}{5}$

[NDA (I) 2022]

Consider the following data for the next three (03) items that follow:

There are 90 applications for the job. Some of them are graduates. Some of them have less than three years experience.

	Number of graduates	Number of non-graduates
At least 3 years Experience	18	9
Less than 3 years experience	36	27

Let G be the event that the first applicant interviewed is a graduate and T be the event that first applicant interviewed has at least 3 years experience.

187. What is $P(G \cap \bar{T})$ equal to?

(a) $\frac{1}{5}$

(b) $\frac{2}{5}$

(c) $\frac{3}{5}$

(d) $\frac{4}{5}$

[NDA (I) 2022]

188. What is $P(G|\bar{T})$ equal to?

(a) $\frac{2}{7}$

(b) $\frac{3}{7}$

(c) $\frac{4}{7}$

(d) $\frac{5}{7}$

[NDA (I) 2022]

189. What is $P(\bar{T}|\bar{G})$ equal to?

(a) $\frac{1}{4}$

(b) $\frac{1}{3}$

(c) $\frac{3}{5}$

(d) $\frac{3}{4}$

[NDA (I) 2022]

Consider the following data for the next three (03) items that follow:

The incidence of suffering from a disease among workers in an industry has a chance of $33\frac{1}{3}\%$.

190. What is the probability that exactly 3 out of 6 workers suffer from a disease?

(a) $\frac{80}{729}$

(b) $\frac{10}{81}$

(c) $\frac{10}{243}$

(d) $\frac{160}{729}$

[NDA (I) 2022]

191. What is the probability that no one out of 6 workers suffers from a disease?

(a) $\frac{665}{729}$

(b) $\frac{64}{729}$

(c) $\frac{4}{243}$

(d) $\frac{1}{729}$

[NDA (I) 2022]

192. What is the probability that at least one out of 6 workers suffer from a disease?

(a) $\frac{728}{729}$

(b) $\frac{665}{729}$

(c) $\frac{653}{729}$

(d) $\frac{596}{729}$

[NDA (I) 2022]

193. What is the probability of getting a composite number in the list of natural numbers from 1 to 50?

(a) $\frac{7}{10}$

(b) $\frac{17}{25}$

(c) $\frac{18}{25}$

(d) $\frac{33}{50}$

[NDA 2022 (II)]

194. If $n > 7$, then what is the probability of $C(n, 7)$ is a multiple of 7?

(a) 0

(b) $\frac{1}{7}$

(c) $\frac{1}{2}$

(d) 1

[NDA 2022 (II)]

195. Two numbers x and y are chosen at random from a set of first 10 natural numbers. What is the probability that $(x+y)$ is divisible by 4?

(a) $\frac{1}{5}$

(b) $\frac{2}{9}$

(c) $\frac{8}{45}$

(d) $\frac{7}{45}$

[NDA 2022 (II)]

196. A number x is chosen at random from first n natural numbers. What is the probability that the number chosen satisfies $x + \frac{1}{x} > 2$?

(a) $\frac{1}{n}$

(b) $\frac{1}{(2n)}$

(c) $\frac{(n-1)}{n}$

(d) 1

[NDA 2022 (II)]

197. Three fair dice are tossed once. What is the probability that they show different numbers that are in AP?

(a) $\frac{1}{12}$

(b) $\frac{1}{18}$

(c) $\frac{1}{36}$

(d) $\frac{1}{72}$

[NDA 2022 (II)]

198. If $P(A) = 0.5$, $P(B) = 0.7$ and $P(A \cap B) = 0.3$, then what is the value of $P(A' \cap B) + P(A \cap B') + P(A \cap B)$?

(a) 0.6

(b) 0.7

(c) 0.8

(d) 0.9

[NDA 2022 (II)]

199. Five coins are tossed once. What is the probability of getting at most four tails?

(a) $\frac{31}{32}$

(b) $\frac{15}{16}$

(c) $\frac{29}{32}$

(d) $\frac{7}{8}$

[NDA 2022 (II)]

200. Three fair dice are thrown. What is the probability of getting a total greater than or equal to 15?

(a) $\frac{19}{216}$

(b) $\frac{1}{12}$

(c) $\frac{17}{216}$

(d) $\frac{5}{54}$

[NDA 2022 (II)]

201. The probability that a person hits a target is 0.5. What is the probability of at least one hit in 4 shots?

(a) $\frac{1}{8}$

(b) $\frac{1}{16}$

(c) $\frac{15}{16}$

(d) $\frac{7}{8}$

[NDA 2022 (II)]

202. A box contains 2 white balls, 3 black balls and 4 red balls. What is the number of ways of drawing 3 balls from the box with at least one black ball?

(a) 84

(b) 72

(c) 64

(d) 48

[NDA 2022 (II)]

203. During war one ship out of 5 was sunk on an average in making a certain voyage. What is the probability that exactly 3 out of 5 ships would arrive safely?

(a) $\frac{16}{625}$

(b) $\frac{32}{625}$

(c) $\frac{64}{625}$

(d) $\frac{128}{625}$

[NDA 2022 (II)]

204. A card is drawn from a pack of 52 cards. A gambler bets that it is either a spade or an ace. The odds against his winning are:

(a) 9:4

(b) 35:17

(c) 17:35

(d) 4:9

[NDA 2022 (II)]

205. The completion of a construction job may be delayed due to strike. The probability of strike is 0.6. The probability that the construction job gets completed on time if there is no strike is 0.85 and the probability that the construction job gets completed on time if there is a strike is 0.35. What is the probability that the construction job will not be completed on time?

(a) 0.35

(b) 0.45

(c) 0.55

(d) 0.65

[NDA 2022 (II)]

206. One bag contains 3 white and 2 black balls, another bag contains 2 white and 3 black balls. Two balls are drawn from the first bag and put it into the second bag and then a ball is drawn from the second bag. What is the probability that it is white?

(a) $\frac{6}{7}$

(b) $\frac{33}{70}$

(c) $\frac{3}{10}$

(d) $\frac{1}{70}$

[NDA – 2023 (1)]

207. Three dice are thrown. What is the probability that each face shows only multiples of 3?

(a) $\frac{1}{9}$

(b) $\frac{1}{18}$

(c) $\frac{1}{27}$

(d) $\frac{1}{3}$

[NDA – 2023 (1)]

208. What is the probability that the month of December has 5 Sundays?

(a) 1

(b) $\frac{1}{4}$

(c) $\frac{3}{7}$

(d) $\frac{2}{7}$

[NDA – 2023 (1)]

209. A natural number n is chosen from the first 50 natural numbers. What is the probability that $n + \frac{50}{n} < 50$?

(a) $\frac{23}{25}$

(b) $\frac{47}{50}$

(c) $\frac{24}{25}$

(d) $\frac{49}{50}$

[NDA – 2023 (1)]

210. Let A and B be two independent events such that $P(\bar{A}) = 0.7$, $P(\bar{B}) = k$, $P(A \cup B) = 0.8$. What is the value of k ?

(a) $\frac{5}{7}$

(b) $\frac{4}{7}$

(c) $\frac{2}{7}$

(d) $\frac{1}{7}$

[NDA – 2023 (1)]

211. A biased coin with the probability of getting head equal to $\frac{1}{4}$ is tossed five times. What is the probability of getting tail in all the four tosses followed by head?

(a) $\frac{81}{512}$

(b) $\frac{81}{1024}$

(c) $\frac{81}{256}$

(d) $\frac{27}{1024}$

[NDA – 2023 (1)]

212. A coin is biased so that heads comes up thrice as likely as tails. In four independent tosses of the coin, what is probability of getting exactly three heads?

(a) $\frac{81}{256}$

(b) $\frac{27}{64}$

- (c) $\frac{27}{256}$ (d) $\frac{9}{256}$

[NDA – 2023 (1)]

213. Let X and Y be two random variables such that $X + Y = 100$. If X follows Binomial distribution with parameters $n = 100$

and $p = \frac{4}{5}$, what is the variance of Y?

- (a) 1 (b) $\frac{1}{2}$
(c) 16 (d) $\frac{1}{16}$

[NDA – 2023 (1)]

214. Three dice are thrown. What is the probability of getting a sum which is a perfect square?

- (a) $\frac{17}{108}$ (b) $\frac{5}{108}$
(c) $\frac{19}{108}$ (d) $\frac{23}{108}$

[NDA-2023 (2)]

215. A, B, C and D are mutually exclusive and exhaustive events. If $2P(A) = 3P(B) = 4P(C) = 5P(D)$, then what is $77P(A)$ equal to?

- (a) 12 (b) 15
(c) 20 (d) 30

[NDA-2023 (2)]

216. Two distinct natural numbers from 1 to 9 are picked at random. What is the probability that their product has 1 in its units place?

- (a) $\frac{1}{81}$ (b) $\frac{1}{72}$
(c) $\frac{1}{18}$ (d) $\frac{1}{36}$

[NDA-2023 (2)]

217. Two dice are thrown. What is the probability that difference of numbers on them is 2 or 3?

- (a) $\frac{7}{36}$ (b) $\frac{7}{18}$
(c) $\frac{5}{18}$ (d) $\frac{11}{36}$

[NDA-2023 (2)]

218. The probability that a person recovers from a disease is 0.8. What is the probability that exactly 2 persons out of 5 will recover from the disease?

- (a) 0.55512 (b) 0.02048
(c) 0.2048 (d) 0.0512

[NDA-2023 (2)]

219. Suppose that there is a chance for a newly constructed building to collapse, whether the design is faulty or not. The chance that the design is faulty is 10%. The chance that the building collapses is 95% if the design is faulty, otherwise it is 45%. If it is seen that the building has collapsed, then what is the probability that it is due to faulty design?

- (a) 0.10 (b) 0.19
(c) 0.45 (d) 0.95

[NDA-2023 (2)]

220. A fair coin is tossed 6 times. What is the probability of getting a result in the 6th toss which is different from those obtained in the first five tosses?

- (a) $\frac{7}{16}$ (b) $\frac{1}{16}$
(c) $\frac{1}{32}$ (d) $\frac{1}{64}$

[NDA-2023 (2)]

221. In a class, there are n students including the students P and Q. What is the probability that P and Q sit together if seats are assigned randomly?

- (a) $\frac{1}{n}$ (b) $\frac{2}{n}$
(c) $\frac{4}{n}$ (d) $\frac{1}{2n}$

[NDA-2023 (2)]

222. In a Binomial distribution $B(n, p)$, $n=6$ and $9P(X=4)=P(X=2)$. What is p equal to?

- (a) $\frac{1}{4}$ (b) $\frac{1}{2}$
(c) $\frac{3}{4}$ (d) $\frac{4}{5}$

[NDA-2023 (2)]

Consider the following for the next (05) items that follow
Three boys P, Q, R and three girls S, T, U are to be arranged in a row for a group photograph.

223. What is the probability that all three boys sit together?

- (a) $\frac{1}{5}$ (b) $\frac{1}{4}$
(c) $\frac{1}{3}$ (d) $\frac{1}{12}$

[NDA-2023 (2)]

224. What is the probability that boys and girls sit alternatively?

- (a) $\frac{4}{5}$ (b) $\frac{1}{10}$
(c) $\frac{5}{6}$ (d) $\frac{1}{7}$

[NDA-2023 (2)]

225. What is the probability that no two girls sit together?

- (a) $\frac{2}{3}$ (b) $\frac{3}{5}$
(c) $\frac{1}{18}$ (d) $\frac{1}{5}$

[NDA-2023 (2)]

226. What is the probability that P and Q take the two end positions?

- (a) $\frac{1}{15}$ (b) $\frac{7}{15}$
(c) $\frac{14}{15}$ (d) $\frac{11}{15}$

[NDA-2023 (2)]

227. What is the probability that Q and U sit together?

- (a) $\frac{2}{3}$ (b) $\frac{1}{4}$
(c) $\frac{5}{6}$ (d) $\frac{1}{3}$

[NDA-2023 (2)]

228. A bag contains 5 black and 4 white balls. A man selects two balls at random. What is the probability that both of these are of the same colour?

- (a) $\frac{1}{6}$ (b) $\frac{5}{108}$
(c) $\frac{4}{9}$ (d) $\frac{5}{18}$

[NDA-2024 (1)]

229. If a random variable (x) following binomial distribution with mean 5 and variance 4, and $5^{23}P(X = 3) = \lambda 4^2$, then what is the value of λ ?

- (a) 3 (b) 5
(c) 23 (d) 25

[NDA-2024 (1)]

230. If two random variables X and y are connected by relation $\frac{2X - 3Y}{5X + 4Y} = 4$ and X follows Binomial distribution with

parameters $n = 10$ and $p = \frac{1}{2}$, then what is the variance of Y?

- (a) $\frac{810}{361}$ (b) $\frac{9}{19}$
(c) $\frac{21}{361}$ (d) $\frac{121}{361}$

[NDA-2024 (1)]

231. If the letters of the word "TIRUPATI" are written down at random, then what is the probability that both Ts are always consecutive?

- (a) $\frac{1}{2}$ (b) $\frac{1}{4}$

(c) $\frac{1}{7}$

(d) $\frac{1}{14}$

[NDA-2024 (1)]

232. Let $m = 77^n$. The index n is given a positive integral value at random. What is the probability that the value of m will have 1 in the units place?

(a) $\frac{1}{2}$

(b) $\frac{1}{3}$

(c) $\frac{1}{4}$

(d) $\frac{1}{n}$

[NDA-2024 (1)]

233. Three different numbers are selected at random from the first 15 natural numbers. What is the probability that the product of two of the numbers is equal to third number?

(a) $\frac{1}{91}$

(b) $\frac{2}{455}$

(c) $\frac{1}{65}$

(d) $\frac{6}{455}$

[NDA-2024 (1)]

Consider the following for the next two (02) items that follow:

Let A and B be two events such that $P(A \cup B) \geq 0.75$ and $0.125 \leq P(A \cap B) \leq 0.375$.

234. What is the minimum value of $P(A) + P(B)$?

(a) 0.625

(b) 0.750

(c) 0.825

(d) 0.875

[NDA-2024 (1)]

235. What is the maximum value of $P(A) + P(B)$?

(a) 0.75

(b) 1.125

(c) 1.375

(d) 1.625

[NDA-2024 (1)]

Consider the following for the next two (02) items that follow:

A , B and C are three events such that $P(A) = 0.6$, $P(B) = 0.4$, $P(C) = 0.5$, $P(A \cup B) = 0.8$, $P(A \cap C) = 0.3$ and $P(A \cap B \cap C) = 0.2$ and $P(A \cup B \cup C) \geq 0.85$

236. What is the minimum value of $P(B \cap C)$?

(a) 0.1

(b) 0.2

(c) 0.35

(d) 0.45

[NDA-2024 (1)]

237. What is the maximum value of $P(B \cap C)$?

(a) 0.1

(b) 0.2

(c) 0.35

(d) 0.45

[NDA-2024 (1)]

Consider the following for the next two (02) items that follow:

An unbiased coin is tossed n times. The probability of getting at least one tail is p and the probability of at least two tails is q and $p - q = \frac{5}{32}$

238. What is the value of n ?

(a) 4

(b) 5

(c) 6

(d) 7

[NDA-2024 (1)]

239. What is the value of $p + q$?

(a) $\frac{57}{32}$

(b) $\frac{53}{32}$

(c) $\frac{51}{32}$

(d) 1

[NDA-2024 (1)]

240. A fair coin is tossed till two heads occur in succession. What is the probability that the number of tosses required is less than 6?

(a) $\frac{5}{64}$

(b) $\frac{15}{32}$

(c) $\frac{31}{64}$

(d) $\frac{19}{32}$

[NDA-2024 (2)]

241. Urn A contains 2 white and 2 black balls while urn B contain 3 white and 2 black balls. One ball is transferred from urn A to urn B and then a ball is drawn out of urn B. What is the probability that the ball is white?

(a) $\frac{11}{20}$

(b) $\frac{7}{12}$

(c) $\frac{3}{5}$

(d) 1

[NDA-2024 (2)]

242. For two events A and B , $P(A) = P(A|B) = 0.25$ and $P(B|A) = 0.5$. Which of the following are correct?

I. A and B are independent.

II. $P(A^c \cup B^c) = 0.875$

III. $P(A^c \cap B^c) = 0.375$

select the answer using the code given below.

(a) I and II only

(b) II and III only

(c) I and III only

(d) I, II and III

[NDA-2024 (2)]

243. Two perfect dice are thrown. What is the probability that the sum of the numbers on the faces is neither 9 nor 10?

(a) $\frac{1}{36}$

(b) $\frac{5}{36}$

(c) $\frac{7}{36}$

(d) $\frac{29}{36}$

[NDA-2024 (2)]

244. The occurrence of a disease in an industry is such that the workers have 20% chance of suffer from it. What is the probability that out of 6 workers chosen at random, 4 or more will suffer from the disease?

(a) $\frac{53}{3125}$

(b) $\frac{63}{3125}$

(c) $\frac{73}{3125}$

(d) $\frac{83}{3125}$

[NDA-2024 (2)]

245. Three perfect dice are rolled. Under the condition that no two dice show the same face, what is the probability that one of the faces shown is an ace (one)?

(a) $\frac{5}{9}$

(b) $\frac{2}{3}$

(c) $\frac{1}{3}$

(d) $\frac{1}{2}$

[NDA-2024 (2)]

246. Three perfect dice D_1 , D_2 and D_3 are rolled. Let x , y , z represents the number on D_1 , D_2 and D_3 respectively. What is the number of possible outcomes such that $x < y < z$?

(a) 20

(b) 18

(c) 14

(d) 10

[NDA-2024 (2)]

247. In a binomial distribution, if mean is 6 and standard deviation is $\sqrt{2}$, then what are the values of the parameters n and p respectively?

(a) 18 and $\frac{1}{3}$

(b) 9 and $\frac{1}{3}$

(c) 18 and $\frac{2}{3}$

(d) 9 and $\frac{2}{3}$

[NDA-2024 (2)]

248. Three distinct natural numbers are chosen at random from 1 to 10. What is the probability that they are consecutive?

(a) $\frac{1}{12}$

(b) $\frac{3}{40}$

(c) $\frac{1}{15}$

(d) $\frac{7}{120}$

[NDA-2024 (2)]

249. A , B , C are three mutually exclusive and exhaustive events associated with a random experiment. If $3P(B) = 4P(A)$ and $3P(C) = 2P(B)$, then what is $P(A)$ equal to?

(a) $\frac{7}{29}$

(b) $\frac{8}{29}$

(c) $\frac{9}{29}$

(d) $\frac{10}{29}$

[NDA-2024 (2)]

250. A die has two faces with number 4, three faces with number 5 and one face with number 6. If the die is rolled once, then what is the probability of getting 4 or 5?
(a) $\frac{1}{3}$ (b) $\frac{2}{3}$
(c) $\frac{5}{6}$ (d) $\frac{1}{2}$ [NDA-2024 (2)]
251. A box contains 2 black, 4 yellow and 6 white balls. Three balls are drawn in succession with replacement. What is the probability that all three are of same colour?
(a) $\frac{1}{6}$ (b) $\frac{1}{36}$
(c) $\frac{1}{12}$ (d) $\frac{5}{12}$ [NDA-2024 (2)]
252. A can hit a target 5 times in 6 shots, B can hit 4 times in 5 shot and C can hit 3 times in 4 shots. What is the probability that A and C may hit but B may lose?
(a) $\frac{1}{8}$ (b) $\frac{1}{6}$
(c) $\frac{1}{4}$ (d) $\frac{1}{3}$ [NDA-2024 (2)]
253. The letter of the word ZOOLOGY are arranged in all possible ways. What is the probability that the consonants and vowels occur alternatively?
(a) $\frac{6}{35}$ (b) $\frac{3}{35}$
(c) $\frac{2}{35}$ (d) $\frac{1}{35}$ [NDA-2024 (2)]
254. A natural number x is chosen at random from the first 100 natural numbers. What is the probability that $x^2 + x > 50$?
(a) $\frac{93}{100}$ (b) $\frac{47}{50}$
(c) $\frac{24}{25}$ (d) $\frac{23}{25}$ [NDA-2024 (2)]
- Consider the following for the two (02) items that follow:**
Let X be a random variable following binomial distribution with parameters $n = 6$ and $p = k$. Further, $9P(X = 4) = P(X = 2)$.
255. What is the value of k ?
(a) $\frac{1}{2}$ (b) $\frac{1}{3}$
(c) $\frac{1}{4}$ (d) $\frac{1}{5}$ [NDA-2025 (1)]
256. What is the value of $P(X = 3)$?
(a) $\frac{135}{1024}$ (b) $\frac{5}{128}$
(c) $\frac{45}{1024}$ (d) $\frac{70}{1024}$ [NDA-2025 (1)]
- Consider the following for the two (02) items that follow:**
A committee of 6 members is formed from a group of 7 gentlemen and 4 ladies.
257. What is the probability that the committee includes exactly 3 gentlemen?
(a) $\frac{10}{33}$ (b) $\frac{30}{77}$
(c) $\frac{100}{231}$ (d) $\frac{5}{11}$ [NDA-2025 (1)]
258. What is the probability that the committee includes at least 2 ladies?
(a) $\frac{41}{66}$ (b) $\frac{47}{66}$
(c) $\frac{49}{66}$ (d) $\frac{53}{66}$ [NDA-2025 (1)]
- Consider the following for the two (02) items that follow:**
The probabilities that A, B and C become managers are $\frac{3}{10}$, $\frac{1}{2}$ and $\frac{4}{5}$ respectively. The probabilities that bonus scheme will be introduced if A, B and C become managers are $\frac{4}{9}$, $\frac{2}{9}$ and $\frac{1}{3}$ respectively.
259. What is the probability that the bonus scheme will be introduced?
(a) $\frac{17}{45}$ (b) $\frac{19}{45}$
(c) $\frac{23}{45}$ (d) $\frac{26}{45}$ [NDA-2025 (1)]
260. If the bonus scheme has been introduced, then what is the probability that the manager appointed was B?
(a) $\frac{5}{23}$ (b) $\frac{6}{23}$
(c) $\frac{7}{23}$ (d) $\frac{8}{23}$ [NDA-2025 (1)]
261. If $P(A) = \frac{1}{3}$, $P(B) = \frac{1}{2}$ and $P(A \cap B) = \frac{1}{4}$, then what is the value of $P(B|A^c)$?
(a) $\frac{1}{8}$ (b) $\frac{3}{8}$
(c) $\frac{5}{8}$ (d) $\frac{7}{8}$ [NDA-2025 (1)]
262. If $P(A) = \frac{1}{3}$, $P(B) = \frac{1}{2}$ and $P(A \cap B) = \frac{1}{4}$, then what is the value of $P(A^c \cap B^c)$?
(a) $\frac{1}{4}$ (b) $\frac{5}{12}$
(c) $\frac{7}{12}$ (d) $\frac{11}{12}$ [NDA-2025 (1)]
263. If two fair dice are tossed, then what is the probability that the sum of the numbers on the faces of the dice is strictly greater than 7?
(a) $\frac{1}{3}$ (b) $\frac{5}{12}$
(c) $\frac{7}{12}$ (d) $\frac{3}{4}$ [NDA-2025 (1)]
264. The probability of a man hitting a target is $\frac{1}{5}$. If the man fires 7 times, then what is the probability that he hits the target at least twice?
(a) $1 - \left(\frac{3}{5}\right)\left(\frac{4}{5}\right)^6$ (b) $1 - \left(\frac{3}{5}\right)\left(\frac{4}{5}\right)^7$
(c) $1 - \left(\frac{11}{5}\right)\left(\frac{4}{5}\right)^6$ (d) $1 - \left(\frac{11}{5}\right)\left(\frac{4}{5}\right)^7$ [NDA-2025 (1)]
265. Let X be a random variable following binomial distribution whose mean and variance are 200 and 160 respectively. What is the value of the number of trials (n)?
(a) 500 (b) 1000
(c) 1500 (d) 2000 [NDA-2025 (1)]
266. In a sample survey of a village, the probability that a farmer is in debt is 0.60. What is the probability that three randomly selected farmers are all in debt (assume independence of events)?
(a) 0.000216 (b) 0.064
(c) 0.216 (d) 0.512 [NDA-2025 (2)]
267. The probability that a family owns a laptop is 0.68; that it also owns a desktop is 0.56. If the probability that it owns both is 0.48, then what is the probability that a randomly selected family owns a laptop or a desktop?
(a) 0.80 (b) 0.76
(c) 0.36 (d) 0.28 [NDA-2025 (2)]
268. An urn contains 10 white and 5 red balls. If two balls are drawn at random, then what is the probability that both the balls are red?
(a) $\frac{2}{21}$ (b) $\frac{1}{7}$
(c) $\frac{4}{21}$ (d) $\frac{3}{7}$ [NDA-2025 (2)]
269. An urn contains 5 white, 6 red and 4 blue balls. Three balls are drawn at random. What is the probability that a white ball, a red ball and a blue ball are drawn?
(a) $\frac{28}{91}$ (b) $\frac{2}{7}$
(c) $\frac{24}{91}$ (d) $\frac{23}{91}$ [NDA-2025 (2)]
270. Under which of the following conditions may binomial distribution be used?

- I. The number of trials is infinite and not fixed.
 II. The trials are independent.
 III. Each trial has two possible outcomes.
 Select the correct answer using the code given below:

(a) I only (b) III only
 (c) I and II (d) II and III

[NDA-2025 (2)]

271. A person X speaks the truth 4 out of 5 times and person Y speaks the truth 5 out of 6 times. What is the probability that they will contradict each other in stating the fact?
 (a) $\frac{3}{10}$ (b) $\frac{1}{15}$
 (c) $\frac{1}{6}$ (d) $\frac{7}{10}$

[NDA-2025 (2)]

272. The probability that a student passes Physics test is $\frac{2}{3}$ and the probability that he passes both Physics test and English test is $\frac{11}{15}$. The probability that he passes at least one test is $\frac{4}{5}$. What is the probability that he passes English test?
 (a) $\frac{11}{15}$ (b) $\frac{13}{15}$
 (c) $\frac{14}{15}$ (d) 1

[NDA-2025 (2)]

273. An event X can happen with probability p and event Y can happen with probability q . Further, X and Y are independent events. Which of the following statements is/are correct?
 I. The probability that exactly one of the events happens is $p + q - pq$.
 II. The probability that at least one of the events happens is $p + q - 2pq$.
 Select the answer using the code given below.
 (a) I only (b) II only
 (c) Both I and II (d) Neither I nor II

[NDA-2025 (2)]

274. Three faces of a die are black, two faces are white and one face is red. The die is tossed three times. What is the probability that the colours black, white and red appear in the first, second and third tosses respectively?
 (a) $\frac{1}{36}$ (b) $\frac{1}{6}$
 (c) $\frac{7}{36}$ (d) $\frac{5}{36}$

[NDA-2025 (2)]

275. A fair coin is tossed 4 times. What is the probability that two heads do not occur consecutively?
 (a) $\frac{1}{8}$ (b) $\frac{3}{8}$
 (c) $\frac{7}{16}$ (d) $\frac{1}{2}$

[NDA-2025 (2)]

276. In a throw of three dice, what is the probability of getting one prime number, one composite number and one number which is neither prime nor composite?
 (a) $\frac{1}{2}$ (b) $\frac{1}{3}$
 (c) $\frac{1}{4}$ (d) $\frac{1}{6}$

[NDA-2025 (2)]

277. An integer is chosen at random from the first 50 integers. What is the probability that the integer is neither divisible by 5 nor 9?
 (a) $\frac{7}{10}$ (b) $\frac{18}{25}$
 (c) $\frac{37}{50}$ (d) $\frac{19}{25}$

[NDA-2025 (2)]

278. Out of 50 consecutive natural numbers, two integers are chosen at random. What is the probability that their sum is odd?
 (a) $\frac{1}{2}$ (b) $\frac{24}{49}$
 (c) $\frac{1}{4}$ (d) $\frac{25}{49}$

[NDA-2025 (2)]

279. Let X be a random variable following binomial distribution with parameters $n = 5$ and $p = k$. Further, $P(X = 1) = 0.4096$ and $P(X = 2) = 0.2048$. What is the value of k ?
 (a) 0.2 (b) 0.25
 (c) 0.3 (d) 0.35

[NDA-2025 (2)]

280. If $P(A) = 0.3$, $P(B) = 0.4$ and $P(A|B) = 0.5$, then what is the value of $P(B|A)$?
 (a) 0.325 (b) 0.333
 (c) 0.375 (d) 0.667

[NDA-2025 (2)]

281. If $P(A) = \frac{1}{3}$, $P(B) = \frac{1}{2}$ and $P(A \cap B) = \frac{1}{4}$, then what is the value of $P(A \cup B)$?
 (a) $\frac{7}{12}$ (b) $\frac{2}{3}$
 (c) $\frac{3}{4}$ (d) $\frac{11}{12}$

[NDA-2025 (2)]

ANSWER KEY

1.	d	2.	a	3.	d	4.	c	5.	b	6.	a	7.	a	8.	a	9.	b	10.	b
11.	a	12.	d	13.	b	14.	a	15.	c	16.	b	17.	d	18.	b	19.	b	20.	c
21.	d	22.	c	23.	d	24.	b	25.	b	26.	c	27.	c	28.	d	29.	a	30.	d
31.	c	32.	c	33.	b	34.	b	35.	b	36.	c	37.	c	38.	b	39.	d	40.	c
41.	a	42.	c	43.	c	44.	b	45.	d	46.	b	47.	c	48.	d	49.	a	50.	d
51.	c	52.	b	53.	a	54.	d	55.	d	56.	b	57.	b	58.	c	59.	a	60.	d
61.	a	62.	d	63.	d	64.	c	65.	d	66.	a	67.	b	68.	c	69.	b	70.	c
71.	b	72.	d	73.	b	74.	c	75.	c	76.	d	77.	a	78.	a	79.	a	80.	a
81.	a	82.	d	83.	a	84.	b	85.	d	86.	b	87.	a	88.	c	89.	a	90.	b
91.	c	92.	b	93.	d	94.	d	95.	a	96.	a	97.	c	98.	a	99.	c	100.	c
101.	d	102.	c	103.	b	104.	a	105.	c	106.	b	107.	d	108.	a	109.	c	110.	b
111.	c	112.	d	113.	c	114.	b	115.	a	116.	c	117.	c	118.	c	119.	b	120.	a
121.	c	122.	b	123.	a	124.	b	125.	b	126.	a	127.	b	128.	b	129.	a	130.	b
131.	b	132.	d	133.	c	134.	c	135.	a	136.	c	137.	a	138.	b	139.	c	140.	c
141.	b	142.	c	143.	c	144.	b	145.	a	146.	b	147.	b	148.	a	149.	b	150.	d
151.	d	152.	a	153.	c	154.	c	155.	d	156.	d	157.	d	158.	a	159.	c	160.	c
161.	b	162.	b	163.	b	164.	a	165.	a	166.	d	167.	c	168.	d	169.	d	170.	b
171.	c	172.	c	173.	c	174.	a	175.	c	176.	d	177.	c	178.	d	179.	a	180.	d
181.	d	182.	a	183.	a	184.	d	185.	a	186.	b	187.	b	188.	c	189.	d	190.	d
191.	b	192.	b	193.	b	194.	b	195.	b	196.	c	197.	b	198.	d	199.	a	200.	d
201.	c	202.	c	203.	d	204.	a	205.	b	206.	*	207.	c	208.	c	209.	b	210.	c
211.	b	212.	b	213.	c	214.	a	215.	d	216.	d	217.	b	218.	d	219.	b	220.	b
221.	b	222.	a	223.	a	224.	b	225.	d	226.	a	227.	d	228.	c	229.	c	230.	a
231.	b	232.	c	233.	c	234.	d	235.	c	236.	b	237.	c	238.	b	239.	a	240.	d
241.	b	242.	d	243.	d	244.	a	245.	d	246.	a	247.	d	248.	c	249.	c	250.	c
251.	a	252.	a	253.	d	254.	b	255.	c	256.	a	257.	a	258.	d	259.	c	260.	a
261.	b	262.	b	263.	b	264.	c	265.	b	266.	c	267.	b	268.	a	269.	c	270.	d
271.	a	272.	b	273.	d	274.	a	275.	a	276.	d	277.	b	278.	d	279.	a	280.	d
281	d																		

Solutions

Sol. 1. (d)

The favorable events to comes head at flipping a first coin = {HT, HH}

The favorable events to comes tail at flipping a first coin = {T1, T2, T3, T4, T5, T6}

The total favorable events
= {HT, HH, T1, T2, T3, T4, T5, T6}

∴ required probability = 1/8

Sol. 2. (a)

Given,

$$P(A/B)=0.4, P(\bar{B})=0.8, P(A \cup B)=0.5$$

$$\Rightarrow P(B) + P(\bar{B}) = 1$$

$$\therefore P(B) = 1 - P(\bar{B}) = 1 - 0.8 = 0.2$$

We know that multiplication theorem of probability,

$$P(A \cap B) = P(B) \cdot P(A/B) = 0.2 \times 0.4 = 0.08$$

Sol. 3. (a)

∴ Favorable event for first doll = 1

∴ Probability for first doll = 1/6

Favorable event for second doll = 5

∴ Probability for second doll = 5/6

Favorable event for third doll = 4

∴ Probability for third doll = 4/6

$$\therefore \text{Required probability} = \frac{1}{6} \cdot \frac{5}{6} \cdot \frac{4}{6} = \frac{5}{54}$$

Sol. 4. (c)

$$\frac{1}{4!} = \frac{1}{24}$$

Sol. 5. (b)

$$P(\text{All three questions are correct}) = \frac{1}{4} \cdot \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{64}$$

$P(\text{All three questions will not correct}) = 1 -$

$$\frac{1}{64} = \frac{63}{64}$$

Sol. 6. (a)

Sol. 7. (a)

$$\frac{{}^2C_2}{{}^4C_2} = \frac{1}{6}$$

Sol. 8. (a)

Let radius of given circle be r . Now, make a concentric circle with radius $r/2$.

The given point is close to the circumference than to the centre, if it lies in the shaded region.

∴ required probability =

$$\frac{\pi \{r^2 - (r/2)^2\}}{\pi r^2} = \frac{3/4 r^2}{r^2} = \frac{3}{4}$$

Sol. 9. (b)

$$\frac{{}^{40}C_1 + {}^{25}C_1}{{}^{125}C_1} = \frac{40 + 25}{125} = \frac{13}{25}$$

Sol. 10. (b)

Probability of selection of husband $P(H) = 1/5$

$$\therefore P(\bar{H}) = 1 - \frac{1}{5} = \frac{4}{5}$$

and probability of selection of wife $P(W) = 1/3$

$$\therefore P(\bar{W}) = 1 - \frac{1}{3} = \frac{2}{3}$$

∴ Probability that only one of them is selected

$$= P(H)P(\bar{W}) + P(\bar{H})P(W)$$

$$= \left(\frac{1}{2}\right)\left(\frac{2}{3}\right) + \left(\frac{4}{5}\right)\left(\frac{1}{3}\right) = \frac{2}{15} + \frac{4}{15} = \frac{6}{15} = \frac{2}{5}$$

Sol. 11. (a)

Total number of ways of the arrangement of the

letters of the word 'UNIVERSITY' = $\frac{10!}{2!}$

Number of ways that both I's come together = 9!

∴ Number of ways that both I's do not come together $\frac{10!}{2} - 9!$

$$\therefore \text{Required probability} = \frac{\left(\frac{10!}{2} - 9!\right)}{\left(\frac{10!}{2}\right)} = \frac{4}{5}$$

Sol. 12. (d)

required probability = $6/216 = 1/36$

Sol. 13. (b)

Sol. 14. (a)

Sol. 15. (b)

For certain event $P(E) = 1$

Sol. 16. (b)

$$\text{Given, } P(A \cup B) = \frac{3}{4}, P(A \cap B) = \frac{1}{4}, P(A^c) = 2/3$$

$$P(A) = 1 - P(A^c) = 1 - \frac{2}{3} = \frac{1}{3}$$

Now we have

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\frac{3}{4} = \frac{1}{3} + P(B) - \frac{1}{4}$$

$$\Rightarrow P(B) = \frac{3}{4} + \frac{1}{4} - \frac{1}{3} = \frac{2}{3}$$

Sol. 17. (d)

$${}^3C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right) = \frac{3}{8}$$

Sol. 18. (b)

Sol. 19. (b)

$$\text{mean} = np = 10 \cdot \frac{2}{3} = 6.67 \approx 7$$

Sol. 20. (c)

Probability that both balls are of same colour =

$P(\text{one green ball from urn I}) \times P(\text{One green ball from urn II})$

$$= \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

Sol. 21. (d)

Probability of getting at least one green ball = $P(\text{one green ball from urn I and one white ball from urn II}) + P(\text{one green ball from urn I and one green ball from urn II}) + P(\text{one black ball from urn I and one green ball from urn II})$

$$= \frac{1}{2} \times \frac{1}{2} + \frac{1}{2} \times \frac{1}{2} + \frac{1}{2} \times \frac{1}{2} = \frac{3}{4}$$

Sol. 22. (c)

Let S be the sample space.

∴ $n(S) = 36$

A: even number on the first die

= {(2,1), (2,2), (2,3), (2,4), (2,5), (2,6), (4,1), (4,2), (4,3), (4,4), (4,5), (4,6), (6,1), (6,2), (6,3), (6,4), (6,5), (6,6)}

B: Number on the second die is greater than 4
= {(1,5), (1,6), (2,5), (2,6), (3,5), (3,6), (4,5), (4,6), (5,5), (5,6), (6,5), (6,6)}

$$n(A) = 18, n(B) = 12, P(A) = \frac{18}{36} = \frac{1}{2}$$

$$\Rightarrow P(B) = \frac{12}{36} = \frac{1}{3}$$

Also, $A \cap B = \{(2,5), (2,6), (4,5), (4,6), (6,5), (6,6)\}$

$$P(A \cap B) = \frac{6}{36} = \frac{1}{6}$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \\ = \frac{1}{2} + \frac{1}{3} - \frac{1}{6} = \frac{3+2-1}{6} = \frac{4}{6} = \frac{2}{3}$$

Sol. 23. (d)

$$\frac{1}{2} \cdot \frac{1}{3} = \frac{1}{6}$$

Sol. 24. (b)

Sol. 25. (b)

We know that, events are said to be mutually exclusive or incompatible, if the happening of anyone of them precludes the happening of all the others i.e., if no two or more of them can happen simultaneously in the same trial.

Two or more events associated to a random experiment are said to be mutually exclusive events, if the occurrence of one of them prevents or denies the occurrence of all others.

Here, A and B are mutually exclusive because A have a set of even numbers and B have a set of odd numbers. Which cannot be happen simultaneously in the same trial.

Sol. 26. (c)

For die, sample space = {1,2,3,4,5,6}

Total number = 6

For a coin, sample space = {H, T}

Total number = 2

For another coin, sample space = {H, T}

Total number = 2

Therefore, associated sample space of throwing together a die and two coins = $6 \times 2 \times 2 = 24$ points.

Which is possible, so, $n = 50$, $p = 1/5$ and $q = 4/5$ may be the parameter of a Binomial distribution.

Sol. 27. (c)

Given that,

$$P(\bar{A}) = 0.4, P(\bar{B}) = 0.3 \text{ and } P(A \cup B) = 0.9$$

By addition theorem of probability.

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\Rightarrow P(A \cup B) = 1 - P(\bar{A}) + 1 - P(\bar{B}) - \{1 - P(\overline{A \cap B})\}$$

$$\Rightarrow P(A \cup B) = 2 - P(\bar{A}) - P(\bar{B}) - 1 + P(\overline{A \cap B})$$

$$P(\overline{A \cap B}) = P(A \cup B) + P(\bar{A}) + P(\bar{B}) - 1$$

$$[\text{by De-Morgan's law, } P(\overline{A \cap B}) = P(\bar{A} \cup \bar{B})]$$

$$\Rightarrow P(\bar{A} \cup \bar{B}) = 0.9 + 0.4 + 0.3 - 1 = 0.6$$

Sol. 28. (d)

Probability of getting head in a single toss,

$P(H) = 1/2$

Probability of getting tail in a single toss,

$P(T) = 1/2$

∴ Required probability – P(HHHHT or TTTTH)

Sol. 29. (a)

$$\frac{6}{36} = \frac{1}{6}$$

Sol. 30. (d)

Probability of getting a head in a single coin, $P = \frac{1}{2}$

$$\frac{1}{2}$$

Probability of getting a tail in a single coin, $q = \frac{1}{2}$

∴ Probability of getting exactly two heads

$$= {}^4C_2 \times \left(\frac{1}{2}\right)^{4-2} \times \left(\frac{1}{2}\right)^2$$

$$= \frac{4 \times 3}{2} \times \left(\frac{1}{2}\right)^2 \times \left(\frac{1}{2}\right)^2 = 6 \times \frac{1}{4} \times \frac{1}{4} = \frac{3}{8}$$

Sol. 31. (c)

$$\frac{{}^3C_2}{{}^8C_2} = \frac{3}{28}$$

Sol. 32. (c)

Sol. 33. (b)

Sol. 34. (b)

There are two binomial perimeter n and p

Sol. 35. (b)

Given that, in a binomial distribution, the occurrence and the non-occurrence of an event are equally likely.

i.e., $p=q = \frac{1}{2}$ and mean of Binomial distribution =

$$np = 6$$

$$\Rightarrow n \times \frac{1}{2} = 6 \Rightarrow n = 12$$

So, the required number of trials is 12.

Sol. 36. (c)

$$\frac{3}{36} = \frac{1}{12}$$

Sol. 37. (c)

Total sample space, $n(S) = 6 \times 6 \times 6 = 216$

Favorably cases = $\{(1,1,1), (2,2,2), (3,3,3), (4,4,4), (5,5,5), (6,6,6)\}$

∴ Total number of favorable cases,

$$n(E) = 6$$

$$\therefore \text{Required probability} = \frac{n(E)}{n(S)} = \frac{6}{216} = \frac{1}{36}$$

Sol. 38. (b)

$\{T, (H,1), (H,2), (H,3), (H,4), (H,5), (H,6)\}$

total 7 outcomes

Sol. 39. (d)

$np=2$ and $np=4 \Rightarrow 2q=4 \Rightarrow q=2$

But $p+q=12$

∴ $q \neq 2$

(b) $n=4$ and $p = 3/2$

∴ $p+q=1$

$$\Rightarrow q = 1 - \frac{3}{2} = -\frac{1}{2}$$

But $p, q \geq 0$

$$\therefore q \neq \frac{1}{2}$$

(c) $n=8$ and $p=1$

$$\therefore p+a=1 \Rightarrow q=1-1=0$$

But $q \neq 0 \therefore 0 < p, q < 1$

(d) $np=10$ and $npq=8$

$$\Rightarrow 10q=8$$

$$\Rightarrow q = \frac{4}{5}$$

$$\text{and } p=1-q=1-4/5=1/5$$

$$\therefore n=50$$

Which is possible, so $n=50$, $p=\frac{1}{5}$ and $q=\frac{4}{5}$ may

be the parameter of a binomial distribution.

Sol. 40. (c)

$$\text{Given that, } P(A) = \frac{1}{5}, P(A \cup B) = \frac{7}{10}$$

Also, A and B are independent events, then

$$P(A \cap B) = P(A) \cdot P(B)$$

$$\Rightarrow P(A) + P(B) - P(A \cup B) = P(A) \cdot P(B)$$

(by addition theorem of probability)

$$\Rightarrow \frac{1}{5} + P(B) - \frac{7}{10} = \frac{1}{5} \times P(B)$$

$$\Rightarrow P(B) + \frac{2-7}{10} = \frac{P(B)}{5} \Rightarrow P(B) - \frac{P(B)}{5} = \frac{5}{10}$$

$$\Rightarrow \frac{4P(B)}{5} = \frac{1}{2} \Rightarrow P(B) = \frac{5}{8}$$

$$\therefore P(\bar{B}) = 1 - P(B) = 1 - \frac{5}{8} = \frac{3}{8}$$

Sol. 41. (a)

I. Two or more events associated to a random experiment are exhaustive, if their union is the sample space.

$$\text{i.e., } A_1 \cup A_2 \cup \dots \cup A_n = S$$

$$\Rightarrow P(A_1 \cup A_2 \cup \dots \cup A_n) = 1$$

II. Two or more events associated to a random experiment are said to be mutually exclusive events, if the occurrence of one of them prevents or denies the occurrence of all others.

$$\text{i.e., } E_1 \cap E_2 \cap \dots \cap E_n = \phi$$

$$\Rightarrow P(E_1 \cap E_2 \cap \dots \cap E_n) = 0$$

Sol. 42. (c)

$$\text{Given that, } P(A) = \frac{2}{3}, P(B) = \frac{2}{5}$$

$$\text{and } P(A \cup B) - P(A \cap B) = \frac{2}{5} \dots (i)$$

[From Eq.(i)]

$$\{P(A) + P(B) - P(A \cap B) - P(A \cap B)\} = \frac{2}{5}$$

(By addition theorem of probability)

$$\Rightarrow \frac{2}{3} + \frac{2}{5} - 2P(A \cap B) = \frac{2}{5}$$

$$\Rightarrow \frac{2}{3} + \frac{2}{5} - P(A \cap B) = \frac{2}{5}$$

$$\Rightarrow 2P(A \cap B) = \frac{2}{3}$$

$$\therefore P(A \cap B) = \frac{1}{3}$$

Sol. 43. (c)

Sol. 44. (b)

Sol. 45. (d)

Required probability = $(1/6)((1/5)(1/4)) = 1/120$

Sol. 46. (b)

$$\therefore P(A \text{ wining}) = \frac{6}{12} = \frac{1}{2}$$

$$\text{and } P(B \text{ wining}) = \frac{4}{12} = \frac{1}{3}$$

∴ Required probability

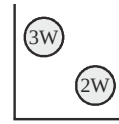
$$= P(A) \cdot P(B) \cdot P(A) + P(B) \cdot P(A) \cdot P(B)$$

$$= \frac{1}{2} \cdot \frac{1}{3} \cdot \frac{1}{2} + \frac{1}{3} \cdot \frac{1}{2} \cdot \frac{1}{3}$$

$$= \frac{1}{12} + \frac{1}{18} = \frac{3+2}{36} = \frac{5}{36}$$

Sol. 47. (c)

Total sample space, $n(S) = {}^5C_2$ because we select two balls out of 5 balls.



Now, Favourable events, $n(E) =$ Two selected balls are black. $= {}^3C_0 \times {}^2C_2$

∴ Required probability =

$$\frac{n(E)}{n(S)} = \frac{{}^3C_0 \times {}^2C_2}{{}^5C_2} = \frac{1 \times 1}{(5 \times 4)/2} = \frac{1}{10}$$

Solution (for next four) Given that,

$X =$ Set of odd numbers from the set A

$Y =$ Set of even numbers from the set A

Let set $A = \{1, 2, 3, 4, 5, 6, 7\}$ and $Z = X + Y$

Sol. 48. (d)

Now, $Z = 5$ is only possible

When $X = 1, 3$ and $Y = 4, 2$

Sample space = $\{(1,2), (1,4), (1,6), (3,2), (3,4), (3,6), (5,2), (5,4), (5,6), (7,2), (7,4), (7,6)\}$

∴ total number of sample space, $n(s) = 12$

and favourable space = $\{(1,4), (3,2)\}$

∴ Total number of favourable cases, $n(E) = 2$

$$\text{So, } p(z=5) = \frac{n(E)}{n(S)} = \frac{2}{12} = \frac{1}{6}$$

Sol. 49. (a)

We know that,

Sum of even and odd numbers never be an even number.

∴ total number of favourable cases, $n(E_2) = 0$

$$\text{So, } P(Z = 10) = \frac{n(E_2)}{n(S)} = \frac{0}{12} = 0$$

Sol. 50. (d)

Now, $Z > 11$ is only possible when $X=7$ and $Y=6$

∴ Favourable case = $(7,6)$

Total number of favourable cases, $n(E_3) = 1$

$$\therefore P(Z > 11) = \frac{n(E_3)}{n(S)} = \frac{1}{12}$$

Sol. 51. (c)

∴ $Z =$ Product of two prime numbers

$= \{2 \times 2, 3 \times 3, 5 \times 5, 7 \times 7, 2 \times 3, 2 \times 5, 2 \times 7, 3 \times 5, 3 \times 7, 5 \times 7\}$

$= \{4, 9, 25, 49, 6, 10, 14, 15, 21, 35\}$

∴ $Z = X + Y = 7 + 6 = 13$

(i.e., maximum of X is 7 and maximum of Y is 6)

So, maximum value of 13 i.e., Z cannot be exceed from 13.

∴ $z \neq 25, 49, 14, 15, 21, 35$

So, $z \in \{4, 9, 6, 10\}$, only the possible values of product of two prime numbers.

Also know that, sum of the odd and even number is always and odd numbers.

(∴ $Z = X + Y$)

So, $Z \neq 4, 6, 10$

∴ $Z = 9$ is only remaining case

Favourable case = $\{(3,6), (5,4), (7,2)\}$

∴ Total number of favourable cases, $n(E_3) = 3$

So, required probability

i.e., $P(Z \text{ is the product of two prime numbers})$
 $= P(Z=9) = \frac{n(E_3)}{n(S)} = \frac{3}{12} = \frac{1}{4}$

Sol. 52. (a)
 $P(A|B') = \frac{P(A \cap B')}{P(B')} = \frac{P(A) - P(A \cap B)}{1 - P(B)}$

Sol. 53. (a)
 $\therefore 1U, 1N, 2I, 2V, 1E, 1R, 1S, 1T, 1Y$
 $\therefore \text{total number of possible arrangements} = \frac{10!}{2!}$

and favourable arrangements = $\frac{10!}{2!} - 9!$

$\therefore \text{Required probability} = \frac{\frac{10!}{2!} - 9!}{\frac{10!}{2!}} = \frac{9!(5-1)}{9! \times 10} \times 2 = \frac{4}{5}$

Sol. 54. (d)
 Box I $\rightarrow 4W, 3B$
 Box II $\rightarrow 3W, 4B$
 Probability for choosing first box = $3/6 = 1/2$
 Probability for choosing second box = $1/2$
 $\therefore \text{Required probability} =$

$$\frac{\frac{1}{2} \times \frac{3}{7}}{\frac{1}{2} \times \frac{3}{7} + \frac{1}{2} \times \frac{4}{7}} = \frac{\frac{3}{14}}{\frac{3}{14} + \frac{4}{14}} = \frac{3}{7}$$

Sol. 55. (d)
 Let A and B be the events that X and Y qualify the examination, respectively.

We have, $P(A) = 0.05$, $P(B) = 0.10$ and $P(A \cap B) = 0.02$

To find $P(\text{only one of A and B will qualify examination})$

Clearly, $P(\text{only one of A and B will qualify the examination})$

$$= P(A \cap \bar{B}) + P(\bar{A} \cap B)$$

$$= P(A) - P(A \cap B) + P(B) - P(A \cap B)$$

$$= P(A) + P(B) - 2P(A \cap B)$$

$$= 0.05 + 0.1 - 2(0.02) = 0.15 - 0.04 = 0.11$$

Sol. 56. (b)
 Let S be the sample space of the experiment and E be the event that the most three tails occur.

Clearly, $n(S) = 2^4 = 16$ and $n(E) = {}^4C_0 + {}^4C_1 + {}^4C_2 + {}^4C_3$

$$= 1 + 4 + \frac{4 \times 3}{2 \times 1} + 4 = 1 + 4 + 6 + 4 = 15$$

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{15}{16}$$

Sol. 57. (b)
 $A = \{10, 12, 14, 16, 18, 20\}$
 $B = \{11, 13, 17, 19\}$
 $C = \{15\}$
 $D = \{10, 11, 12, 13, 14, 15, 16\}$
 $E = \{16\}$

A, B and D are Exhaustive events but not mutually exclusive events

Sol. 58. (c)
 A, B and C are mutually exclusive and exhaustive events

Sol. 59. (a)

B and C are mutually exclusive events but not exhaustive events

Sol. 60. (d)
 C and E are Elementary events

Sol. 61. (a)
 $P(A \cap B) \leq P(A) \leq P(A \cup B) \leq P(A) + P(B)$

Sol. 62. (d)
 We have, box B_1 , containing 2 red and 3 black balls and box B_2 containing 3 red and 2 black balls.

Let E_1 be the event that box B_1 is chosen, E_2 be the event that box B_2 is chosen and A be the event that balls are of different colours.

Clearly, $P(E_1) = P(E_2) = \frac{1}{2}$

$$P\left(\frac{A}{E_1}\right) = \frac{{}^2C_1 \times {}^3C_1}{{}^5C_2} = \frac{6}{10} = \frac{3}{5}$$

$$\text{and } P\left(\frac{A}{E_2}\right) = \frac{{}^3C_1 \times {}^2C_1}{{}^5C_2} = \frac{6}{10} = \frac{3}{5}$$

Now, by the theorem of total probability, we have

$$P(A) = P(E_1) \cdot P\left(\frac{A}{E_1}\right) + P(E_2) \cdot P\left(\frac{A}{E_2}\right)$$

$$= \frac{1}{2} \cdot \frac{3}{5} + \frac{1}{2} \cdot \frac{3}{5} = \frac{1}{2} \left(\frac{2 \times 3}{5} \right) = \frac{3}{5}$$

Sol. 63. (d)
 Let solving a question is success and not solving a question is failure.

Let probability of success = $p = \frac{1}{2}$

And probability of failure = $q = \frac{1}{2}$

Let X be random variable that denotes the number of success in 5 trials.

Clearly, $X \sim B \text{ in } \left(5, \frac{1}{2}\right)$

$$\therefore P(X = x) = {}^5C_x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{5-x}$$

$$\Rightarrow x = 0, 1, 2, 3, 4, 5 = {}^5C_x \left(\frac{1}{2}\right)^5$$

Required probability = $P(X \geq 2)$

$$= 1 - [P(x=0) + P(x=1)] = 1$$

$$- \left[{}^5C_0 \left(\frac{1}{2}\right)^5 + {}^5C_1 \left(\frac{1}{2}\right)^5 \right] = 1 - \frac{6}{32} = \frac{26}{32} = \frac{13}{16}$$

Sol. 64. (c)
 As, $A \subseteq B$, then $A \cup B = B$ and $A \cap B = A$

Clearly, $P(A \cap \bar{B}) = P(\phi) = 0$

$$\text{Now, } P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} = \frac{P(A)}{P(B)}$$

$$\Rightarrow P\left(\frac{A}{A \cup B}\right) = P\left(\frac{A}{B}\right) = \frac{P(A)}{P(B)}$$

$$\text{but } P\left(\frac{B}{A}\right) = \frac{P(B \cap A)}{P(A)} = \frac{P(A)}{P(A)} = 1$$

Hence, option (c) is not correct.

Sol. 65. (d)
 Let X be a random variable that follows binomial distribution with parameter n and p.

Then, mean = $E(x) = np = 2 \dots (i)$

and variance = $\text{var}(x) = npq = 1 \dots (ii)$

On dividing Eq. (ii) by Eq. (i), we get $q = 1/2$

$$\Rightarrow p = 1/2 \quad (\because p+q=1)$$

Now, on substituting the value of p in Eq. (i), we get $n = 4$

Thus, $X \sim B \text{ in } \left(\frac{1}{2}, 4\right)$ and

$$P(X=x) = {}^4C_x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{4-x}, x = 0, 1, 2, 3, 4$$

$$= P(x=0) = {}^4C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^{4-0}$$

$$\Rightarrow {}^4C_0 \left(\frac{1}{2}\right)^4 = 1 \times \frac{1}{16} = \frac{1}{16}$$

Sol. 66. (a)
 exactly one will hit $\frac{1}{2} \cdot \frac{2}{3} + \frac{1}{3} \cdot \frac{1}{2} = \frac{1}{2}$

Sol. 67. (b)
 We have, $P(A \cup B) = \frac{1}{3}$, $P(A \cap B) =$

$$\frac{1}{4} \text{ and } P(\bar{A}) = \frac{2}{3}$$

To find: $P(B)$

We know that,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\Rightarrow \frac{3}{4} = (1 - P(\bar{A})) + P(B) - \frac{1}{4}$$

$$\Rightarrow \frac{3}{4} + \frac{1}{4} = 1 - P(\bar{A}) + P(B)$$

$$\Rightarrow 1 = 1 - P(\bar{A}) + P(B) \Rightarrow P(B) = P(\bar{A}) = \frac{2}{3}$$

Sol. 68. (c)
 Clearly, the possible outcomes in throwing of two dice are:

$$\begin{pmatrix} (1,1) & (1,2) & (1,3) & - & - & (1,5) & (1,6) \\ (2,1) & (2,2) & (2,3) & - & - & (2,5) & (2,6) \\ - & - & - & - & - & - & - \\ - & - & - & - & - & - & - \\ (5,1) & (5,2) & (5,3) & - & - & (5,5) & (5,6) \\ (6,1) & (6,2) & (6,3) & - & - & (6,5) & (6,6) \end{pmatrix}$$

Since, 5 will never appear on any one of the dice, so number of exhaustive events are:

$$= 36 - 6 - 6 + 1 = 36 - 11 = 25$$

Sol. 69. (b)
 Total number of ways drawing 2 cards successively without replacement = ${}^{52}C_2$

Number of aces = 4

and number of ways 2 aces without replacement = 4C_2

$\therefore \text{Required probability} =$

$$\frac{{}^4C_2}{{}^{52}C_2} = \frac{4 \times 3}{52 \times 51} = \frac{1}{13 \times 17} = \frac{1}{221}$$

Sol. 70. (c)
 Probability for head, $P(A) = 1/2$

$$\Rightarrow P(\bar{A}) = 1 - \frac{1}{2} = \frac{1}{2}$$

1.No head in four trials

$$\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{16}$$

2.Success in third attempt

$$\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{8}$$

$\therefore$ Both statements are correct.

Sol. 71. (b)

Ist Bag, IInd Bag

B-2 W-3

W-4 B-5

Total 6 Total 8

Taking one ball from each bag, the required probability

$$\frac{{}^2C_1 \times {}^3C_1}{{}^6C_1} + \frac{{}^4C_1 \times {}^5C_1}{{}^8C_1}$$

$$= \frac{2}{6} \times \frac{3}{8} + \frac{4}{6} \times \frac{5}{8} = \frac{6+20}{48} = \frac{26}{48} = \frac{13}{24}$$

Sol. 72. (d)

$$P(A) = \frac{1}{2} \Rightarrow P(\bar{A}) = 1 - \frac{1}{2} = \frac{1}{2}$$

$$P(B) = \frac{1}{3}$$

$$\Rightarrow P(\bar{B}) = 1 - \frac{1}{3} = \frac{2}{3}$$

$$P(C) = \frac{1}{4} \Rightarrow P(\bar{C}) = 1 - \frac{1}{4} = \frac{3}{4}$$

Probability for not solving the question.

$$\Rightarrow \frac{1}{2} \times \frac{2}{3} \times \frac{3}{4} = \frac{1}{4}$$

∴ The probability that the problem is solved = 1 -

$$\frac{1}{4} = \frac{3}{4}$$

Sol. 73. (b)

Single digit numbers (1, 2, 3, 4, 5, 6, 7, 8)

Pairs with Prime sum:

{(1,2) (1,4) (1,6) (2,1) (2,3) (2,5) (2,9) (3,2) (3,4) (3,8) (4,1) (4,3) (4,7) (4,9) (5,2) (5,6) (5,8) (6,1) (6,5) (6,7) (7,4) (7,6) (8,3) (8,5) (8,9) (9,2) (9,4) (9,8)}

⇒ Total = 28 pairs

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{28}{72} = \frac{7}{18}$$

Sol. 74. (c)

Digits are: 1, 2, 3, ..., 9

We know that the product can be odd only when all the three digits chosen are odd.

$$\therefore n(E) = {}^5C_3 = \frac{5!}{3!2!} = \frac{5 \times 4}{2} = 10$$

$$\therefore \text{Also, } n(S) = {}^9C_3 = \frac{9!}{3!6!} = \frac{9 \times 8 \times 7 \times 6!}{3! \times 6!} = 84$$

$$\therefore P(E) = \frac{10}{84} = \frac{5}{42}$$

Sol. 75. (c)

$$P(\bar{B}) = 0.8 \Rightarrow P(B) = 0.2$$

$$P(A \cup B) = 0.5 \text{ and } P\left(\frac{A}{B}\right) = 0.4$$

Now, $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$$\text{or } 0.5 = x + 0.2 - P(A \cap B)$$

$$\text{or } 0.3 - x = -P(A \cap B)$$

$$\text{or } x - 0.3 = P(A \cap B)$$

$$\text{or } P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} \Rightarrow 0.4 = \frac{x - 0.3}{0.2}$$

$$\Rightarrow x = 0.08 + 0.3 \Rightarrow P(A) = 0.38$$

Sol. 76. (d)

Here, $np = 2$... (i)

And $npq = 1$... (ii)

$$\Rightarrow 2q = 1 \text{ {from (i) & (ii)}}$$

$$\Rightarrow q = 1/2$$

$$\therefore p = (1 - q) = \left(1 - \frac{1}{2}\right) = \frac{1}{2}$$

$$\text{Thus, } n = \frac{2}{p} = 4 \quad \text{{from (i)}}$$

So, the binomial distribution is given by

$${}^nC_r \left(\frac{1}{2}\right)^r$$

$$\therefore p(x > 1)$$

$$= {}^4C_2 \left(\frac{1}{2}\right)^2 + {}^4C_3 \left(\frac{1}{2}\right)^3 + {}^4C_4 \left(\frac{1}{2}\right)^4$$

$$= \left(\frac{1}{2}\right)^4 [{}^4C_2 + {}^4C_3 + {}^4C_4] = \frac{1}{16} [6 + 4 + 1] = \frac{11}{16}$$

Sol. 77. (a)

Let x = number of heads obtain in a throw of 7 coins

$$\text{Then } x \text{ is } B \left(n = 7, p = \frac{1}{2} \right)$$

The probability mass function is

$$p(x) = {}^7C_x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{7-x}, x = 0, 1, \dots, 7$$

$$p(\text{at least 3 heads}) = 1 - p(\text{at most 2 heads})$$

$$= 1 - \{p(0) + p(1) + p(2)\}$$

$$= 1 - \left\{ {}^7C_0 \left(\frac{1}{2}\right)^0 + {}^7C_1 \left(\frac{1}{2}\right)^1 + {}^7C_2 \left(\frac{1}{2}\right)^2 \right\}$$

$$= 1 - \left(\frac{1}{2}\right)^7 \{ {}^7C_0 + {}^7C_1 + {}^7C_2 \}$$

$$= 1 - \left(\frac{1}{2}\right)^7 [1 + 7 + 21] = \frac{99}{128}$$

Number of tosses is 128

$$\therefore \text{The required no. of thrown} = 128 \times \frac{99}{128} = 99$$

Sol. 78. (a)

Head for single toss:

$$P(A) = \frac{1}{2} \Rightarrow P(\bar{A}) = 1 - \frac{1}{2} = \frac{1}{2} \quad (p: x > 3)$$

∴ required no. of times

$${}^5C_4 \left(\frac{1}{2}\right)^{5-4} \left(\frac{1}{2}\right)^4 + {}^5C_5 \left(\frac{1}{2}\right)^{5-5} \left(\frac{1}{2}\right)^5$$

$$= \frac{5}{4} \left(\frac{1}{2}\right)^5 + {}^5C_5 \left(\frac{1}{2}\right)^5 = \left(\frac{1}{2}\right)^5 [5 + 1] = \frac{6}{32} = \frac{3}{16}$$

Sol. 79. (a)

$$x^2 - 3x + 2 \geq 0 \Rightarrow (x-2)(x-1) \geq 0$$

Case I. $(x-2) \geq 0$ and $(x-1) \geq 0$

(when both the factors are +ve)

$$\Rightarrow x \geq 2 \text{ and } x \geq 1$$

Case II. $(x-2) \leq 0$ and $(x-1) \leq 0$

(when both the factors are -ve)

$$\Rightarrow x \leq 2 \text{ and } x \leq 1$$

Clearly, when $x \leq 1$ or $x \geq 2$, the expression $x^2 - 3x + 2 \geq 0$

⇒ 0 to 1 and 2 to 5 there are four intervals are in our favour out of 5 so probability is 4/5

Sol. 80. (a)

$$P(E_1) = \frac{2000}{12000} = \frac{1}{6}$$

$$P(E_2) = \frac{4000}{12000} = \frac{1}{3}$$

$$P(E_3) = \frac{6000}{12000} = \frac{1}{2}$$

$$\Rightarrow P\left(\frac{E}{E_1}\right) = \frac{1}{10} \Rightarrow P\left(\frac{E}{E_2}\right) = \frac{3}{100}$$

$$\Rightarrow P\left(\frac{E}{E_3}\right) = \frac{15}{100} = \frac{3}{20}$$

By Bayes' theorem:

$$P\left(\frac{E_1}{E}\right) = \frac{P\left(\frac{E}{E_1}\right) \cdot P(E_1)}{P\left(\frac{E}{E_1}\right) \cdot P(E_1) + P\left(\frac{E}{E_2}\right) \cdot P(E_2) + P\left(\frac{E}{E_3}\right) \cdot P(E_3)}$$

$$= \frac{\frac{1}{100} \times \frac{1}{6}}{\left(\frac{1}{100} \times \frac{1}{6}\right) + \left(\frac{3}{100} \times \frac{1}{3}\right) + \left(\frac{3}{20} \times \frac{1}{2}\right)} = \frac{1}{52}$$

Sol. 81. (a)

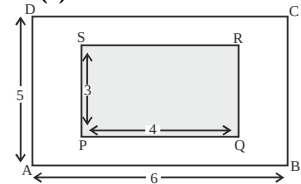
$$P(E) = \frac{1}{2} \text{ (for head in single toss)}$$

$$P(\bar{E}) = 1 - \frac{1}{2} = \frac{1}{2}$$

$$= {}^5C_1 \left(\frac{1}{2}\right)^5 + {}^5C_3 \left(\frac{1}{2}\right)^5 + {}^5C_5 \left(\frac{1}{2}\right)^5$$

$$= \left(\frac{1}{2}\right)^5 [{}^5C_1 + {}^5C_3 + {}^5C_5] = \left(\frac{1}{2}\right)^5 [5 + 10 + 1] = \frac{16}{32} = \frac{1}{2}$$

Sol. 82. (d)



Required probability =

$$\frac{\text{Area of rectangle PQRS}}{\text{Area of Rectangle ABCD}} = \frac{12}{30} = \frac{2}{5}$$

Sol. 83. (a)

The probability when one time comes head out of 100 times = ${}^{100}C_1$.

$$\left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^{100-1} = {}^{100}C_1 \left(\frac{1}{2}\right)^{100}$$

The probability when three times comes head out

$$\text{of 100 times} = {}^{100}C_3 \left(\frac{1}{2}\right)^{100} \dots\dots\dots$$

The required probability

$$= ({}^{100}C_1 + {}^{100}C_2 + {}^{100}C_3 + \dots\dots\dots + {}^{100}C_{99}) \left(\frac{1}{2}\right)^{100}$$

$$= (2^{100} - 1) \cdot \left(\frac{1}{2}\right)^{100} = 2^{99} \cdot \frac{1}{2^{100}} = \frac{1}{2}$$

Sol. 84. (b)

There are four combinations which have the sum less than 5.

(1,1,1), (1,1,2), (1,2,1), (2,1,1)

$$S(E) = 216 - 4 = 212, S(P) = 216$$

$$\text{Probability} = \frac{S(E)}{S(P)} = \frac{212}{216}$$

$$\text{Required probability} = \frac{53}{54}$$

Sol. 85. (d)

$$P(A) = 1/3$$

$$P(B) = 3/4$$

$$P(\bar{A}) = \frac{2}{3}$$

$$P(\bar{B}) = \frac{1}{4}$$

The required probability = $P(A) \cdot P$

$$(\bar{B}) + P(B) - P(\bar{A})$$

$$= \left(\frac{1}{3}\right)\left(\frac{1}{4}\right) + \left(\frac{3}{4}\right)\left(\frac{2}{3}\right)$$

$$= \frac{1}{12} + \frac{6}{12} = \frac{7}{12}$$

Sol. 86. (a)

The required probability = $\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{8}$

Sol. 87. (a)

The total number of possible outcomes = 52
Number of favorable outcomes i.e., queen of spade is 1 out of 52 cards. Therefore, probability of getting queen of spade,

$$P(A) = \frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}} = \frac{1}{52}$$

Sol. 88. (c)

Total sample space = 36 = S(P)

(1,3), (1,4), (1,5), (1,6)

(2,2), (2,3), (2,4), (2,5), (2,6)

(3,1), (3,2), (3,3), (3,4), (3,5), (3,6)

(4,1), (4,2), (4,3), (4,4), (4,5), (4,6)

(5,1), (5,2), (5,3), (5,4), (5,5), (5,6)

(6,1), (6,2), (6,3), (6,4), (6,5), (6,6)

S(E) = 33

$$\text{Probability} = \frac{S(E)}{S(P)} = \frac{33}{36}$$

$$\text{Probability} = \frac{11}{12}$$

Sol. 89. (d)

Let the number of missiles be 'n'

Then, $(0.3)^n \geq 0.8$

Therefore no n exists such that $(0.3)^n \geq 0.8$

$\Rightarrow n = 8$

Sol. 90. (b)

$$P(\bar{A} \cap B) = P(B) - P(A \cap B)$$

$$0.3 = P(B) - P(A \cap B)$$

$$= P(B) - [(A + P(B) - P(A \cup B))]$$

$$= P(A \cup B) - P(A)$$

$$P(A \cup B) = 0.3 + P(A)$$

$$= 0.3 + 0.2$$

$$= 0.5$$

$$\text{Now, } P\left(\frac{A}{A \cup B}\right) = \frac{P(A \cap (A \cup B))}{P(A \cup B)}$$

$$= \frac{P(A \cap B) \cup (A \cap B)}{P(A \cup B)} \quad \because A \text{ \& B are mutually Exclusive}$$

$$= \frac{0.2}{0.5}$$

$$= \frac{2}{5}$$

$$= 2/5$$

$$= 2/5$$

Sol. 91. (c)

Total days in the month of December = 31

There are 4 full weeks and 3 days in the month of December.

These 3 days will make these combinations

(i) Mon, Tue, Wed

(ii) Tue, Wed, Thu

(iii) Wed, Thu, Fri

(iv) Thu, Fri, Sat

(v) Fri, Sat, Sun

(vi) Sat, Sun, Mon

(vii) Sun, Mon, Tue

There are 3 likely events and 7 total events. Then required probability,

$$P = \frac{S(E)}{S(P)}$$

$$P = 3/7$$

Sol. 92. (b)

Probability of salesman making a sale = $\frac{70}{100} = 0.70$

Let A and B be the events that salesman sell the product to A and B respectively.

$$\therefore P(A) = 0.7 \text{ and } P(B) = 0.7$$

Since, A and B are independent

$$\therefore P(A \cap B) = P(A) \cdot P(B) = 0.7 \times 0.7 = 0.49$$

$$\text{Now, } P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$0.7 + 0.7 - 0.49 = 0.91$$

Sol. 93. (d)

Let p, q and r be the numbers on the upper faces in first, second and third thrown respectively.

Now, number of favorable cases such that sum of upper faces is zero equals to number of integral solution of $p+q+r=0$ subject to condition $-2 \leq p, q, r \leq 3$. i.e. equal to coefficient of t^0 in $(t^{-2} + t^{-1} + \dots + t^3)^3$

$$= \text{coefficient of } t^6 \text{ in } (1+t+\dots+t^5)^3$$

$$= \text{coefficient of } t^6 \text{ in } \left(\frac{1-t^6}{1-t} \right)^3$$

$$= \text{coefficient of } t^6 \text{ in } (1+t)^3 (1-t)^{-3}$$

$$= \text{coefficient of } t^6 \text{ in } (1+t)^3 {}^3C_1 t^6 + \dots (t-t)^3$$

$$= \text{coefficient of } t^6 \text{ in } (1+t)^{-3} - {}^3C_1 \times \text{coefficient of } t^0 \text{ in } (1-t)^{-3} = {}^8C_2 - 3 = 25$$

$$\therefore \text{Required probability} = \frac{25}{216}$$

Sol. 94. (d)

Given,

$$P(\text{chance if it rains}) = 25\%$$

$$\Rightarrow P = \frac{25}{100} = \frac{1}{4} \Rightarrow q = 1 - p = 1 - \frac{1}{4} = \frac{3}{4}$$

P(at least on rainy day with in a period of 7 days)

$$= 1 - {}^7C_0 p^0 q^7 = 1 - \left(\frac{3}{4} \right)^7$$

Sol. 95. (a)

Let the events be A=Passes in test I

B= passes in test II

C= passes in test III

Given, P(A) = m, $P(\bar{A}) = 1 - m$

$$P(B) = n, \quad P(\bar{B}) = 1 - n$$

$$P(C) = \frac{1}{2}, \quad P(\bar{C}) = \frac{1}{2}$$

Students is considered successful if he passes in test, I, II or I, III or all the three.

$\therefore$ Required probability

$$= P(ABC) + P(\bar{A}BC) + P(ABC)$$

$$P(A) \times P(B) \times P(C) + P(\bar{A}) \times P(B) \times P(C) + P(A) \times P(\bar{B}) \times P(C) + P(A) \times P(B) \times P(\bar{C})$$

$$= \frac{mn}{2} + \frac{m(1-n)}{2} + \frac{mn}{2} = \frac{mn+m-mn+mn}{2} = \frac{m(n+1)}{2}$$

$$\Rightarrow \frac{1}{2} = \frac{m(n+1)}{2} \Rightarrow m(n+1) = 1$$

Sol. 96. (a)

Let the students be E_1, E_2 and E_3 and $P(E_1), P(E_2)$ and $P(E_3)$ denotes the probabilities of the students to solve question correctly.

$\therefore$ Odds in favour of the correct answer are 5:2, 4:3 and 3:4, respectively.

$$\therefore P(E_1) = \frac{5}{5+2} = \frac{5}{7}$$

$$P(E_2) = \frac{5}{4+3} = \frac{4}{7}$$

$$P(E_3) = \frac{3}{3+4} = \frac{3}{7}$$

$\therefore$ Probability that at least two students solve question correctly.

= P

$$(E_1 E_2 \bar{E}_3) + P(\bar{E}_1 E_2 E_3) + P(E_1 \bar{E}_2 E_3) + P(E_1 E_2 E_3)$$

$$= P(E_1) \cdot P(E_2) \cdot P(\bar{E}_3) + P(\bar{E}_1) \cdot P(E_2) \cdot P(E_3)$$

$$+ P(E_1) \cdot P(\bar{E}_2) \cdot P(E_3) + P(E_1) \cdot P(E_2) \cdot P(E_3)$$

[$\because E_1, E_2$ and E_3 are independent to each other]

$$= \frac{5}{7} \cdot \frac{4}{7} \left(1 - \frac{3}{7} \right) + \left(1 - \frac{5}{7} \right) \cdot \frac{4}{7} \cdot \frac{3}{7} + \frac{5}{7} \left(1 - \frac{4}{7} \right) \cdot \frac{3}{7} + \frac{5}{7} \cdot \frac{4}{7} \cdot \frac{3}{7}$$

$$= \frac{5}{7} \cdot \frac{4}{7} \cdot \frac{4}{7} + \frac{2}{7} \cdot \frac{4}{7} \cdot \frac{3}{7} + \frac{5}{7} \cdot \frac{3}{7} \cdot \frac{3}{7} + \frac{5}{7} \cdot \frac{4}{7} \cdot \frac{3}{7}$$

$$= \frac{1}{7 \times 7 \times 7} (80 + 24 + 45 + 60) = \frac{209}{343}$$

Sol. 97. (c)

Let p denotes the probability that medicine to be 75% effective to a cure a patient, then

$$p = 75\% = \frac{75}{100} = \frac{3}{4}$$

$$q = 1 - p = \frac{3}{4} - \frac{1}{4} = \frac{1}{4}$$

Let x denotes the probability that at least one patient is cured by the medicine.

$$= 1 - {}^5C_0 (p)^0 (q)^5 = 1 - {}^5C_0 \left(\frac{3}{4} \right)^0 \left(\frac{1}{4} \right)^5$$

$$= 1 - \left(\frac{1}{4} \right)^5 = 1 - \frac{1}{1024} = \frac{1023}{1024}$$

Sol. 98. (a)

$$\text{Given, } P(A) = \frac{3}{5}, P(B) = \frac{3}{10}$$

$$\text{and } P\left(\frac{A}{B}\right) = \frac{2}{3} \Rightarrow \frac{P(A \cap B)}{P(B)} = \frac{2}{3}$$

$$\Rightarrow \frac{P(A \cap B)}{\frac{3}{10}} = \frac{2}{3} \Rightarrow P(A \cap B) = \frac{2}{3} \times \frac{3}{10} = \frac{1}{5}$$

$$\text{Now, } P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\Rightarrow P(A \cup B) = \frac{3}{5} + \frac{3}{10} - \frac{1}{5} = \frac{6+3-2}{10} = \frac{7}{10}$$

$$\Rightarrow P(\bar{A} \cup \bar{B}) = 1 - P(A \cup B) = 1 - \frac{7}{10} = \frac{3}{10}$$

$$\text{and } P(\bar{B}) = 1 - P(B) = 1 - \frac{3}{10} = \frac{7}{10}$$

$$\therefore P\left(\frac{\bar{A}}{\bar{B}}\right) = \frac{P(\bar{A} \cap \bar{B})}{P(\bar{B})} = \frac{P(\bar{A} \cup \bar{B})}{P(\bar{B})} = \frac{\frac{3}{10}}{\frac{7}{10}} = \frac{3}{7}$$

Sol. 99. (c)

$$\text{Given, } P(A) = 0.02 = \frac{2}{100} = \frac{1}{50}$$

$$P(B) = 0.10 =$$

$$\frac{10}{100} = \frac{1}{10} \text{ and } P(C) = 0.05 = \frac{5}{100} = \frac{1}{20}$$

Probability of the machine will stop, if at least one of its three components fails _____

$$P = (A \cup B \cup C) = 1 - P$$

$$(\bar{A} \cup \bar{B} \cup \bar{C}) = 1 - P(\bar{A} \cap \bar{B} \cap \bar{C})$$

$$= 1 - P(\bar{A}) P(\bar{B}) P(\bar{C}) = 1 - \left(1 - \frac{1}{50} \right) \left(1 - \frac{1}{10} \right) \left(1 - \frac{1}{20} \right)$$

$$= 1 - \left(\frac{49}{50} \right) \left(\frac{9}{10} \right) \left(\frac{19}{20} \right) = 1 - \frac{8379}{10000} = 1 - 0.8379 = 0.1621$$

Now, required probability of that the machine will not stop working = $1 - 0.1621 = 0.8379 = 0.84$.

Sol. 100. (c)

Given, A_1, A_2 and A_3 are independent events

such that $P(A_i) = \frac{1}{1+i}$, where $i = 1, 2, 3$

The probability that at least one of the three events occurs

$$P(A_1 \cup A_2 \cup A_3) = 1 - P(\bar{A}_1)P(\bar{A}_2)P(\bar{A}_3)$$

$$= 1 - \left(1 - \frac{1}{2} \right) \left(1 - \frac{1}{3} \right) \left(1 - \frac{1}{4} \right) = 1 - \left(\frac{1}{2} \right) \left(\frac{2}{3} \right) \left(\frac{3}{4} \right) = 1 - \frac{1}{4} = \frac{3}{4}$$

$$\frac{3}{4}$$

Sol. 101. (d)

We have, $P(\text{winning}) = 1/3 \Rightarrow P(\text{drawing}) = 1/6$

$$P(\text{loss}) = 1 - \left(\frac{1}{3} + \frac{1}{6} \right) = \frac{1}{2}$$

Point of win, loss or draw are 2, 0, 1 respectively, 5 points are possible, in three cases (2, 2, 1) (2, 1, 2) (1, 2, 2)

$\therefore$ Required probability = $P(WWD) + P(WDW) + P(DWW)$

$$= 3 \times \frac{1}{3} \times \frac{1}{3} \times \frac{1}{6} = \frac{1}{18}$$

Sol. 102. (c)

Given, binomial distribution $B(6, p)$

Here, $n = 6$

According to the question, $16P(X=4) = P(X=2)$

$$\Rightarrow 16 {}^6C_4 p^4 q^2 = {}^6C_2 p^2 q^4$$

$$\Rightarrow 16 {}^6C_2 p^4 q^2 = {}^6C_2 p^2 q^4$$

$$\Rightarrow 16p^2 = q^2 \Rightarrow 16p^2 = (1-p)^2$$

$$\Rightarrow 16p^2 = 1 + p^2 - 2p \Rightarrow 15p^2 + 2p - 1 = 0$$

$$\Rightarrow 15p^2 + 5p - 3p - 1 = 0$$

$$\Rightarrow 5p(3p+1) - 1(3p+1) = 0 \Rightarrow (3p+1)(5p-1) = 0$$

$$\Rightarrow 3p+1 = 0 \text{ or } 5p-1 = 0 \Rightarrow p = -\frac{1}{3} \text{ or } p = \frac{1}{5}$$

$\therefore p = \frac{1}{5}$, since $-\frac{1}{3}$ is not possible.

Sol. 103. (b)

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{0.2}{0.5}$$

$$\{ \therefore A \subseteq B \Rightarrow P(A \cap B) = P(A) = \frac{1}{2} \}$$

$$\text{and } P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{0.2}{0.2} = 1$$

Sol. 104. (a)

$n(E) = 4$ (number of aces)

$n(S) = 52$ (total number of cards)

$$\therefore P(E) = \frac{4}{52} = \frac{1}{13}$$

Sol. 105. (c)

Both the statements are correct.

Sol. 106. (b)

E = the sum of the faces is either 8 or 9

$= \{(2,6) (6,2) (3,5) (5,3) (4,4) (3,6) (6,3) (4,5) (5,4)\}$

$$\Rightarrow n(E) = 9 \text{ and}$$

$$n(S) = 6 \times 6 = 36$$

$$\therefore P(E) = \frac{9}{36} = \frac{1}{4}$$

Sol. 107. (d)

A and B are mutually exclusive events, so

$$P(A \cup B) = P(A) + P(B) = \frac{1}{3} + \frac{1}{4} = \frac{7}{12}$$

$$\Rightarrow P(\overline{A \cup B}) = 1 - \frac{7}{12}$$

$$\Rightarrow P(\overline{A \cap B}) = \frac{5}{12}$$

$$\{ \therefore \overline{A \cup B} = \overline{A \cap B} \}$$

Sol. 108. (a)

Total number of ways to make a committee of two persons from 2 men and 2 women = ${}^4C_2 = 6$

Number of ways such that both the persons chosen are women = 1.

$\therefore$ Required probability = $1/6$

Sol. 109. (c)

The probability that question will be solved = 1 - probability that the question is not solved by any of the students.

$$= 1 - \left(1 - \frac{1}{2} \right) \left(1 - \frac{1}{3} \right) \left(1 - \frac{1}{4} \right)$$

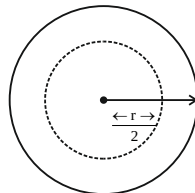
$$= 1 - \frac{1}{2} \times \frac{2}{3} \times \frac{3}{4} = 1 - \frac{1}{4} = \frac{3}{4}$$

Sol. 110. (b)

If the point lies closer to the centre then it should lie inside the region which has radius $\left(\frac{r}{2} \right)$

$$\therefore \text{Required probability} = \frac{P(E)}{P(S)}$$

$$= \frac{\pi \left(\frac{r}{2} \right)^2}{\pi r^2} = \frac{1}{4}$$



Sol. 111. (c)

mean = $np = 12$ (i)

variance = $npq = 4$

$$12q = 4$$

Then $q = 1/3$

We know that $p + q = 1$

So $p = 2/3$

Now by (i)

$$n = 18$$

Sol. 112. (d)

There are total 51 non negative integers up to 50

Out of these 51, 16 are multiple of 3

So required probability $P(X) = 16/51$

Statement 1 is wrong.

Out of these 51, 25 are odd numbers

So required probability $P(Y) = 25/51$

Statement 2 is wrong.

Both statements are incorrect.

Sol. 113. (c)

The triangle is formed when (3,5,7), (3,7,9) or (5,7,9) are chosen.

$$\therefore \text{The required probability} = \frac{3}{{}^5C_3} = \frac{3}{\frac{5 \times 4}{2}} = 0.3$$

Sol. 114. (b)

The probability of selecting two persons out of four, such that one out of two men and one out of two women is to be chosen, is:

$$\frac{{}^2C_1 \times {}^2C_1}{{}^4C_2} = \frac{2 \times 2}{\frac{4 \times 3}{2}} = \frac{2}{3}$$

Sol. 115. (a)

$$P(A \cup B) = P(A) + (A \cap B) - P(A \cap B)$$

$$\text{or } P(B) = P(A \cup B) + P(A \cap B) - P(A)$$

$$\frac{2}{3} + \frac{1}{6} - \frac{1}{2} = \frac{1}{3}$$

$$\text{Now, } P(\overline{A \cap B}) = \frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$$

$$\{ \therefore P(\overline{A}) = 1 - \frac{1}{2} = \frac{1}{2} \}$$

Sol. 116. (c)

$$P(A) = \frac{1}{3}, P(B) = \frac{1}{6}, P(A \cap B) = \frac{1}{12}$$

$$P(B|A) = \frac{P(B \cap A)}{P(A)}$$

$$P(B \cap A) = P(B) - P(A \cap B) = \frac{1}{12}$$

$$P(\overline{A}) = \frac{2}{3}$$

$$\therefore P\left(\frac{B}{\overline{A}}\right) = \frac{12}{\frac{2}{3}} = \frac{18}{2} = 9$$

Sol. 117. (c)

$$np = \frac{2}{3}, npq = \frac{5}{9}$$

$$\Rightarrow q = \frac{5}{9} \times \frac{3}{2} = \frac{5}{6}$$

$$\Rightarrow p = \frac{1}{6}, n = 4$$

$$p(x=2) = {}^4C_2 \left(\frac{1}{6} \right)^2 \times \left(\frac{5}{6} \right)^2 = 6 \times \frac{1}{36} \times \frac{25}{36} = \frac{25}{216}$$

Sol. 118. (c)

If at least 4 out of 5 ships reach safely, then the required probability is:

$$= 5 \times \left(\frac{1}{3} \right)^4 \times \frac{2}{3} + \left(\frac{1}{3} \right)^5 \{ \text{either any 4 or all 5 reach} \}$$

safely}

$$= 5 \times \frac{2}{3^5} + \frac{1}{3^5} = \frac{11}{243}$$

Sol. 119. (b)

$$p(\text{none born in same month}) = \frac{12 \times 11 \times 10}{12 \times 12 \times 12}$$

$p(\text{at least two born in same month})$

$$= 1 - \frac{12 \times 11 \times 10}{12 \times 12 \times 12} = \frac{144 - 110}{144} = \frac{17}{36}$$

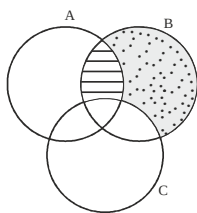
Sol. 120. (a)

From the figure,

$$P(B \cap C) = P(B) - P(A \cap B \cap \overline{C}) + P(\overline{A} \cap B \cap \overline{C})$$

$$= \frac{3}{4} - \left[\frac{1}{3} + \frac{1}{3} \right]$$

$$= \frac{3}{4} - \frac{2}{3} = \frac{1}{12}$$



Sol. 121. (c)

The probability of appearing any odd number = 1/9

∴ the probability of appearing any even number = 2/9

When the dice is tossed the prime numbers that can appear are 2, 3, 5.

Probability of appearing 3, 5 or 2 = $\frac{2+1+1}{9} = \frac{4}{9}$

Sol. 122. (b)

If the student answers a question correctly, the either he/she knows of he/she guesses correct,

i.e. $P + (1-p) \frac{1}{m}$

So, the probability that he/she really knows the answer is

$$\frac{P}{P + (1-p) \frac{1}{m}} = \frac{mp}{mp + 1 - p} = \frac{mp}{(m-p)p + 1}$$

Sol. 123. (a)

Mean = 3 variance

$np = 3npq$

$q = 1/3$

then $p = 2/3$

probability of exactly three success

$$= {}^5C_3 \left(\frac{2}{3}\right)^3 \left(\frac{1}{3}\right) = \frac{80}{243}$$

Sol. 124. (b)

These are direct formulae

Only 2nd formula in incorrect.

Sol. 125. (b)

Probability of solving at least one question

$$= 1 - (0.6)(0.5) = 0.7$$

Sol. 126. (a)

$$n(E) = C(4, 2) \times C(5, 1) = 6 \times 5 = 30$$

$$n(S) = C(9, 3) = 84$$

$$P(E) = \frac{30}{84} = \frac{5}{14}$$

Sol. 127. (b)

It $B \subset A$, then $P(A-B) =$

$$= P(A) - P(A \cap B) = P(A) - P(B)$$

Statement 1 is correct

$P(A \text{ alone or } B \text{ alone})$

$$= P(A) - P(A \cap B) + P(B) - P(A \cap B)$$

$$= P(A) + P(B) - 2P(A \cap B)$$

Statement 2 is false

It A and B are mutually exclusive, then

$$P(A \cap B) = 0$$

Statement 3 is correct

Sol. 128. (b)

$$P = \frac{1}{2} \left[\frac{3}{5} + \frac{5}{8} \right] = \frac{1}{2} \times \frac{49}{40} = \frac{49}{80}$$

Sol. 129. (c)

$$n(E) = 1, n(S) = 10 \times 9 = 90$$

$$P(E) = 1/90$$

Sol. 130. (b)

$$2P(A) = 3P(B)$$

$$\Rightarrow \frac{2P(A)}{P(A \cap B)} = \frac{3P(B)}{P(A \cap B)}$$

$$\Rightarrow \frac{1}{2} \times \frac{P(A \cap B)}{P(A)} = \frac{1}{3} \times \frac{P(A \cap B)}{P(B)}$$

$$\Rightarrow \frac{1}{2} P(B/A) = \frac{1}{3} P(A/B)$$

Sol. 131. (b)

$$E = \{(1,6), (2,5), (3,4), (4,3), (5,2), (6,1)\}$$

$$n\{E\} = 6, n(S) = 36 \Rightarrow P(E) = \frac{6}{36} = \frac{1}{6}$$

Sol. 132. (d)

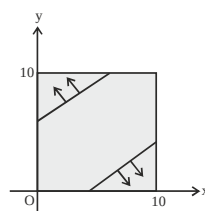
No. of possible out comes

$$= 10 + 6 + 3 + 1 = 20$$

Sol. 133. (c)

$$\text{Probability} = \frac{2 \times \frac{1}{2} \times 5 \times 6}{11 \times 11}$$

$$= \frac{30}{121}$$



Sol. 134. (c)

$$P(A) + P(B) - 2P(A \cap B) = q$$

$$P(A \cap B) = p$$

$$P(A) + P(B) = 2p + q$$

$$1 - P(\bar{A}) + 1 - P(\bar{B}) = p + q$$

$$P(\bar{A}) + P(\bar{B}) = 2 - 2p - q$$

$$P(\bar{A} \cap \bar{B}) = 1 - P(A \cup B)$$

$$= 1 - (p + q) = 1 - p - q$$

Sol. 135. (a)

Required probability

$$= \frac{3}{4} \times \frac{2}{4} \times \frac{3}{4} + \frac{1}{4} \times \frac{2}{4} \times \frac{3}{4} + \frac{1}{4} \times \frac{2}{4} \times \frac{1}{4}$$

$$= \frac{26}{64} = \frac{13}{32}$$

Sol. 136. (c)

Required probability

$$= ({}^8C_6 + {}^8C_7 + {}^8C_8) \left(\frac{1}{2}\right)^8$$

$$= (28 + 8 + 1) \times \frac{1}{256} = \frac{37}{256}$$

Sol. 137. (a)

Required probability =

$$\frac{25 \times 2}{25 \times 2 + 35 \times 4 + 40 \times 5} = \frac{5}{39}$$

Sol. 138. (b)

$$P(A) + P(B) + P(C) = 1$$

$$\Rightarrow P(A) + \frac{3}{2}P(A) + \frac{1}{2} \times \frac{3}{2}P(A) = 1$$

$$\Rightarrow \frac{13}{4}P(A) = 1 \Rightarrow P(A) = \frac{4}{13}$$

Sol. 139. (c)

$$n(S) = 36$$

$$A = \{(1,5), (2,5), (3,5), (4,5), (5,5), (6,5), (5,1),$$

$$(5,2), (5,3), (5,4), (5,6)\}$$

$$B = \{(5,5), (6,4), (4,6), (6,5), (5,6), (6,6)\}$$

$$A \cap B = \{(5,6), (6,5), (5,6)\}$$

$$P(B/A) = \frac{3}{11}$$

Sol. 140. (c)

Sol. 141. (b)

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\frac{2}{3} = P(A) + P(B) - \frac{1}{6}$$

$$P(A) + P(B) = \frac{5}{6}$$

$$P(A) \times P(B) = \frac{1}{6}$$

by solving these two equations

$$P(A) = \frac{1}{2}, P(B) = \frac{1}{3}$$

Sol. 142. (c)

for given condition sum of numbers on dice is 8 sample space is $\{(2,6), (3,5), (4,4), (5,3), (6,2)\}$

out of this sample space probability that the first dice lands on 6 is 1/5.

Sol. 143. (c)

$$= \frac{1}{3} \times \frac{1}{3} + \frac{1}{3} \times \frac{1}{3} + \frac{1}{3} \times \frac{1}{6} + \frac{1}{6} \times \frac{1}{6} + \frac{1}{6} \times \frac{1}{6} = \frac{5}{18}$$

Sol. 144. (b)

$$\frac{{}^3C_2}{{}^n C_2} = \frac{1}{2} \Rightarrow n = 4$$

Sol. 145. (a)

$$\frac{13^4 C_2}{{}^{52} C_2} = \frac{1}{17}$$

Sol. 146. (a)

$$n = 8, p = 1/3, q = 2/3$$

$$\text{mean} = np = 8/3$$

$$\text{variance} = npq = 16/9$$

$$\text{standard deviation} = 4/3$$

Sol. 147. (b)

$$P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} = \frac{P(A) + P(B) - P(A \cup B)}{P(B)}$$

$$= \frac{0.5 + 0.6 - 1}{0.6} = \frac{1}{6}$$

Sol. 148. (a)

$$\frac{\pi \left(\frac{r}{2}\right)^2}{\pi r^2} = \frac{1}{4}$$

Sol. 149. (b)

Sol. 150. (d)

fourteenth card is independent from last 13 card so probability is $4/52 = 1/13$

Sol. 151. (d)

$$P(A \cup B) = 1 - P(A \cap B) = 1 - [1.1 - 0.4] = 1 - 0.7 = 0.3$$

Sol. 152. (a)

$$1 - \frac{1}{2} \times \frac{1}{4} \times \frac{3}{4} = \frac{29}{32}$$

Sol. 153. (c)

Sample space $\{(1,2), (1,3), (1,4), (1,5), (1,6), (2,3), (2,4), (2,5), (2,6), (3,4), (3,5), (3,6), (4,5), (4,6), (5,6)\}$

$$\frac{15}{36} = \frac{5}{12}$$

Sol. 154. (c)

$$\frac{3}{6} \times \frac{1}{2} = \frac{1}{4}$$

Sol. 155. (d)

$$P(H) = 3P(T)$$

$$\text{so } P(H) = 3/4 \text{ and } P(T) = 1/4$$

probability of at most two tails =

$$1 - \frac{1}{4} \times \frac{1}{4} \times \frac{1}{4} = \frac{63}{64} = 0.98$$

Sol. 156. (d)

all three books are defective by multiplication theorem

$$P(D) = \frac{5}{20} \cdot \frac{4}{19} \cdot \frac{3}{18} = 0.009$$

Sol. 157. (d)

Sol. 158. (a)

sample space for

$$2 = \{(1,1)\}$$

$$3 = \{(1,2), (2,1)\}$$

$$5 = \{(1,4), (4,1), (3,2), (2,3)\}$$

$$7 = \{(1,6), (2,5), (3,4), (4,3), (5,2), (6,1)\}$$

$$11 = \{(6,5), (5,6)\}$$

total 15 elements in favour

$$\text{so probability} = \frac{15}{36} = \frac{5}{12}$$

Sol. 159. (c)

probability that none of the trucks chosen meet emission standards?

by multiplication theorem

$$\frac{5}{10} \cdot \frac{4}{9} \cdot \frac{3}{8} = \frac{1}{12}$$

Sol. 160. (c)

by bayes' theorem

$$P(E) = \frac{1}{1 + \frac{1}{2} + \frac{3}{4}} = \frac{4}{9}$$

Sol. 161. (b)

If A and B are mutually exclusive then $P(A) + P(B) \leq 1$

but here $P(A) = P(B) = 0.6$

so first statement is incorrect.

$$P\left(\frac{A}{B}\right) = 1 \Rightarrow P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} = 1$$

$$P(A \cap B) = P(B)$$

it means B is subset of A.

so $P(A \cup B) = P(A)$

now

$$P\left(\frac{B}{A}\right) = \frac{P(A \cap B)}{P(A)} = \frac{P(A \cup B)}{P(A)} = \frac{1 - P(A \cup B)}{1 - P(A)}$$

$$= \frac{1 - P(A \cup B)}{1 - P(A)} = \frac{1 - P(A)}{1 - P(A)} = 1$$

second statement is correct.

Sol. 162. (b)

$${}^4C_2 \left(\frac{1}{6}\right)^2 \left(\frac{5}{6}\right) = \frac{25}{216}$$

Sol. 163. (b)

Sol. 164. (a)

Probability of At least one of them will be selected = $1 - \frac{24}{35} = \frac{11}{35}$

Probability of Only one of them will be selected

$$\frac{1}{7} \cdot \frac{4}{5} + \frac{6}{7} \cdot \frac{1}{5} = \frac{10}{35}$$

Probability of None of them will be selected

$$= \frac{6}{7} \cdot \frac{4}{5} = \frac{24}{35}$$

Probability of both of them will be selected

$$= \frac{1}{7} \cdot \frac{1}{5} = \frac{1}{35}$$

Sol. 165. (a)

$$\frac{{}^6C_4}{{}^{15}C_4} = \frac{6 \times 5 \times 4 \times 3}{15 \times 14 \times 13 \times 12} = \frac{1}{91}$$

Sol. 166. (d)

$$\frac{{}^2C_2 \cdot {}^2C_1}{{}^4C_3} = \frac{1}{2}$$

Sol. 167. (c)

Prime numbers = 2, 3, 5, 7

$$\frac{{}^4C_2}{{}^{10}C_2} = \frac{4 \times 3}{10 \times 9} = \frac{2}{15}$$

Sol. 168. (d)

$$Y = 10 - X$$

number of parameters will be same

$$n = 10$$

$$P = 1 - 1/5 = 4/5$$

Sol. 169. (a)

$$P(\bar{A} \cup \bar{B}) = 1 - [P(B) - P(A \cap B)]$$

$$= 1 - [0.5 - 0.4] = 0.9$$

$$P(\bar{B} / \bar{A}) = \frac{P(\bar{B} \cap \bar{A})}{P(\bar{A})} = \frac{1 - P(A \cup B)}{1 - P(A)}$$

$$\frac{1 - P(A \cup B)}{1 - P(A)} = \frac{1 - [0.6 + 0.5 - 0.4]}{1 - 0.4} = 0.5$$

Sol. 170. (b)

failures caused by X

$$\frac{0.02 \times 0.05}{0.02 \times 0.05 + 0.03 \times 0.03 + 0.05 \times 0.02} = \frac{10}{29}$$

Sol. 171. (c)

No two dice show same colour = $6 \times 5 \times 4 = 120$

cases out of them one die show 6 on it, possible

number of such cases is $3 \times 5 \times 4 = 60$

Probability = $60/120 = 1/2$

Sol. 172. (c)

$$P(A \cup B) = 5/6$$

$$P(A \cap B) = 1/3$$

$$P(\text{not } A) = 1/2$$

$$P(A) = 1/2$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$5/6 = 1/2 + P(B) - 1/3$$

$$P(B) = 2/3$$

$$P(A)P(B) = 1/3$$

$$P(A) + P(B) = 1/2 + 2/3 = 7/6 > P(A \cup B)$$

third statement is incorrect

Sol. 173. (c)

$$P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{L + M - P(A \cup B)}{M} \quad \text{we}$$

know that

$$P(A \cup B) \leq 1$$

$$-P(A \cup B) \geq -1$$

$$L + M - P(A \cup B) \geq L + M - 1$$

$$\frac{L + M - P(A \cup B)}{M} \geq \frac{L + M - 1}{M}$$

$$P(A/B) \geq \frac{L + M - 1}{M}$$

Sol. 174. (a)

$$P(\bar{A}) = 1/2, \text{ then } P(A) = 1/2$$

$$P(A \cup B) = 5/6, P(A \cap B) = 1/3$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\frac{5}{6} = \frac{1}{2} + P(B) - \frac{1}{3}$$

$$P(B) = 2/3$$

$P(A)P(B) = 1/3 = P(A \cap B)$ so events are mutually independent

$P(A \cap B)$ is not zero so events are not mutually exclusive.

Sol. 175. (c)

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(E \cup F) = \frac{1}{2} + \frac{1}{2} - \frac{1}{4} = \frac{3}{4}$$

Sol. 176. (d)

$$np = 2/3, npq = 5/9$$

$$\text{then } q = 5/6 \text{ then } p = 1/6$$

$$\text{and } n = 4$$

$$X = 2$$

$$P(X=2) = {}^4C_2 \left(\frac{5}{6}\right)^2 \left(\frac{1}{6}\right) = \frac{25}{216}$$

Sol. 177. (c)

Sol. 178. (d)

$$1. P(E \cup F) \leq 1$$

$$P(E) + P(F) - P(E \cap F) \leq 1$$

$$P(E \cap F) \geq P(E) + P(F) - 1$$

Statement I is correct

$$2. P(E \cup F) = P(E) + P(F) - P(E \cap F)$$

$$3. P(E \cup F) + P(A \cap B) = P(E) + P(F)$$

$$P(E \cup F) \leq P(E) + P(F)$$

Statement 1, 2, 3 all are correct.

Sol. 179. (a)

$$P\left(\frac{A}{B}\right) < P(A)$$

$$\frac{P(A \cap B)}{P(B)} < P(A)$$

$$\Rightarrow P(A \cap B) < P(A)P(B)$$

$$\Rightarrow \frac{P(A \cap B)}{P(A)} < P(B)$$

$$\Rightarrow P\left(\frac{B}{A}\right) < P(B)$$

Sol. 180. (d)

Probability of solving

$$\text{Problem} = 1 - P(\bar{A})P(\bar{B})P(\bar{C})$$

$$\frac{29}{32} = 1 - \frac{1}{2} \cdot \frac{1}{4} \cdot (1 - p)$$

$$\frac{29}{32} = 1 - \frac{1}{8} + \frac{p}{8}$$

$$\frac{p}{8} = \frac{29}{32} - 1 + \frac{1}{8}$$

$$\frac{p}{8} = \frac{29 - 32 + 4}{32}$$

$$\frac{p}{8} = \frac{1}{32} \Rightarrow p = \frac{1}{4}$$

$$\frac{p}{8} = \frac{1}{32} \Rightarrow p = \frac{1}{4}$$

Sol. 181. (d)

Probability of hitting a six

$$= \frac{8}{60} = \frac{2}{15}$$

$$\text{then probability of not hitting a six} = \frac{13}{15}$$

Sol. 182. (a)

3 digit no's are formed using 1, 3, 7 without

repetition.

$$+ + +$$

$$1 \ 2 \ 3$$

$$= 3 \times 2 \times 1 = 6 \text{ (By counting principle)}$$

Probability that no. is divisible by 3

i.e. favourable outcomes (ie, no. formed is

divisible by 3)

total no. of outcomes (3 digit no without

repetition

for a no. to be divisible by sum of digits must be

divisible by 3 but here $1 + 3 + 7 = 11$ which is

not divisible by 3.

So

$\Rightarrow$ favourable outcomes comes out to be zero

ie. $0/6 = 0$

Sol. 183. (a)

Probability that the roots of the equation $x^2 + x +$

$n = 0$ are real $n \in \mathbb{N}$ & $n < 4$

For roots to be real $b^2 - 4ac \geq 0$ (discriminate)

$$(1)^2 - 4(1)(n) \geq 0$$

$$1 - 4n \geq 0$$

$$1 \geq 4n$$

$$n \leq \frac{1}{4}$$

But it is given that $n < 4$ and belongs to $\mathbb{N}$ is

natural no

So there is no such no. exist

So probability will come out to zero

Sol. 184. (d)

$$P(\bar{A}) = \frac{7}{10} \quad P(\bar{B}) = \frac{3}{10}$$

$$P(A/B) = \frac{7}{14} \text{ T.F. } P(B/A)$$

$$P(B/A) = \frac{P(A \cap B)}{P(A)}$$

$$P(A/B) = \frac{3}{14}$$

$$\frac{P(A \cap B)}{P(B)} = \frac{3}{14}$$

$$\frac{P(A \cap B)}{\frac{7}{10}} = \frac{3}{14}$$

$$P(A \cap B) = \frac{3}{14} \times \frac{7}{10} = \frac{3}{20}$$

$$P(B/A) = \frac{\frac{3}{20}}{\frac{3}{10}} = \frac{1}{2}$$

$$P(A) = \frac{3}{10}$$

As we know

$$P(B) + P(\bar{B}) = 1$$

$$P(\bar{B}) = 1 - P(B)$$

$$\text{Or } P(B) = 1 - P(\bar{B})$$

$$1 - \frac{3}{10} = \frac{7}{10}$$

Sol. 185. (a)

Seven white balls and three black balls are randomly placed in a row (no 2 black balls together)

$$\frac{7!}{7!} \times \frac{{}^8P_3}{3!} = \frac{8!}{5! \times 3!}$$

$$= \frac{8 \times 7 \times 6}{3 \times 2} = 56$$

$$\text{Total no. of arrangement} = \frac{10!}{7! \times 3!}$$

P(no 2 black balls together)

$$= \frac{8!}{5! \times 3!} = \frac{8!}{5! \times 3!} \times \frac{7! \times 3!}{10!}$$

$$= \frac{7 \times 6}{10 \times 9} = \frac{7}{15}$$

Sol. 186. (b)

0 can be formed at unit place

When we multiply

5 with an even number

Probability of even

$$\text{Number} = \frac{{}^1C_1 \times {}^2C_1}{{}^5C_2} = \frac{2}{10} = \frac{1}{5}$$

Sol. 187. (b)

$$P(G \cap \bar{T}) = \frac{36}{90} = \frac{2}{5}$$

Sol. 188. (c)

$$P\left(\frac{G}{\bar{T}}\right) = \frac{P(G \cap \bar{T})}{P(\bar{T})} = \frac{\frac{2}{5}}{\frac{63}{90}} = \frac{4}{7}$$

Sol. 189. (d)

$$P\left(\frac{\bar{T}}{G}\right) = \frac{P(\bar{T} \cap G)}{P(G)} = \frac{\frac{27}{90}}{\frac{36}{90}} = \frac{3}{4}$$

Sol. 190. (d)

$$p = 1/3$$

Exactly 3 out of 6

$$\Rightarrow {}^6C_3 \left(\frac{1}{3}\right)^3 \left(\frac{2}{3}\right)^3$$

$$\Rightarrow 20 \times \frac{8}{3^6} = \frac{160}{729}$$

Sol. 191. (b)

no one out of 6

$$\Rightarrow {}^6C_0 \left(\frac{1}{3}\right)^0 \left(\frac{2}{3}\right)^6$$

$$= \frac{64}{729}$$

Sol. 192. (b)

At least one suffer

$$\Rightarrow 1 - \frac{64}{729} = \frac{665}{729}$$

Sol. 193. (b)

There are 34 composite numbers from 1 to 50

$$\text{so required probability} = \frac{34}{50} = \frac{17}{25}$$

Sol. 194. (b)

put $n = 8, 9, 10, \dots, 47, 48$

we will not get a multiple of 7 in nC_7

put $n = 49, 50, 51, 52, 53, 54, 55$

we will get a multiple of 7 in nC_7

now put $n = 56, 57, 58, \dots, 96, 97$

we will not get a multiple of 7 in nC_7

i.e. one group of 7 consecutive numbers will give a multiple of 7 than 6 groups of 7 consecutive numbers will not give a multiple of 7.

so required probability = $1/7$

Sol. 195. (b)

$x + y = 4$ (2 cases)

$x + y = 8$ (6 cases)

$x + y = 12$ (8 cases)

$x + y = 16$ (4 cases)

$x + y = 20$ (0 cases)

total possible outcomes for $x + y$ are $10 \times 9 = 90$

so required probability = $20/90 = 2/9$

Sol. 196. (c)

$$x + \frac{1}{x} > 2, \text{ this expression is always correct}$$

for any real number greater than 1

$$\text{so required probability is } \frac{(n-1)}{n}$$

Sol. 197. (b)

AP of common difference 1

(1, 2, 3), (2, 3, 4), (3, 4, 5), (4, 5, 6)

AP of common difference 2

(1, 3, 5), (2, 4, 6),

same above 6 results will be there in descending AP's

so 12 results are in favour, so required probability = $12/216 = 1/18$

Sol. 198. (d)

$$P(A' \cap B) + P(A \cap B) + P(A \cap B') = P(A \cup B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cup B) = 0.5 + 0.7 - 0.3 = 0.9$$

Sol. 199. (a)

probability of at most 4 tail = $1 - \text{probability of all 5 tails}$

$$= 1 - \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{31}{32}$$

Sol. 200. (d)

cases of getting sum 15 = 10

cases of getting sum 16 = 6

cases of getting sum 17 = 3

cases of getting sum 18 = 1

required probability = $20/216 = 5/54$

Sol. 201. (c)

probability of at least one hit = $1 - \text{probability of no hit}$

$$= 1 - \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{15}{16}$$

Sol. 202. (c)

number of ways of drawing 3 balls from the box with at least one black ball = number of ways of drawing one black ball + number of ways of drawing of two black balls + number of ways of drawing of three black balls

$$= {}^3C_1 {}^6C_2 + {}^3C_2 {}^6C_1 + {}^3C_3 {}^6C_0 = 45 + 18 + 1 = 64$$

Sol. 203. (d)

probability of sinking a ship = $1/5$

probability arriving a ship safely = $4/5$

probability that exactly 3 out of 5 ships would arrive safely

$$= {}^5C_3 \left(\frac{4}{5}\right)^3 \left(\frac{1}{5}\right)^2 = 10 \times \frac{64}{3125} = \frac{128}{625}$$

Sol. 204. (a)

number of outcomes in favour of spade or ace = 16

number of outcomes not in favour of spade or ace = 36

odds against his winning = $36 : 16 = 9 : 4$

Sol. 205. (b)

probability of construction job completed on time = probability when there is no strike + probability when there is a strike

$$= P(S) \cdot P\left(\frac{C}{S}\right) + P(\bar{S}) \cdot P\left(\frac{C}{\bar{S}}\right)$$

$$= 0.6 \times 0.35 + 0.4 \times 0.85 = 0.45$$

Sol. 206. (*)

If both drawn balls are white then there will be 4 white balls and 3 black balls in bag, then probability of white ball from 2nd bag

$$= \frac{{}^3C_2 \cdot 4}{{}^5C_2 \cdot 7} = \frac{12}{70}$$

If both drawn balls are black then there will be 2 white balls and 5 black balls in bag, then probability of white ball from 2nd bag

$$= \frac{{}^2C_2 \cdot 2}{{}^5C_2 \cdot 7} = \frac{2}{70}$$

If drawn balls are one black and one white, then there will be 3 white balls and 4 black balls in bag, then probability of white ball from 2nd bag

$$= \frac{{}^2C_1 {}^3C_1 \cdot 3}{{}^5C_2 \cdot 7} = \frac{18}{70}$$

$$\text{so required probability} = \frac{12}{70} + \frac{2}{70} + \frac{18}{70} = \frac{32}{70}$$

Sol. 207. (c)

probability of appearing multiple of 3 = $1/3$

From the concept of independent event

probability that each face shows only multiples of 3 where three dice are thrown

$$= \frac{1}{3} \cdot \frac{1}{3} \cdot \frac{1}{3} = \frac{1}{27}$$

Sol. 208. (c)

there are 31 days in Decembers

from 1 dec to 28 dec each day will appear 4 times,

probability of 5th Sunday on remaining 3 days = $\frac{3}{7}$

Sol. 209. (b)

$$n + \frac{50}{n} < 50$$

above inequality satisfied or $x = 2, 3, 4, 5, 6, 7, \dots, 48$

there are 47 values of x satisfies the inequality

$$\text{so required probability} = \frac{47}{50}$$

Sol. 210. (c)

for independent events

$$P(A \cup B) = P(A) + P(B) - P(A)P(B)$$

$$0.8 = 0.3 + 1 - k - 0.3(1 - k)$$

$$0.8 = 0.3 + 1 - k - 0.3 + 0.3k$$

$$0.7k = 0.2$$

$$k = \frac{2}{7}$$

Sol. 211. (c)

probability of getting tail in all the four tosses followed by head

$$= P(T).P(T).P(T).P(T).P(H)$$

$$= \frac{3}{4} \cdot \frac{3}{4} \cdot \frac{3}{4} \cdot \frac{3}{4} \cdot \frac{1}{4} = \frac{81}{1024}$$

Sol. 212. (b)

probability of head on one toss = $\frac{3}{4}$

and tail = $\frac{1}{4}$

probability of getting exactly three heads

$$= {}^4C_3 \left(\frac{3}{4} \right)^3 \left(\frac{1}{4} \right)^1 = 4 \times \frac{27}{256} = \frac{27}{64}$$

Sol. 213. (c)

$$Y = 100 - X$$

$$\text{variance of } X = npq = 100 \times \frac{4}{5} \times \frac{1}{5} = 16$$

now we know that variance is independent from change of origin but depends on change of scale. after multiplying each data of X with -1 , variance will be multiplied by $(-1)^2$, then addition of 100 will not affect the value of variance.

so variance of data Y will be 16.

Sol. 214. (a)

results in favour are 4, 9, 16

there are 3 cases for sum 4

and 25 cases for sum 9

and 6 cases for sum 16

so required probability = $\frac{34}{216} = \frac{17}{108}$

Sol. 215. (d)

A, B, C and D are mutually exclusive and exhaustive events

$$P(A) + P(B) + P(C) + P(D) = 1$$

$$\text{let } 2P(A) = 3P(B) = 4P(C) = 5P(D) = k$$

$$\frac{k}{2} + \frac{k}{3} + \frac{k}{4} + \frac{k}{5} = 1$$

$$\frac{77k}{60} = 1$$

$$77k = 60$$

$$77P(A) = \frac{77k}{2} = 30$$

Sol. 216. (d)

any two numbers can be picked so total

$$\text{outcomes} = {}^9C_2 = 36$$

there is only one case in favour

both required different numbers must be 3 and 7.

probability that their product has 1 in its units place = $\frac{1}{36}$

Sol. 217. (b)

difference of numbers is 2

cases : (1,3), (2,4), (3,5), (4,6), (3,1), (4,2), (5,3), (6,4)

difference of numbers is 3

cases : (1,4), (2,5), (3,6), (4,1), (5,2), (6,3)

required probability = $\frac{14}{36} = \frac{7}{18}$

Sol. 218. (d)

probability that a person recovers from a disease is 0.8

probability that exactly 2 persons out of 5 will recover from the disease

$$= {}^5C_2 \left(\frac{4}{5} \right)^2 \left(\frac{1}{5} \right)^3 = 10 \times \frac{16}{3125} = \frac{32}{625} = 0.0512$$

Sol. 219.

Question is based on bay's theorem

Required probability =

$$\begin{aligned} & \frac{P(F)P\left(\frac{C}{F}\right)}{P(F)P\left(\frac{C}{F}\right) + P(\bar{F})P\left(\frac{C}{\bar{F}}\right)} \\ &= \frac{\frac{10}{100} \times \frac{95}{100}}{\frac{10}{100} \times \frac{95}{100} + \frac{90}{100} \times \frac{45}{100}} = \frac{95}{500} = 0.19 \end{aligned}$$

Sol. 220. (c)

6th toss which is different from those obtained in the first five tosses

HHHHHT + TTTTTH

$$= \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{32}$$

Sol. 221. (b)

n students can sit in a row with n

$$(n-1)!$$

$$\text{required probability} = \frac{(n-1)!}{n!} = \frac{1}{n}$$

Sol. 222. (a)

$$9P(X=4) = P(X=4)$$

$$9 \times {}^6C_4 (p)^4 (q)^2 = {}^6C_2 (p)^2 (q)^4$$

$$9 \times p^2 = q^2$$

$$3p = q$$

$$3p = 1 - p$$

$$p = \frac{1}{4}$$

Sol. 223. (a)

all six persons can sit in a row with 6

number of cases of all three boys sitting together

$$= 4$$

Sol. 224. (b)

all six persons can sit in a row with 6

number of cases of boys and girls sit

alternatively (BGBGBG or GBGBGB)

$$= 3 \times 3 \times 3 \times 3 \times 3 \times 3$$

Sol. 225. (d)

all six persons can sit in a row with 6

number of cases of no girl sit

$$\text{together}(XBXBXBX) = 3$$

Sol. 226. (a)

all six persons can sit in a row with 6

number of cases when P and Q takes end

positions (PXXXXQ or QXXXXP)

$$= 4$$

Sol. 227. (d)

all six persons can sit in a row with 6

number of cases when Q and U sit together

$$= 5$$

Sol. 228. (c)

both balls can be black or white

so required probability =

$$= \frac{{}^5C_2 + {}^4C_2}{{}^9C_2} = \frac{10+6}{36} = \frac{4}{9}$$

Sol. 229. (c)

$$np = 5 \quad \dots \dots \dots (i)$$

$$npq = 4 \quad \dots \dots \dots (ii)$$

by solving (i) and (ii)

$$q = \frac{4}{5}$$

$$\text{than } p = \frac{1}{5}$$

now by eq (i)

$$n = 25$$

$$P(X=3) = {}^{25}C_3 \left(\frac{1}{5} \right)^3 \left(\frac{4}{5} \right)^{22}$$

$$P(X=3) = \frac{25 \cdot 24 \cdot 23}{6} \left(\frac{1}{5} \right)^3 \left(\frac{4}{5} \right)^{22}$$

$$P(X=3) = 23 \cdot \frac{4^{23}}{5^{23}}$$

$$5^{23} P(X=3) = 23 \times 4^{23} = \lambda \times 4^4$$

$$\lambda = 23$$

Sol. 230. (a)

$$\frac{2X-3Y}{5X+4Y} = 4$$

$$20X+16Y=2X-3Y$$

$$Y = -\frac{18}{19}X \quad \dots \dots \dots (i)$$

variance of data X

$$= npq = 10 \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{5}{2}$$

variance of data Y will be

$$\frac{5}{2} \times \left(-\frac{18}{19} \right)^2 = \frac{810}{361}$$

Sol. 231. (a)

Total number of words formed by TIRUPATI

$$= \frac{8!}{2! \cdot 2!}$$

number of words with both T together = $7 \times 6!$

required probability

$$= \frac{7 \times 6!}{8!} = \frac{1}{2}$$

Sol. 232. (c)

$$\text{unit digit of } 7^{4n+1} = 7$$

$$\text{unit digit of } 7^{4n+2} = 9$$

$$\text{unit digit of } 7^{4n+3} = 3$$

$$\text{unit digit of } 7^{4n} = 1$$

so unit place of 7^{7n} can be 7, 9, 3, 1

probability of each above digit is $\frac{1}{4}$

Sol. 233. (c)

$$\text{total outcomes} = {}^{15}C_3$$

7 outcomes are in favour

(2,3,6), (2,4,8), (2,5,10), (2,6,12), (3,4,12),

(2,7,14), (3,5,15)

$$\text{so required probability} = \frac{7}{{}^{15}C_3} = \frac{1}{65}$$

Sol. 234. (d)

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A) + P(B) = P(A \cup B) + P(A \cap B)$$

minimum of $[P(A) + P(B)] = \text{minimum of } P(A$

$\cup B) + \text{minimum of } P(A \cap B)$

$$= 0.75 + 0.125 = 0.875$$

Sol. 235. (c)

$$P(A) + P(B) = P(A \cup B) + P(A \cap B)$$

maximum of $[P(A) + P(B)] = \text{maximum of } P(A$

$\cup B) + \text{maximum of } P(A \cap B)$

$$= 1 + 0.375 = 1.375$$

Sol. 236. (b)

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$$

$$P(A \cup B \cup C) = P(A \cup B) + P(C) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$$

$$P(A \cup B \cup C) = 0.8 + 0.5 - P(B \cap C) - 0.3 + 0.2$$

$$P(A \cup B \cup C) = 1.2 - P(B \cap C)$$

$$1 \geq P(A \cup B \cup C) \geq 0.85$$

$$1 \geq 1.2 - P(B \cap C) \geq 0.85$$

$$0.35 \geq P(B \cap C) \geq 0.2$$

minimum value of $P(B \cap C)$ is 0.2

Sol. 237. (c)

by above solution maximum value of minimum value of $P(B \cap C)$ is 0.35

Sol. 238. (b)

probability of at least one tail

$$p = 1 - {}^nC_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^n$$

probability of at least two tail

$$q = 1 - {}^nC_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^n - {}^nC_1 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^{n-1}$$

$$\text{given that } p - q = \frac{5}{32}$$

$${}^nC_1 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^{n-1} = \frac{5}{32}$$

$$n \left(\frac{1}{2}\right)^n = \frac{5}{32}$$

$$n = 5$$

Sol. 239. (a)

$$p = 1 - {}^5C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^5 = \frac{31}{32}$$

$$q = 1 - {}^5C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^5 - {}^5C_1 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^{5-1}$$

$$q = 1 - \frac{1}{32} - \frac{5}{32} = \frac{26}{32}$$

$$p + q = \frac{31}{32} + \frac{26}{32} = \frac{57}{32}$$

Sol. 240. (d)

Results in favour

HH its probability = $1/4$

THH its probability = $1/8$

TTHH its probability = $1/16$

HTHH its probability = $1/16$

TTTHH its probability = $1/32$

HTTHH its probability = $1/32$

THTHH its probability = $1/32$

Required probability

$$= \frac{1}{4} + \frac{1}{8} + \frac{2}{16} + \frac{3}{32} = \frac{19}{32}$$

Sol. 241. (b)

If drawn ball is white then there will be 4 white balls and 2 black balls in bag, then probability of white ball from 2nd bag = $\frac{{}^2C_1 \cdot 4}{{}^4C_1 \cdot 6} = \frac{1}{3}$

If drawn ball is black then there will be 3 white balls and 3 black balls in bag, then probability of white ball from 2nd bag = $\frac{{}^2C_1 \cdot 3}{{}^4C_1 \cdot 6} = \frac{1}{4}$

so required probability

$$= \frac{1}{3} + \frac{1}{4} = \frac{7}{12}$$

Sol. 242. (d)

$$P(A) = 0.25$$

$$P\left(\frac{B}{A}\right) = \frac{P(A \cap B)}{P(A)} = \frac{P(A \cap B)}{0.25} = 0.5$$

$$P(A \cap B) = \frac{1}{8}$$

$$P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} = \frac{1/8}{P(B)} = 0.25$$

$$P(B) = 0.5$$

by above data we can check that

$$P(A \cap B) = P(A)P(B) = \frac{1}{8}$$

so A and B are independent.

$$P(\bar{A} \cup \bar{B}) = P(\overline{A \cap B}) = 1 - P(A \cap B) = \frac{7}{8} = 0.875$$

$$P(\bar{A} \cap \bar{B}) = P(\overline{A \cup B}) = 1 - P(A \cup B)$$

$$= 1 - [0.25 + 0.50 - 0.125] = 0.375$$

Sol. 243. (d)

there are 4 cases of sum 9 and 3 cases of sum 10, so probability of sum 9 or 10 will be $7/36$. and probability of sum neither 9 nor 10 will be $29/36$.

Sol. 244. (a)

$$p = 20\% = 1/5$$

$$P(X = 4) + P(X = 5) + P(X = 6)$$

$$= {}^6C_4 \left(\frac{1}{5}\right)^4 \left(\frac{4}{5}\right) + {}^6C_5 \left(\frac{1}{5}\right)^5 \left(\frac{4}{5}\right)^1 + {}^6C_6 \left(\frac{1}{5}\right)^6 \left(\frac{4}{5}\right)^0$$

$$= \frac{240}{5^6} + \frac{24}{5^6} + \frac{1}{5^6} = \frac{265}{5^6} = \frac{53}{3125}$$

Sol. 245. (d)

Total cases in favour = $6 \times 5 \times 4 = 120$

results in favour

one on first dice = $1 \times 5 \times 4 = 20$

one on second dice = $5 \times 1 \times 4 = 20$

one on third dice = $4 \times 4 \times 1 = 20$

total results in favour = 60

so required probability = $60/120 = 1/2$

Sol. 246. (a)

$z = 6, y = 5$, then 4 values of x are possible

$z = 6, y = 4$, then 3 values of x are possible

$z = 6, y = 3$, then 2 values of x are possible

$z = 6, y = 2$, then 1 values of x are possible

$z = 5, y = 4$, then 3 values of x are possible

$z = 5, y = 3$, then 2 values of x are possible

$z = 5, y = 2$, then 1 value of x are possible

$z = 4, y = 3$, then 2 value of x are possible

$z = 4, y = 2$, then 1 value of x are possible

$z = 3, y = 2$, then 1 value of x are possible

total 20 outcomes are possible.

Sol. 247. (d)

$$\text{mean} = np = 6$$

$$\text{variance} = \sigma^2 = npq = 2$$

$$6q = 2$$

$$q = 1/3 \text{ then } p = 2/3$$

$$\text{then } n = 9$$

Sol. 248. (c)

total number of results = ${}^{10}C_3 = 120$

outcomes in favour are (1,2,3), (2,3,4), (3,4,5)

(4,5,6), (5,6,7), (6,7,8), (7,8,9), (8,9,10)

required probability = $8/120 = 1/15$

Sol. 249. (c)

$$3P(B) = 4P(A) \text{ and } 3P(C) = 2P(B)$$

for three mutually exclusive and exhaustive events

$$P(A) + P(B) + P(C) = 1$$

$$\frac{3}{4}P(B) + P(B) + \frac{2}{3}P(B) = 1$$

$$P(B) = 12/29$$

$$\text{then } P(A) = 9/29$$

Sol. 250. (c)

faces of dice are 4,4,5,5,5,6

probability of 4 or 5 = $5/6$

Sol. 251. (a)

there are three possibilities that all balls can be of same colour

= probability of all black + probability of all yellow + probability of all white

required probability

$$= \frac{2}{12} \cdot \frac{2}{12} \cdot \frac{2}{12} + \frac{4}{12} \cdot \frac{4}{12} \cdot \frac{4}{12} + \frac{6}{12} \cdot \frac{6}{12} \cdot \frac{6}{12} = \frac{1}{6}$$

Sol. 252. (a)

$$P(A)P(\bar{B})P(C) = \frac{5}{6} \cdot \frac{1}{5} \cdot \frac{3}{4} = \frac{1}{8}$$

Sol. 253. (d)

results in favour (CVCVCVC)

$$\frac{4! \times 3!}{3!} = \frac{4! \times 3!}{7!} = \frac{1}{35}$$

Sol. 254. (b)

1, 2, 3, 4, 5, 6 are not in favour

so required probability = $94/100 = 47/50$

Sol. 255. (c)

$$9P(X = 4) = P(X = 2).$$

By binomial distribution formula

$$9 \times {}^6C_4 (p)^4 (q)^2 = {}^6C_2 (p)^2 (q)^4$$

$$9 \times (p)^2 = (q)^2$$

$$3p = q$$

We know that $p + q = 1$

i.e. $p = 1/4$ and $q = 3/4$

so $p = k = 1/4$

Sol. 256. (a)

$$P(X = 3)$$

$$= {}^6C_3 (p)^3 (q)^3 = 20 \left(\frac{1}{4}\right)^3 \left(\frac{3}{4}\right)^3 = \frac{135}{1024}$$

Sol. 257. (a)

Out of total 11 members, any 6 members can be selected as ${}^{11}C_6$ types.

Number of cases in favour of selection of exactly 3 gentlemen = ${}^7C_3 \cdot {}^4C_3$

$$\text{So required probability} = \frac{{}^7C_3 \cdot {}^4C_3}{{}^{11}C_6} = \frac{10}{33}$$

Sol. 258. (d)

Out of total 11 members, any 6 members can be selected as ${}^{11}C_6$ types.

Number of cases in favour of selection of at least 2 ladies = exactly 2 ladies + exactly 3 ladies + exactly 4 ladies

$$= {}^7C_4 \cdot {}^4C_2 + {}^7C_3 \cdot {}^4C_3 + {}^7C_2 \cdot {}^4C_4 = 210 + 140 + 21 = 370$$

$$\text{So required probability} = \frac{370}{{}^{11}C_6} = \frac{53}{66}$$

Sol. 259. (c)

By the concept of theorem of total probability

Let probability of bonus scheme = $P(X)$

$$P(X) = P(A)P\left(\frac{X}{A}\right) + P(B)P\left(\frac{X}{B}\right) + P(C)P\left(\frac{X}{C}\right)$$

$$P(X) = \left(\frac{3}{10}\right)\left(\frac{4}{9}\right) + \left(\frac{1}{2}\right)\left(\frac{2}{9}\right) + \left(\frac{4}{5}\right)\left(\frac{1}{3}\right) = \frac{23}{45}$$

Sol. 260. (a)

Question is based on bays' theorem

Required probability =

$$= \frac{P(B)P\left(\frac{X}{B}\right)}{P(A)P\left(\frac{X}{A}\right) + P(B)P\left(\frac{X}{B}\right) + P(C)P\left(\frac{X}{C}\right)}$$

$$= \frac{\binom{1}{2} \binom{2}{2}}{\binom{3}{10} \binom{4}{9} + \binom{1}{2} \binom{2}{9} + \binom{4}{5} \binom{1}{3}} = \frac{5}{23}$$

Sol. 261. (b)

$$P\left(\frac{B}{A^c}\right) = \frac{P(B \cap A^c)}{P(A^c)} = \frac{P(B) - P(A \cap B)}{1 - P(A)}$$

$$= \frac{\frac{1}{3} - \frac{1}{4}}{1 - \frac{1}{3}} = \frac{\frac{4}{12} - \frac{3}{12}}{\frac{2}{3}} = \frac{\frac{1}{12}}{\frac{2}{3}} = \frac{3}{8}$$

Sol. 262. (b)

$$P(\overline{A \cap B}) = P(\overline{A \cup B}) = 1 - P(A \cup B)$$

$$= 1 - [P(A) + P(B) - P(A \cap B)]$$

$$= 1 - \left[\frac{1}{3} + \frac{1}{2} - \frac{1}{4} \right] = \frac{5}{12}$$

Sol. 263. (b)

When two dice are thrown together than no of cases

Sum eight = 5 cases

Sum nine = 4 cases

Sum ten = 3 cases

Sum eleven = 2 cases

Sum twelve = 1 cases

Total cases = 15

So required probability = $15/36 = 5/12$

Sol. 264. (c)

Probability of hitting twice

= 1 - probability of no hit - probability of one hit

$$= 1 - {}^7C_0 \left(\frac{4}{5}\right)^7 - {}^7C_1 \left(\frac{1}{5}\right)^1 \left(\frac{4}{5}\right)^6$$

$$= 1 - \left(\frac{4}{5}\right)^7 - 7 \left(\frac{1}{5}\right) \left(\frac{4}{5}\right)^6 = 1 - \left(\frac{4}{5}\right)^6 \left[\frac{4}{5} + \frac{7}{5}\right]$$

$$= 1 - \left(\frac{11}{5}\right) \left(\frac{4}{5}\right)^6$$

Sol. 265. (b)

Mean = np = 200

Variance = npq = 160

q = $160/200 = 4/5$

p + q = 1

so p = $1/5$

Then n = 1000

Sol. 266. (c)

For independent event

$$P(A \cap B \cap C) = P(A)P(B)P(C)$$

$$= 0.6 \times 0.6 \times 0.6 = 0.216$$

Sol. 267. (b)

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= 0.68 + 0.56 - 0.48 = 0.76$$

Sol. 268. (a)

Probability of selection of 2 red balls

$$= \frac{{}^5C_2}{{}^{15}C_2} = \frac{5 \times 4}{15 \times 14} = \frac{2}{21}$$

Sol. 269. (c)

Probability of a white, a red and a blue ball

$$= \frac{{}^5C_1 \cdot {}^6C_1 \cdot {}^4C_1}{{}^{15}C_3} = \frac{5 \times 6 \times 4}{15 \times 14 \times 13} = \frac{24}{91}$$

Sol. 270. (d)

Number of trials is finite for binomial distribution.

Each trials are independent.

Each trial has two possible outcomes.

Statement 2 and 3 are correct.

Sol. 271. (a)

Probability of contradict each other =

$$P(A \cap \overline{B}) + P(\overline{A} \cap B) = \frac{4}{5} \times \frac{1}{6} + \frac{1}{5} \times \frac{5}{6} = \frac{3}{10}$$

Sol. 272. (b)

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\frac{4}{5} = \frac{2}{3} + P(B) - \frac{11}{15}$$

$$P(B) = \frac{13}{15}$$

Sol. 273. (d)

Probability of exactly one event

$$P(A \cap \overline{B}) + P(\overline{A} \cap B) = (p - pq) + (q - pq)$$

$$= p + q - 2pq$$

Probability of atleast one event

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= p + q - pq$$

Both statements are wrong.

Sol. 274. (a)

3 faces are black, 2 are white, one is red

Probability that black, white and red

$$P(A \cap B \cap C) = P(A)P(B)P(C)$$

$$\frac{3}{6} \cdot \frac{2}{6} \cdot \frac{1}{6} = \frac{1}{36}$$

Sol. 275. (a)

Outcomes in favour

HTHT, THTH, HTTH, TTTH

TTHT, THTT, HTTT, TTTT

So required probability = $8/16 = 1/2$

Sol. 276. (d)

We need exactly one prime, one composite, and one neither in three throws.

Number of ways to assign which throw shows which type = $3! = 6$.

For each assignment the number of choices = (choices for prime) $\times$ (choices for composite) $\times$ (choices for neither) = $3 \times 2 \times 1 = 6$

So total favorable outcomes = $6 \times 6 = 36$

Total possible outcomes for three dice = $6^3 = 216$.

Probability = $36/216 = 1/6$

Sol. 277. (b)

Out of first 50 integers 10 are divisible by 5, and 5 are divisible by 9 and only 1 is divisible by 45

$$P(A \cup B) = \frac{10}{50} + \frac{5}{50} - \frac{1}{50} = \frac{7}{25}$$

Probability of either divisible by 5 nor by 9

$$1 - \frac{7}{25} = \frac{18}{25}$$

Sol. 278. (d)

For odd sum, one number should be even and one odd

So required probability

$$= \frac{{}^{25}C_1 \cdot {}^{25}C_1}{{}^{50}C_2} = \frac{25 \times 25}{50 \times 49} \times 2 = \frac{25}{49}$$

Sol. 279. (a)

By binomial distribution

$$P(X = 1) = 0.4096$$

$${}^{25}C_1 p q^4 = 0.4096 \quad \dots\dots(i)$$

$$P(X = 2) = 0.2048$$

$${}^{25}C_2 p^2 q^3 = 0.2048 \quad \dots\dots(ii)$$

Divide (i) by (ii)

$$\frac{5q}{10p} = 2$$

$$q = 4p$$

we know that p + q = 1

so p = 0.2 = k and q = 0.8

Sol. 280. (d)

By multiplication theorem

$$P\left(\frac{A}{B}\right)P(B) = P(A \cap B) = P\left(\frac{B}{A}\right)P(A)$$

$$0.5 \times 0.4 = P\left(\frac{B}{A}\right) \times 0.3$$

$$P\left(\frac{B}{A}\right) = 0.667$$

Sol. 281. (d)

$$P(\overline{A \cup B}) = 1 - P(A \cap B)$$

$$= 1 - [P(A) - P(A \cap B)]$$

$$= 1 - \left[\frac{1}{3} - \frac{1}{4} \right] = \frac{11}{12}$$